SEC P vs NP log 2026-05-08

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-17 23:46 UTC

SEC P vs NP — daily log 2026-05-08

118 cycles recorded

Möbius Mass of Gate-Support Meet-Semilattice Bounds AC^0 PARITY Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Rota incidence algebra (Möbius functions of finite meet-semilattices) — a corner of order theory rarely formalized for circuit complexity, with < 5 papers crossing it with AC^0 or communication-matrix lower bounds × AC^0 circuit size for $PARITY_n$, viewed via the Sherstov pattern-matrix lift to deterministic two-party communication complexity (so the S_n action on inputs lifts to a row/column symmetry of the communication matrix)
- **Recorded**: 2026-05-08 00:24 UTC
- **Entry ID**: `15f8d5df3519`

Statement

For an AC^0 circuit C on n inputs, let P_C be the meet-semilattice generated by the gate variable-supports $\{S(g) : g \in gates(C)\}$ together with □ and [n], closed under set intersection and ordered by $\subseteq$, and let μ_{P_C} denote its Möbius function. Define the Möbius-mass invariant $\psi(C) := \log_2(1 + \Sigma_{S \in P_C} |\mu_{P_C}(□, S)|)$; this is naturally invariant under the S_n action permuting input variables, hence descends to the Sherstov pattern-matrix symmetry group. We conjecture that for every AC^0 circuit C of depth d that computes $PARITY_n$ exactly, $\psi(C) \geq c \cdot n^{1/(d-1)}$ for an absolute constant $c > 0$ independent of d, n; a single counterexample (one PARITY-correct circuit with $\psi(C)$ below this threshold) refutes it.

Rationale

Random restrictions are blind to the global meet-structure of gate supports, but the Möbius function on the support lattice encodes all higher-order intersection inclusion-exclusion in one rational number — exactly what shellability/Cohen-Macaulay arguments leverage. Because PARITY genuinely depends on all n inputs, any AC^0 circuit computing it must populate rank-strata of its support meet-semilattice in a way that prevents the alternating sum from collapsing, forcing $|\mu|$-mass to accumulate as the depth tightens the fan-in budget per layer.

Novelty

- Judge: `NOVEL` over 7 arXiv hits

Top hits consulted: - [2304.02770v2] Tight Correlation Bounds for Circuits Between AC0 and TC0 - [1410.8702v3] The Möbius function of the small Ree groups - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [0805.2135v1] Communication Lower Bounds Using Dual Polynomials - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity?

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out (returncode=124) after 240s without producing any RESULT line or data, so neither the support nor falsification criteria can be evaluated. | next: Reduce the parameter grid (e.g., cap n at 20 and d at 3) or precompute the meet-semilattice incrementally to fit within the time budget, then rerun the pre-registered protocol.

Secant Variety Dimension Lower-Bounds Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry of Secant Varieties × Randomized Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-08 00:38 UTC
- **Entry ID**: `9499045b5ede`

Statement

For the communication matrix M of the DISJOINTNESS problem on n elements, the dimension of the secant variety sec(M) satisfies dim(sec(M)) ≥ c·n for some absolute constant c > 0. This implies R(DISJ_n) = Ω(n) via the dimension-communication complexity duality conjecture.

Rationale

Secant variety dimension captures the 'spread' of matrix rank in projective space. For DISJOINTNESS, the sparse structure of M forces its secant variety to have high dimension, which correlates with the need for Ω(n) communication to distinguish disjointness queries.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.14s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3087e973.py", line 82, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3087e973.py", line 63, in run_trial
    matrix = generate_disjointness_matrix(n)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3087e973.py", line 19, in generate_disj
    matrix = [[0] * (2**n) for _ in range(2**n)]
              ~~~~^~~~~~~~
MemoryError
```

Judge reasoning

Test crashed with MemoryError, preventing evaluation of the conjecture's validity | next: Optimize matrix generation for sparse representations or use incremental testing with smaller n

---

Stern-Brocot Rank of Truth Tables Lifts Average-to-Worst MCSP

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Continued-fraction / Stern-Brocot combinatorics (Hensley sums of partial quotients, Gauss–Kuzmin statistics) × Average-case to worst-case hardness lifting for gap-MCSP via Yao-XOR amplification on truth-table integers
- **Recorded**: 2026-05-08 00:42 UTC
- **Entry ID**: `7d4339776448`

Statement

For f: {0,1}^k → {0,1}, encode T(f) as the bigint p_f = Σ_x f(x)·2^{val(x)} ∈ [0, $2^{\{2}$k}) and define the Stern-Brocot rank SB(f) = Σ_j a_j, the sum of partial quotients of the continued-fraction expansion of p_f / $2^{\{2}$k} (equivalently, the depth of p_f/$2^{\{2}$k} in the Stern-Brocot tree). We conjecture: (i) for uniformly random f, E[SB(f)] = Θ(2^k · k) with concentration |SB − E[SB]| = O(2^k · k^{1/2}) whp; (ii) every f computed by a Boolean circuit of size s ≤ 2^{k/4} satisfies SB(f) ≤ C · s · 2^{k/2} · k for an absolute constant C; (iii) SB is XOR-additive up to slack: SB(f ⊕ g) ≥ SB(f) + SB(g) − O(k+□), so any worst-case lower bound SB(h) = ω(2^{k/2}·k) on a target h amplifies via t-fold XOR into a 2^{Ω(t)} lower bound on gap-MCSP for the hardness-amplified family — a concrete avg-to-worst transfer.

Rationale

Sums of continued-fraction partial quotients are arithmetic invariants reflecting the multiplicative structure of the integer p_f, which is exactly the regime in which Yao-XOR composes truth-table integers; truth tables of low-complexity circuits inherit short Euclidean reductions while Gauss–Kuzmin / Hensley statistics force random TTs to have SB □ 2^k·k. The k/4 circuit-size threshold sits a polynomial factor below the Razborov–Rudich largeness threshold 2^{k/2}, so the predicted gap can in principle evade natural proofs while still yielding sub-exponential lower bounds. Continued-fraction combinatorics has essentially never been used as a circuit-lower-bound proxy, so the bridge could expose new arithmetic/structural invariants of MCSP-hard truth tables.

Novelty

- Judge: " over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by NATURAL_PROOFS barrier

Slice Rank of Layer-Variable Tensor Lower-Bounds Read-Twice BP Size for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Slice rank / partition rank in additive combinatorics (Croot-Lev-Pach 2016, Tao 2016, Naslund 2020) × Read-twice branching programs computing IP_2 = ⊕_i x_i·y_i (Borodin–Razborov–Smolensky 1993 read-twice barrier)
- **Recorded**: 2026-05-08 01:14 UTC
- **Entry ID**: `f9268bf85309`

Statement

For a read-k branching program P with vertex set V(P), define the layer-variable tensor T_P ∈ F_2^{|V|×|V|×n} by T_P[u,v,j] := (number of edges u→v labelled by a literal of x_j) mod 2, and let ρ(P) := slice_rank_{F_2}(T_P). Conjecture (A) For every read-twice branching program P, ρ(P) ≤ 6·log_2 |V(P)|. Conjecture (B) For every (unrestricted) branching program P that computes IP_2 on n variables, ρ(P) ≥ n/8; in particular any read-twice P for IP_2 satisfies |V(P)| ≥ 2^{n/48}, and a single (P,ρ) pair refuting either inequality falsifies the claim.

Rationale

Slice rank is the F_2 invariant that drove the cap-set collapse: it measures how far a 3-tensor is from being decomposable along axis-aligned slabs, and read-twice forces the variable axis of T_P to be a sparse, low-multiplicity fiber pattern — exactly the regime where Tao-style slice arguments give logarithmic bounds. Conversely, IP_2 is the canonical bilinear obstruction, and any BP encoding it must spread the x_j-fibers of T_P across many slices because each x_j actually multiplies a distinct y_j, conjecturally producing a partition-rank-style anti-cap configuration with linear slice rank. The bridge is novel because slice rank has been deployed for cap sets, sunflower bounds, and matrix multiplication exponent, but never (to <5 papers) as a BP-size lower-bound invariant.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ebc5334.py", line 231, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ebc5334.py", line 192, in run_trial
    T_P = generate_read_twice_bp(n, random.choice(w_values))
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ebc5334.py", line 104, in generate_rea
    u = random.choice(V)
        ^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 347, in choice
    raise IndexError('Cannot choose from an empty sequence')
IndexError: Cannot choose from an empty sequence
```

Judge reasoning

The test crashed with an IndexError in generate_read_twice_bp (empty sequence passed to random.choice) before producing any data, so neither the support nor falsification conditions were evaluated. | next: Fix generate_read_twice_bp to guarantee a non-empty vertex set V (e.g., enforce a minimum width and seed initial layers) before sampling, then re-run the 240-sample sweep.

Polymatroid Rank Lower Bound for Monotone DNF Representing k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Polymatroid Theory × Monotone DNF Size
- **Recorded**: 2026-05-08 01:18 UTC
- **Entry ID**: `c3ca2e711eef`

Statement

For any k-CLIQUE indicator function on n vertices, its minimal monotone DNF size is Ω(ρ(n,k)^2), where ρ(n,k) is the rank of the polymatroid defined by the intersection lattice of k-element subsets. For random 3-CNF formulas with clause-density 4, the polymatroid rank ρ(n,k) ≤ log n implies DNF size ≥ n^{1.5}.

Rationale

Polymatroid rank captures the combinatorial geometry of inclusion-exclusion constraints in DNFs. By linking it to k-CLIQUE's lattice structure, we expose inherent complexity in representing clique indicators, which are known to resist efficient monotone circuit representations.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b8682b4f.py", line 106, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b8682b4f.py", line 83, in run_trial
    rho_n_k = rank([[1 if i in clause else 0 for i in range(1, n+1)] for clause in clauses])
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b8682b4f.py", line 44, in rank
    rref = gaussian_elimination(A, b)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b8682b4f.py", line 29, in gaussian_elim
    augmented_matrix[j][k] -= factor * augmented_matrix[i][k]
    ~~~~~~~~~~~~~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError during rank computation, preventing data collection | next: Fix the gaussian_elimination implementation to handle matrix dimensions properly

---

Free Cumulant Gap in Read-Twice BP for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Read-Twice Branching Programs
- **Recorded**: 2026-05-08 01:28 UTC
- **Entry ID**: `adf4036d951b`

Statement

For any read-twice branching program P computing IP_2, the free cumulant κ_4 of its adjacency matrix satisfies $\kappa_4(P) \geq \Omega(n)$, whereas for read-once BPs computing IP_2, $\kappa_4(P) = O(\log n)$.

Rationale

Free cumulants capture non-commutative dependencies in branching programs. IP_2's read-once lower bound stems from its high correlation with parity functions, which manifest as divergent free cumulants in non-commutative models. Read-twice BPs may exhibit structured dependencies that collapse cumulant growth.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6ef5c7d1.py", line 97, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6ef5c7d1.py", line 73, in run_trial
    kappa_4 = free_cumulant(r_transform(A))
                            ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6ef5c7d1.py", line 53, in r_transform
    R[k][0] = determinant(A ** k)
                          ~~^^~~
TypeError: unsupported operand type(s) for ** or pow(): 'list' and 'int'
```

Judge reasoning

Test crashed due to TypeError, preventing data collection. Support fraction cannot be evaluated without successful runs. | next: Fix the r_transform function's matrix exponentiation implementation to handle list-based matrices properly

---

Persistent Homology Barcodes and Communication Complexity of Boolean Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Persistent Homology × Communication Complexity
- **Recorded**: 2026-05-08 01:44 UTC
- **Entry ID**: `ceaca8c608a7`

Statement

For any Boolean function f: $\{0,1\}^n \to \{0,1\}$, the sum of the lengths of the persistent homology bars in its communication matrix's Vietoris-Rips complex satisfies $\Sigma L_i = \Theta(\log CC(f))$, where CC(f) is the deterministic communication complexity. The barcode lengths L_i are measured in the scale of the input space's diameter.

Rationale

Persistent homology captures topological features of the communication matrix's geometry, which may reflect the minimal 'information bottleneck' required for coordination between parties. The barcode lengths could quantify how the function's structure resists simplification via partitioning, aligning with KW's protocol cost-to-circuit-size equivalence.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `-9`, elapsed: `46.80s`

```
(no output)
```

Judge reasoning

Test crashed before producing data, preventing evaluation of support/falsification criteria | next: Re-run test with debugging enabled to identify crash cause

---

Jacobi Recurrence b_2 Floor Constrains SoS-2 Max-CUT Ratio

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Markov–Krein moment theory: Jacobi (three-term recurrence) parameters of empirical spectral measures via the Chebyshev/Wheeler algorithm × Sum-of-Squares degree-2 (Goemans–Williamson SDP) Max-CUT integrality ratio on 3-regular graphs
- **Recorded**: 2026-05-08 01:52 UTC
- **Entry ID**: `6a8b922a8da3`

Statement

For a connected 3-regular graph G on n vertices with adjacency matrix A, let $\mu_G=(1/n)\Sigma\delta_{\lambda_i(A)}$ be the empirical spectral measure and let $\{a_k,b_k\}$*$\{k\geq 0\}$ be its Jacobi parameters (so the orthonormal polynomials satisfy $x\ p_k = b_k\ p$*$\{k+1\} + a_k\ p_k + b_{k-1}\ p_{k-1}$, $b_k>0$). Define the spectral-SDP ratio $\rho(G) := \mathrm{MaxCUT}(G)/[(n/4)(3 - \lambda_{min}(A))]$ and the Jacobi defect $J(G) := b_2(G) - \sqrt{2}$ (where $\sqrt{2}$ is the Kesten–McKay limit for 3-regular trees). Conjecture: there is an absolute constant $c \leq 4$ such that for every 3-regular G on $n \geq 8$ vertices, $\rho(G) \geq 0.879 \Longrightarrow J(G) \geq -c\cdot\log(n)/\sqrt{n}$; a single instance with $\rho(G) \geq 0.879$ and $b_2(G) < \sqrt{2} - 4\cdot\log(n)/\sqrt{n}$ refutes it.

Rationale

Jacobi parameters reduce a spectral measure to its scattering data, and the Kesten–McKay law (the universal local limit for random 3-regular spectra) pins $\{a_k=0, b_0=\sqrt{3}, b_k=\sqrt{2}$ for $k\geq 1\}$; deviations of b_2 measure how far G's local spectrum departs from the free-probabilistic ideal. SoS-2 for Max-CUT is exactly the GW SDP whose tightness for 3-regular graphs is governed by spectral concentration around Kesten–McKay, so a downward push on b_2 should signal exactly the moment-matrix obstruction that lets the SDP overshoot the integral cut. This bridges Markov–Krein scattering data (rare in complexity theory, <5 papers crossed with SDP relaxations) to the open SoS-2 integrality-gap question via a quantity computable in poly(n) without any SDP solver.

Novelty

- Judge: `NOVEL` over 2 arXiv hits

Top hits consulted: - [s2:2512.04951] MAX BISECTION might be harder to approximate than MAX CUT - [380752.380837] On the integrality ratio of semidefinite relaxations of MAX CUT

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9752c379.py", line 98, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9752c379.py", line 67, in run_trial
    v, H = lanczos(G, v0, 12)
           ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9752c379.py", line 23, in lanczos
    w = A @ v[-1]
        ~~^~~~~~~
TypeError: unsupported operand type(s) for @: 'list' and 'list'
```

Judge reasoning

The test crashed with a TypeError in the lanczos routine (using Python lists instead of numpy arrays for matrix-vector multiplication), so no data was produced and the pre-registered support condition cannot be evaluated. | next: Fix the lanczos implementation to use numpy arrays (np.asarray on v0 and A, or store v as an ndarray) and rerun the 120-instance protocol.

Orbit Closure Dimension Bounds Permanent Circuit Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Prehomogeneous Vector Spaces × Boolean Circuit Complexity of Permanent
- **Recorded**: 2026-05-08 02:11 UTC
- **Entry ID**: `165008182215`

Statement

For a random n×n matrix M with entries ±1, let d(M) be the dimension of the orbit closure of M under GL_n×GL_n action. The permanent's minimal circuit size over GF(2) satisfies Circuit-Size(Perm_n) ≥ Ω(d(M) / log n) for all n ≤ 40.

Rationale

Prehomogeneous spaces encode hidden symmetry structures that constrain algebraic complexity. By linking orbit closure dimensions (a geometric invariant) to circuit size (a computational invariant), we expose how algebraic structure might inherently limit or amplify complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.02s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before collecting data; support fraction cannot be evaluated | next: Increase time limit and re-run test with $n \leq 40$

Lie Stabilizer Dimension Linearly Bounds Arithmetic Formula Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric Invariant Theory (sl_N Lie-algebra stabilizers of homogeneous forms) × Arithmetic formula size for homogeneous polynomials (GCT-style hardness for det vs perm)
- **Recorded**: 2026-05-08 02:17 UTC
- **Entry ID**: `44fe923c126a`

Statement

Let $f \in Q[x_1,\ldots,x_N]$ be a homogeneous polynomial of degree $d \geq 3$ that uses every variable ($\partial f/\partial x_i \neq 0\ \forall$ i), and define its reduced Lie stabilizer $rLS(f) := \dim \{ A \in sl_N : \Sigma_{i,j} A_{ij} x_j \partial f/\partial x_i \equiv 0$ in $Q[x] \}$. We conjecture that for every such f computable by an arithmetic formula F of size s (counted as the number of internal +/× gates), $rLS(f) \leq 4 \cdot s$. A single (formula F, expanded polynomial f, computed rLS) triple with $rLS(f) > 4 \cdot s$ refutes the conjecture; conversely, since $rLS(det_n) = 2n^2-2$, any formula for det_n has size $\geq (n^2-1)/2$.

Rationale

Mulmuley–Sohoni's program isolates polynomials whose stabilizer groups are large (det_n is 'characterized by its stabilizer') and predicts their circuit-rigidity reflects that algebraic symmetry, but no explicit numeric bridge from stabilizer dimension to formula size has been proven or empirically pinned down. A linear inequality $rLS \leq 4s$ would give a missing GCT sub-lemma that is computable, sidesteps the natural-proofs barrier (rLS is not large for random truth tables of Boolean functions and is defined on algebraic, not Boolean, objects), and dodges algebrization (it is a property of f as a point in P(Sym^d), not of an oracle access pattern). It targets the permanent-vs-determinant axis directly through the Lie-algebra side of stabilizer theory.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5b89c1e8.py", line 129, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5b89c1e8.py", line 110, in run_trial
    f = expand_polynomial(formula)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5b89c1e8.py", line 79, in expand_polyno
    for var, exp in term:
        ^^^^^^^^
TypeError: cannot unpack non-iterable Fraction object
```

Judge reasoning

The test crashed with a TypeError in expand_polynomial before any of the 840 trials or calibration checks could run, so neither the support nor falsification condition can be evaluated. | next: Fix the polynomial expansion routine (term iteration unpacks a Fraction) so monomials are represented as (coefficient, [(var, exp), …]) tuples, then rerun the 840-trial sweep plus the det_2/det_3/perm_3 calibration.

Finite Plane Line Count Bounds Tseitin Resolution Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Finite Geometry × Resolution Proof Size
- **Recorded**: 2026-05-08 02:26 UTC
- **Entry ID**: `595a1117d469`

Statement

For a Tseitin formula derived from a finite projective plane of order $q \geq 2$, the minimal resolution proof size is $\Theta(q^3)$. The number of lines in the plane is $q^2 + q + 1$, and the conjecture asserts that the resolution size scales proportionally to q^3 for all valid q.

Rationale

Finite geometry provides structured incidence graphs with inherent combinatorial rigidity. Tseitin formulas on such graphs inherit this rigidity, forcing resolution proofs to traverse a super-linear number of clauses. The cubic scaling arises from the interplay between the plane's line count and the expansion properties of its incidence graph.

Novelty

- Judge: `NOVEL` over 3 arXiv hits

Top hits consulted: - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [2111.12700v2] Universality in long-distance geometry and quantum complexity - [0711.0446v1] Asymptotic Properties of Hilbert Geometry

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
(no output)
```

Judge reasoning

Critic challenges conjecture; support fraction >=0.8 not unambiguously confirmed | next: Verify actual support fraction from test metrics

---

Tree-Depth of Tseitin Graph Bounds Resolution Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric Group Theory × Resolution Proof Length
- **Recorded**: 2026-05-08 02:37 UTC
- **Entry ID**: `07532f1e404c`

Statement

For a Tseitin formula on a graph G with tree-depth τ(G), the minimal resolution refutation length is ≥ 2^{Ω(τ(G))}. For graphs with τ(G) ≤ 2 (e.g., planar graphs), the refutation length is polynomial.

Rationale

Tree-depth captures hierarchical complexity of graphs, which may mirror the recursive structure of resolution proofs. High tree-depth implies deep nested dependencies, requiring exponential proof steps. The geometric group theory perspective links graph embeddings to group actions, suggesting structural barriers to efficient refutations.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.14s`

```
tion_length=0"}
TRIAL: {"seed": 593, "metric_name": "refutation_length", "metric_value": 0, "instances_tested": 1, "c
TRIAL: {"seed": 631, "metric_name": "refutation_length", "metric_value": 0, "instances_tested": 1, "c
TRIAL: {"seed": 677, "metric_name": "refutation_length", "metric_value": 0, "instances_tested": 1, "c
TRIAL: {"seed": 727, "metric_name": "refutation_length", "metric_value": 0, "instances_tested": 1, "c
TRIAL: {"seed": 773, "metric_name": "refutation_length", "metric_value": 0, "instances_tested": 1, "c
TRIAL: {"seed": 821, "metric_name": "refutation_length", "metric_value": 0, "instances_tested": 1, "c
TRIAL: {"seed": 877, "metric_name": "refutation_length", "metric_value": 0, "instances_tested": 1, "c
TRIAL: {"seed": 929, "metric_name": "refutation_length", "metric_value": 0, "instances_tested": 1, "c

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_19ccd50b.py", line 181, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_19ccd50b.py", line 181, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                              ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed before producing valid results, preventing reliable assessment of conjecture support/falsification. | next: Fix test code to handle edge cases and re-run with proper error handling

---

Projective Plane Incidence Matrix Deterministic Communication Complexity

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Finite Geometry × Deterministic Communication Complexity
- **Recorded**: 2026-05-08 03:03 UTC
- **Entry ID**: `710cd3eca0ca`

Statement

For a function f whose communication matrix M_f is the incidence matrix of a projective plane PG(2, q), the deterministic communication complexity D(f) is $\Theta(q^2)$. This holds for all q where PG(2, q) exists (q a prime power).

Rationale

The incidence matrix of a projective plane imposes a highly symmetric, structured correlation between inputs. This regularity forces parties to exchange $\Omega(q^2)$ bits to resolve ambiguities in the matrix's entries, as the dual structure of points and lines creates a combinatorial barrier to efficient coordination.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.12s`


```
ut of range'}, {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 1, 'conjectur
TRIAL: {'seed': 167, 'trials': [{'metric_name': 'discrepancy', 'metric_value': None, 'instances_teste
TRIAL: {'seed': 191, 'trials': [{'metric_name': 'discrepancy', 'metric_value': None, 'instances_teste
TRIAL: {'seed': 211, 'trials': [{'metric_name': 'discrepancy', 'metric_value': None, 'instances_teste
TRIAL: {'seed': 233, 'trials': [{'metric_name': 'discrepancy', 'metric_value': None, 'instances_teste
```


Judge reasoning

Test crashed before producing reliable data; multiple counterexamples suggest potential issues but insufficient evidence due to test failure | next: Implement robust error handling for prime power validation and re-run tests with q=2,3,4 (smallest prime powers) to verify conjecture

---

Multilinear Catalecticant Rank Scales Polynomially in ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Apolarity / classical invariant theory (Sylvester catalecticant matrices and partial-derivative ranks of multilinear polynomials) × ACC^0[m] Boolean circuit size and depth (Williams-Beigel-Tarui SYM∘AND decomposition framework, target NEXP ⊄ ACC^0)
- **Recorded**: 2026-05-08 03:10 UTC
- **Entry ID**: `fec763a69d10`

Statement

For a Boolean function f:{0,1}^n→{0,1}, let α_U ∈ GF(2) be its algebraic-normal-form (ANF) coefficient at U⊆[n], and define the level-k multilinear catalecticant matrix M^{(k)}(f) ∈ GF(2)^{C(n,k)×C(n,k)} by M^{(k)}*{S,T}* = α{S∪T} if S∩T=∅ else 0; let MCR_k(f) be its GF(2)-rank. Conjecture: for every f computable by an ACC^0[m] circuit of size s, depth d, and modulus m≤6, MCR_⌊n/2⌋(f) ≤ (s+1)^{2d}; whereas a uniformly random Boolean f satisfies MCR_⌊n/2⌋(f) ≥ ½·C(n,⌊n/2⌋) with probability ≥ 1−2^{-n}. A single ACC^0[m] circuit (s,d) whose MLE has MCR_⌊n/2⌋ > (s+1)^{2d} falsifies it.

Rationale

Williams’ route to NEXP ⊄ ACC^0 hinges on Beigel–Tarui’s representation of ACC^0 functions as low-rank sums of simple monomial terms, and the catalecticant / partial-derivative rank is the canonical apolarity measure of exactly such low-rank decompositions of polynomials — yet apolarity has been used almost exclusively for arithmetic circuit lower bounds (Landsberg, Teitler, Shpilka–Wigderson), not for Boolean ACC^0 over GF(2). If the bound holds, the same SAT-style derandomization that powers Williams’ NEXP separation gains a concrete algebraic certificate; if it fails, we obtain an explicit ACC^0 circuit whose MLE has anomalously large catalecticant rank, which itself is informative. The mapping is purely computational (Möbius transform + GF(2) rank) so it sidesteps the relativization barrier and is not a naturalization of any pseudo-random-function-defeating property because the bound is one-sided on circuits, not on truth-tables.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_eae0541d.py", line 141, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_eae0541d.py", line 95, in run_trial
   truth_table = [evaluate_circuit(circuit, input_values[:i] + [val] + input_values[i+1:]) for i in ra
                                                                 ^^^
NameError: name 'val' is not defined. Did you mean: 'eval'?
```

Judge reasoning

The test crashed with a NameError (‘val’ is not defined’) before any trials produced data, so neither the support nor falsification conditions could be evaluated. | next: Fix the truth-table construction bug in run_trial (properly bind ‘val’ when iterating over input assignments) and re-run the pre-registered protocol across n∈{8,10,12}.

Asymptotic Dimension of KW Space for PARITY Matches Optimal Formula Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse-Geometric KW Bounds (CG-KW); coarse geometry of metric spaces and asymptotic dimension theory (Gromov, Yu, Nowak-Yu) × De Morgan formula depth D(f) for f in `{0,1}^n -> {0,1}`; the `NC^1` / formula-size complexity class measured by the Karchmer-Wigderson communication game
- **Recorded**: 2026-05-08 03:25 UTC
- **Entry ID**: `b0765766a778`

Statement

For the PARITY function PAR_n on n bits, the asymptotic dimension of the KW metric space satisfies asdim(X_{PAR_n}) >= n - c for some absolute constant c >= 0, where X_{PAR_n} = PAR_n^{-1}(0) x PAR_n^{-1}(1) is equipped with the disagreement-Hamming metric d((x,y),(x’,y’)) = #{i : x_i != x’_i or y_i != y’*i, i in disagree(x,y) U disagree(x’,y’)}. Combined with axiom A1 (Tree-to-Dimension),* *this yields D(PAR_n) >= n - c, matching the known tight bound D(PAR_n) = n. Equivalently, no AsymDim_cover of X*{PAR_n} with bounded diameter can have multiplicity smaller than n - c + 1.

Rationale

This sub-conjecture directly stress-tests axiom A1 on the canonical function whose exact De Morgan formula depth (n) is known via Khrapchenko's bound. If A1 is to be a viable route to depth lower bounds, the coarse-geometric invariant asdim(X_f) must at minimum recover this textbook tight case. Failure (asdim(X_{PAR_n}) = o(n)) would refute A1 quantitatively; success (asdim grows linearly in n) provides the first nontrivial empirical evidence that the coarse-geometric KW pipeline recovers Khrapchenko-strength bounds without invoking the quadratic measure, and it would license the use of asdim as a substitute invariant in the KRW composition program (axiom A3).

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [0906.0693v3] An improved lower bound on the counterfeit coins problem - [2401.14623v1] Structure in Communication Complexity and Constant-Cost Complexity Classes - [0711.0446v1] Asymptotic Properties of Hilbert Geometry

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3ec93f4c.py", line 108, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3ec93f4c.py", line 83, in run_trial
    asdim_value = asdim(X)
                  ^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3ec93f4c.py", line 67, in asdim
    multiplicity = linear_programming(metric, r)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3ec93f4c.py", line 33, in linear_progra
    if hamming_distance(i, j) <= r:
       ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3ec93f4c.py", line 22, in hamming_dista
    return sum(1 for a, b in zip(x, y) if a != b)
                             ^^^^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

The test crashed with a TypeError ('int' object is not iterable) in hamming_distance before any L(n) values were computed, so the pre-registered support condition cannot be evaluated. | next: Fix the type bug by ensuring elements of X are passed as tuples/sequences (e.g., flatten (x,y) pairs into bit-vectors) before computing Hamming distance, then rerun for n in {2,...,6}.

Free Logarithmic Energy Deficit Lower-Bounds R(DISJ)

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Free probability — Voiculescu free logarithmic energy $\iint \log|x-y|^{-1}\, d\mu\otimes d\mu$ of empirical singular-value measures (microstate-free entropy χ^*, NOT free cumulants) × Randomized two-party communication complexity $R_{1/3}(f)$ of sign matrices $M\_f \in \{-1,+1\}^{2^n\times 2^n}$, with DISJOINTNESS as the canonical $\Omega(n)$ target and discrepancy disc(M_f) as the computable proxy via $R_{1/3}(f) \geq (1/2)\cdot\log_2(1/\mathrm{disc}(M\_f)) - O(1)$
- **Recorded**: 2026-05-08 03:37 UTC
- **Entry ID**: `ea69d0cf842b`

Statement

Let $M\_f \in \{-1,+1\}^{N\times N}$ with $N=2^n$ be the sign communication matrix of a total Boolean function f, let $\sigma_1 \geq \ldots \geq \sigma\_N$ be its singular values, and let $\mu\_f := (1/N)\ \Sigma\_i\ \delta\_\{\sigma\_i/\sqrt{N}\}$ be the normalized empirical measure on $[0, \sqrt{N}]$. Define the free logarithmic energy deficit $\Delta E(M\_f) := ½\cdot\log(\sigma_1^2/N) - (2/(N(N-1)))\ \Sigma\_\{i<j\} \log|\sigma\_i/\sqrt{N} - \sigma\_j/\sqrt{N}|$. We conjecture (i) for every total f, $\log_2(1/\mathrm{disc}(M\_f)) \geq (1/16)\cdot n\cdot\Delta E(M\_f) - 2$, and (ii) $\Delta E(M\_DISJ\_n) \geq ¼\cdot\log((1+\sqrt{5})/(\sqrt{5}-1)) - o(1)$, so part (i) yields $R_{1/3}(DISJ\_n) = \Omega(n)$ tightly via Lean-provable tensor-product spectral arithmetic of the 2×2 atom [[1,1],[1,0]].

Rationale

DISJ's communication matrix factors as an n-fold Kronecker power of a fixed 2×2 atom whose two singular values are golden-ratio reciprocals, so its empirical spectrum lives on a multiplicative cascade with anomalously high logarithmic repulsion; free-probabilistic log-energy is exactly the right invariant to detect such cascade-spread spectra and is invisible to single-moment bounds (Frobenius, operator-norm) used in classical discrepancy proofs. The deficit ΔE penalizes both spectral concentration (large σ_1) and clustering (small pairwise log-distance), aligning with Forster-type sign-rank arguments without requiring sign-rank itself. Free log-energy avoids the natural-proofs barrier because it is not closed under random restrictions and not large-monotone — random sign matrices have $\Delta E \approx 0$ by Marchenko–Pastur log-energy saturation, so the invariant is sparse on the average function class.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [1211.0310v1] Large Synoptic Survey Telescope: Dark Energy Science Collaboration - [0510346v1] The Dark Energy Survey - [2401.14623v1] Structure in Communication Complexity and Constant-Cost Complexity Classes - [1201.0716v2] Concavification of free entropy - [1412.0452v3] Entropy discrepancy and total derivatives in trace anomaly

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f0f26f5f.py", line 94, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f0f26f5f.py", line 59, in run_trial
    M_f = sign_communication_matrix(f)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f0f26f5f.py", line 18, in sign_communic
    n = len(f)
        ^^^^^^

TypeError: object of type 'function' has no len()
```

Judge reasoning

The test crashed with a TypeError (len() called on a function object) before producing any data, so neither support nor falsification criteria could be evaluated. | next: Fix sign_communication_matrix to derive n from the function's domain (e.g., pass n explicitly or enumerate inputs in {0,1}^n) rather than calling len(f), then rerun the n∈{2..8}×30 protocol.

Free Cumulant Gap in Read-Twice BP for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Read-Twice Branching Programs
- **Recorded**: 2026-05-08 03:38 UTC
- **Entry ID**: `6e9e752ae070`

Statement

For a read-twice BP P computing IP_2, the fourth free cumulant $\kappa_4(\mu_P)$ of its transition matrix eigenvalue distribution satisfies $\kappa_4(\mu_P) \geq \Omega(n)$, while for general read-twice BPs computing constant-depth circuits, $\kappa_4(\mu_P) \leq O(\log \text{size}(P))$

Rationale

Free cumulants capture non-commutative dependencies in BP transition matrices. IP_2's exponential lower bound stems from its inherent correlation structure, which should manifest as divergent free cumulants compared to structured circuits with polynomial size

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b4b37d8d.py", line 157, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b4b37d8d.py", line 138, in run_trial
    IP2_cumulant = free_cumulant(IP2_eigenvalues, n)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b4b37d8d.py", line 48, in free_cumulant
    raise ValueError("Not enough eigenvalues to compute fourth cumulant")
ValueError: Not enough eigenvalues to compute fourth cumulant
```

Judge reasoning

Test crashed due to insufficient eigenvalues for cumulant computation, preventing result validation | next: Verify eigenvalue sampling adequacy in test setup

---

Second-Read Commutator Frobenius Excess Bounds Read-Twice BP Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free probability / non-commutative L^2: Haagerup-style commutator inequalities and the asymptotic-freeness defect between layer transition matrices that touch the same variable twice × Read-twice branching program size for IP_2 = ⊕_{i=1}^n x_i·y_i (Borodin–Razborov–Smolensky 1993 read-twice barrier)
- **Recorded**: 2026-05-08 04:12 UTC
- **Entry ID**: `9b0a625047c3`

Statement

For a read-twice BP P of size S over variables v_1,…,v_m, let v_i be read at layers t_i < t'_i with 0/1-transition matrices A_i^{0,A_i}1 ∈ {0,1}^{S×S} (first read) and B_i^{0,B_i}1 (second read). Define the variable-symmetrized second-read commutator C_i := (A_i$^{0+A_i}$1)(B_i$^{0+B_i}$1) − (B_i$^{0+B_i}$1)(A_i$^{0+A_i}$1), and the invariant ρ(P) := log_2(1 + max_i ||C_i||_F^2). Conjecture: there exists a universal constant α ≤ 4 such that (i) every read-twice BP of size S satisfies ρ(P) ≤ α·log_2 S, while (ii) every read-twice BP computing IP_2 satisfies ρ(P) ≥ n/4; together these would force S ≥ 2^{n/(4α)} on IP_2.

Rationale

In free probability, two operator families that act on disjoint variable supports become asymptotically free and have small commutator Frobenius norm; reading a variable twice forces non-trivial algebraic interaction whose Hilbert–Schmidt size is, in Cook–Reckhow style, bookkeeping for how much 'fresh' state the BP must allocate to remember the first read. The commutator C_i is exactly the obstruction that any extension of bounded arithmetic capturing read-twice BPs would have to certify, so a polylogarithmic upper bound in S on ρ would translate (via Buss-style witnessing) to a witnessing protocol of size O(log S), giving the IP_2 lower bound by mismatch with Forster-style discrepancy. The bridge is computationally light because C_i is a single 4-term matrix product on 0/1 matrices.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_00546418.py", line 104, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_00546418.py", line 70, in run_trial
    C = [matrix_multiply(A[i], B[i]) - matrix_multiply(B[i], A[i]) for i in range(m)]
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_00546418.py", line 39, in matrix_multip
    C[i][j] += A[i][k] * B[k][j]
               ~~~~^^^
TypeError: 'int' object is not subscriptable
```

Judge reasoning

The test crashed with a TypeError in matrix_multiply (indexing an int as a 2D matrix), so no data was produced and neither the upper-bound nor the IP_2 lower-bound criteria could be evaluated. | next: Fix matrix_multiply to correctly handle the 2D list-of-lists transition matrices (verify A[i] and

B[i] are matrices, not flattened ints) and rerun the 480 random trials plus the 4 IP_2 calibration points.

Additive Energy of Fourier Coefficients Bounds SOS Refutation Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive Combinatorics × SOS Refutation Size for CSPs
- **Recorded**: 2026-05-08 04:33 UTC
- **Entry ID**: `16b99cc1e016`

Statement

For a random 3-CNF formula Φ with n variables and m clauses, the additive energy E of its Fourier coefficients satisfies $E = \Theta(m^2 / k)$, where k is the SOS refutation size. If $E < m^2 / (2k)$, then Φ has a SOS refutation of size ≤ k.

Rationale

Additive energy captures the structure of Fourier coefficients, which encode the correlation between clauses. Lower energy implies more structured interactions, enabling faster SOS refutations. This bridges additive combinatorics with CSP proof complexity via Fourier analysis.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.09s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing validation of support fraction or counterexample detection. | next: Increase timeout duration and re-run experiment with identical parameters

---

A_5 Word-Embedding Spectral Norm Lower-Bounds R(DISJ)

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation theory of Barrington-active A_5 word-program embeddings: replacing each ±1 entry of a sign matrix by a fixed unitary 4×4 image of a Barrington-active 5-cycle pair (α,β) under the standard 4-dim irrep of A_5, then reading off the operator norm of the resulting block matrix as a tensor invariant × Randomized two-party communication complexity R_{1/3}(f) of partial Boolean functions f:$\{0,1\}^{k\times\{0,1\}}$k→{-1,+1}, accessed through the spectral/discrepancy proxy R_{1/3}(f) ≥ (1/2)·log_2(2^{2k}/σ_max(M_f)) − O(1), with DISJ_k as the canonical Ω(k) target
- **Recorded**: 2026-05-08 04:46 UTC
- **Entry ID**: `951fcc823dd8`

Statement

Fix the Barrington-active pair α=(1 2 3 4 5), β=(1 3 5 2 4) in A_5 (their commutator is again a 5-cycle) and let ρ:A_5→U(4) be the standard 4-dim irreducible representation, realized as the orthogonal complement of the all-ones vector inside the 5×5 permutation representation. For any f with sign matrix M_f∈{-1,+1}^{2}k×2^k} define W_f∈C$^{4\cdot 2}$k×4·2^k} by the block rule W_f[x,y]=ρ(α) if M_f[x,y]=+1 and W_f[x,y]=ρ(β) if M_f[x,y]=−1, and put η(f):=log_2(σ_max(W_f)/2^k) and η_0(f):=log_2(σ_max(M_f)/2^k). Conjecture: (i) for every f, η(f) ≥ (1/2)·η_0(f) − 1, and (ii) η(DISJ_k) ≥ 0.51·k for all k≥3; a single (k,seed) instance violating (i), or any k≥3 with η(DISJ_k)<0.51·k, falsifies the claim.

Rationale

Barrington’s theorem already shows that non-abelian A_5 word products refuse to collapse Boolean computation, and the same algebraic non-collapse should preserve bilinear-form mass when applied entrywise to a communication matrix because every nontrivial A_5 irrep has no scalar fixed direction under (α,β). If true, σ_max(W_f) becomes a Forster-style tensor invariant computable in 4·2^k complex dimensions whose logarithm sits between log-rank and randomized communication, and inherits Ω(k) directly on DISJ from the Barrington-active hypothesis on (α,β). This routes the canonical NC^1 = formula = BP technique into the COMM_DISJ pipeline through a single SVD.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3aee443a.py", line 144, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3aee443a.py", line 90, in run_trial
    W_f_random = sign_matrix_to_block_matrix(random_sign_matrix, rho_alpha, rho_beta)
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3aee443a.py", line 45, in sign_matrix_t
    W_f[4*i:4*(i+1), 4*j:4*(j+1)] = rho_alpha
    ~~~^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: list indices must be integers or slices, not tuple
```

Judge reasoning

The test crashed with a TypeError (list indexed as a NumPy array) before any trials completed, so no data was produced to evaluate the pre-registered support or falsification conditions. | next: Fix sign_matrix_to_block_matrix to allocate W_f as a NumPy array (e.g., np.zeros((4*2**k, 4*2**k), dtype=complex)) rather than a list, then rerun the full 360-trial sweep.

Algebraic Connectivity Inverse Lower-Bounds Tseitin Resolution Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Spectral Graph Theory × Resolution Proof Length
- **Recorded**: 2026-05-08 05:10 UTC
- **Entry ID**: `a7704bd11a4d`

Statement

For a Tseitin formula on a graph G with n vertices, the resolution length is at least $2^{\Omega(1/\lambda_2)}$, where λ_2 is the Fiedler value (second smallest eigenvalue of the Laplacian) of G.

Rationale

The Fiedler value λ_2 measures the graph's expansion properties. Expanders have small λ_2, leading to Tseitin formulas requiring exponential resolution length. This conjecture links the algebraic connectivity to the complexity of resolution proofs.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_56d0f092.py", line 90, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_56d0f092.py", line 73, in run_trial
    conjecture_holds = resolution_length >= 2 ** (0.5 / fiedler_val)
                                                  ~~~~^~~~~~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with division by zero, likely due to $\lambda_2 = 0$ in disconnected graph. Support condition not met. | next: Verify test case uses connected graphs to avoid $\lambda_2 = 0$, then re-run with proper parameters

---

Metric Dimension Lower Bound on Tseitin Resolution Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Metric Dimension Theory × Resolution Proof Length
- **Recorded**: 2026-05-08 05:18 UTC
- **Entry ID**: `07a9b4327000`

Statement

For a Tseitin formula derived from a graph G, the Resolution length is at least $2^{\Omega(\nu(G))}$, where $\nu(G)$ is the metric dimension of G. Non-expander graphs (with $\nu(G) = O(1)$) admit polynomial-length Resolution proofs.

Rationale

Metric dimension captures the minimum number of nodes needed to uniquely identify all others via distance. High metric dimension implies structural complexity, which correlates with exponential Resolution lower bounds for Tseitin formulas. This bridges combinatorial graph theory with proof complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.31s`

```
''}
TRIAL: {'metric_name': 'Resolution Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Resolution Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Resolution Length', 'metric_value': 16, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'Resolution Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Resolution Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'ν(G)', 'metric_value': inf, 'instances_tested': 1, 'conjecture_holds': False
TRIAL: {'metric_name': 'Resolution Length', 'metric_value': 4, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Resolution Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Resolution Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Resolution Length', 'metric_value': 2, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Resolution Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Resolution Length', 'metric_value': 2, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Resolution Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Resolution Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
RESULT: FALSIFIED counterexample="Resolution length is less than 2^Ω(ν(G))" first_failing_seed=11
```

Judge reasoning

parse_fail | next: NONE

Witness-Complex Betti Sum Lower-Bounds Formula Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Forman discrete Morse theory / mod-2 simplicial homology of the minimal-witness complex of a Boolean function × Karchmer-Wigderson De Morgan formula depth D(f) for f:{0,1}^n→{0,1}
- **Recorded**: 2026-05-08 05:24 UTC
- **Entry ID**: `7887f040a80a`

Statement

For every non-constant Boolean function f:{0,1}^n→{0,1}, let Σ_f^min be the simplicial complex on vertex set [n] obtained as the down-closure of the inclusion-MINIMAL witness sets W(x,y)={i:x_i≠y_i} ranging over pairs (x,y)∈f$^{\{-1\}(1)\times f}${-1}(0), and let β(f)=Σ_{k≥0} dim_{F_2} H_k(Σ_f^min;F_2) be its total mod-2 Betti number. Then the De Morgan (Boolean) formula depth satisfies D(f) ≥ ⌈log_2(1+β(f))⌉. A single instance with decision-tree depth DT(f) < ⌈log_2(1+β(f))⌉ (which would imply D(f)<⌈log_2(1+β(f))⌉) falsifies the conjecture.

Rationale

A KW protocol of depth d partitions the relation R_f into ≤2^d monochromatic rectangles, each labeled by a single witness coordinate i; restricting Σ_f^min to the vertices used by these labels yields a cover by ≤2^d stars (cones over a single vertex), and a Mayer-Vietoris / nerve argument forces β(f)≤2^d-1 whenever such a cover exists. The minimal-witness complex thus encodes a 'shape obstruction' that is invisible to rank-, Fourier-, or sensitivity-based KW techniques. Discrete Morse / simplicial F_2 homology of witness complexes has appeared in <5 papers crossed with formula depth, giving a genuinely fresh combinatorial-topological angle on KW games.

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1802.03129v3] Rank Selection and Depth Conditions for Balanced Simplicial Complexes - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2003.09703v1] Variance function of boolean additive convolution - [2511.07739v3] A Lower Bound for the Fourier Entropy of Boolean Functions on the Biased Hypercube

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d5c959c1.py", line 155, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d5c959c1.py", line 124, in run_trial
    dt_depth = decision_tree_depth(f, n)
               ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d5c959c1.py", line 48, in decision_tree
    return dp(sigma)
           ^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d5c959c1.py", line 37, in dp
    depth_0 = dp(tuple(new_sigma))
              ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d5c959c1.py", line 37, in dp
    depth_0 = dp(tuple(new_sigma))
              ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d5c959c1.py", line 37, in dp
    depth_0 = dp(tuple(new_sigma))
              ^^^^^^^^^^^^^^^^^^^^
  [Previous line repeated 994 more times]
RecursionError: maximum recursion depth exceeded
```

Judge reasoning

The test crashed with a RecursionError in the decision_tree_depth function before any trial data was produced, so neither the support nor falsification condition can be evaluated. | next: Rewrite decision_tree_depth iteratively (or raise sys.setrecursionlimit and add memoization on the restricted truth-table) and rerun the pre-registered trials for n∈{3..8}.

Finite Field Representation Complexity and ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × ACC^0 Circuit Complexity
- **Recorded**: 2026-05-08 05:35 UTC
- **Entry ID**: `25fee6b0408d`

Statement

For a matroid M representable over GF(q), let r(M) denote its rank and s(M) the minimal number of bits required to encode its representation. Then, the size of the smallest ACC^0 circuit computing the matroid's characteristic function is at most O(s(M) * poly(r(M))), with equality holding for all matroids representable over GF(2).

Rationale

Efficient matroid representations over finite fields imply structured combinatorial properties that can be exploited by ACC^0 circuits. The conjecture links representation complexity (s(M)) to circuit size via the interplay between algebraic structure and computational efficiency, leveraging the inherent regularity of matroid representations.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_325883bf.py", line 78, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_325883bf.py", line 56, in run_trial
    matroid = generate_random_matroid(n, q)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_325883bf.py", line 48, in generate_rand
    if all(row[j] == 0 or row[i] != 0 for i in range(j)):
                                                     ^
NameError: name 'j' is not defined
```

Judge reasoning

Test crashed due to NameError, preventing data collection. Support condition cannot be evaluated without successful trials. | next: Fix the NameError in generate_random_matroid by defining 'j' in the loop scope

---

Matroid Connectivity and ACC^0 Circuit Size for 3-SAT Instances

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: MATROID_CONNECTIVITY × ACC^0_CIRCUIT_SIZE
- **Recorded**: 2026-05-08 05:45 UTC
- **Entry ID**: `66fa281d7a4e`

Statement

For any 3-SAT instance Φ with n variables, the matroid connectivity of the incidence matrix of Φ is Θ(log n) if and only if Φ can be expressed by an ACC^0 circuit of size O(n^k) for some constant k.

Rationale

Matroid connectivity measures the structural complexity of the incidence matrix, which could be related to the circuit complexity. Higher connectivity might indicate a more complex structure, requiring larger circuits.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4843ca4e.py", line 78, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4843ca4e.py", line 27, in run_trial
    connectivity = matroid_connectivity(incidence_matrix)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4843ca4e.py", line 71, in matroid_conne
    return gaussian_elimination(matrix)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4843ca4e.py", line 66, in gaussian_elim
    if j != i and A[j][i] != 0:
                  ~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError, preventing data collection. Pre-registered support condition cannot be evaluated. | next: Debug and fix the Gaussian elimination implementation to handle matrix dimensions properly

---

Sparsest-Cut LP Value Lower-Bounds Tseitin Resolution Length

- **Verdict**: INCONCLUSIVE
- **Bridge**: Leighton-Rao sparsest-cut LP relaxation and metric (semimetric) embedding theory; multicommodity flow LP duality (Leighton-Rao 1988, Linial-London-Rabinovich 1995) — a flow/metric certificate of expansion that is computationally distinct from λ_2 and rarely linked to proof complexity × Resolution refutation length $L_R(\text{Tseitin}(G,\sigma))$ of odd-charged Tseitin CNFs on 3-regular graphs G (the canonical sub-Frege gateway target for EF lower bounds)
- **Recorded**: 2026-05-08 05:56 UTC
- **Entry ID**: 4d59152d4522

Statement

For every connected 3-regular graph G on n vertices and odd parity vector $\sigma \in \{0,1\}^V$ with $\Sigma\, \sigma \equiv 1 \pmod 2$, define $\nu(G) := n^2 \cdot \Phi_{LP}(G)$, where $\Phi_{LP}(G) := \min \{ \Sigma_{(u,v)\in E}\, d(u,v) : d$ a semimetric on V with $\Sigma_{u<v}\, d(u,v) = 1 \}$ is the Leighton-Rao sparsest-cut LP value (a polynomial-size LP). Then there is an absolute constant $c > 0$ with $\log_2 L_R(\text{Tseitin}(G,\sigma)) \geq c \cdot \nu(G)$ for every such (G,σ), and moreover $\nu(G) \leq 8$ whenever G has tree-width ≤ 3 (so ν is $O(1)$ on every non-expander family).

Rationale

Tseitin Resolution lower bounds traditionally invoke spectral expansion, but the LP-dual sparsest-cut certificate is a structurally different — flow-theoretic — witness of expansion that decouples from the second Laplacian eigenvalue and survives spectral perturbations. A direct quantitative link from Φ_{LP} to Resolution length would import the Leighton-Rao / metric-embedding toolkit into proof complexity and give a poly-computable, integrality-gap-aware lower-bound certificate. The cross sparsest-cut-LP × Tseitin-Resolution has, to my knowledge, fewer than five papers despite both fields being well developed.

Novelty

- Judge: `NOVEL` over 2 arXiv hits

Top hits consulted: - [978-3-642-38536-0_14] Graph Expansion, Tseitin Formulas and Resolution Proofs for CSP - [s2:1b3c2faed501856fd5b542e3b6f6185f210f9] Expander Flows and Sparsest Cut Linear Programming

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
Error solving linear system for n=8: division by zero
Error solving linear system for n=8: division by zero
Error solving linear system for n=8: division by zero
Error solving linear system for n=8: division by zero
Error solving linear system for n=8: division by zero
Error solving linear system for n=8: division by zero
Error solving linear system for n=8: division by zero
Error solving linear system for n=8: division by zero

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_711af89a.py", line 185, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_711af89a.py", line 157, in run_trial
    clauses = tseitin_formula(G, sigma)
              ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_711af89a.py", line 75, in tseitin_formu
    clauses.append([-literals[u]])
                    ^^^^^^^^^^^^
TypeError: bad operand type for unary -: 'str'
```

Judge reasoning

The test crashed with a TypeError in tseitin_formula (unary minus on a string literal) before any trials completed, so neither the support nor falsification criteria could be evaluated. No data was produced. | next: Fix the literal-encoding bug (ensure literals[u] is an int, not a str) in tseitin_formula and re-run; also handle the n=8 ‘division by zero’ in the LP solver (likely degenerate semimetric normalization).

Real Dimension Lower Bound for AC^0 PARITY Circuits

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × AC^0 Circuit Complexity
- **Recorded**: 2026-05-08 06:04 UTC
- **Entry ID**: `c9f273ee89ca`

Statement

For any AC^0 circuit C computing PARITY on n variables, the real dimension of the variety defined by the equations of C is at least $\log_2(\text{size}(C))$

Rationale

The real dimension of the variety captures the geometric complexity of the circuit's polynomial representation. A lower bound on this dimension would imply a lower bound on the circuit size, offering an alternative to the switching lemma's random restriction argument.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
d=39.98232491929524'}
TRIAL: {'metric_name': 'real_dimension', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'real_dimension', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'real_dimension', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'real_dimension', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'real_dimension', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'real_dimension', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'real_dimension', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'real_dimension', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'real_dimension', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'real_dimension', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'real_dimension', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_hol
RESULT: FALSIFIED counterexample="dim < lower_bound" first_failing_seed=11
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: Multiple counterexamples show real dimension 1.0 < claimed lower bounds (e.g., 2.32, 39.9). | next: Analyze the specific AC^0 PARITY circuit structures producing dim=1.0 to identify structural flaws in the conjectured lower bound.

Free Cumulant Gap in Read-Twice Branching Programs for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Read-Twice Branching Programs (BP)
- **Recorded**: 2026-05-08 06:11 UTC
- **Entry ID**: `e97847babd53`

Statement

For a read-twice BP P computing IP_2, the free cumulant gap $\rho(P) = \Omega(n)$, whereas for any read-once BP Q, $\rho(Q) = O(\log n)$.

Rationale

Free cumulants capture the non-commutative structure of the BP's transitions, and the exponential gap in IP_2's complexity suggests a corresponding gap in free cumulant measures. This invariant distinguishes read-twice BPs from read-once ones by leveraging the algebraic properties of non-commutative random variables.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c2fd3bf7.py", line 106, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c2fd3bf7.py", line 79, in run_trial
    rho_Q = free_cumulant_gap(read_once_bp)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c2fd3bf7.py", line 66, in free_cumulant
    v /= math.sqrt(sum(x**2 for x in v))
TypeError: unsupported operand type(s) for /=: 'list' and 'float'
```

Judge reasoning

Test crashed due to TypeError, preventing data collection. Pre-registered support condition cannot be evaluated. | next: Debug the free_cumulant_gap function's division operation to handle list types properly

---

Plunnecke Doubling of Top-Fourier Spectrum Bounds ACC0 Size

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Additive combinatorics (Plunnecke-Ruzsa sumset doubling applied to the large-Fourier-spectrum of a Boolean function in F_2^n) × ACC^0[m] circuit size for explicit Boolean functions, accessed through a Williams-style natural property R: f -> {0,1} computable in 2^O(n) time
- **Recorded**: 2026-05-08 06:17 UTC
- **Entry ID**: `23b7dc950026`

Statement

Let f:{0,1}^n -> {-1,+1} with Walsh-Hadamard coefficients f_hat(a) = 2^{-n} sum_x f(x)(-1)^{a.x}, and let S(f) = {a in F_2^n : |f_hat(a)| >= 1/n} be the 1/n-large spectrum. Define the Plunnecke doubling pi(f) = log_2 |S(f) XOR S(f)| / log_2 max(2,|S(f)|), where XOR-sumset is taken in the additive group F_2^n. Conjecture: every f computed by an ACC^0[m] circuit of size s and depth d satisfies pi(f) <= 1 + (d log s)/sqrt(n) + o(1), and hence any explicit family with pi(f_n) >= 1.5 + Omega(1) for infinitely many n is not in ACC^0[m]; the property R := [pi >= 1.5] is constructive (computable in 2^O(n) time via FWHT) yet excludes ACC^0, threading the natural-proofs needle because R fails on sparse-spectrum PRG candidates.

Rationale

Linial-Mansour-Nisan and Tal's Fourier-concentration bounds for AC^0, together with Razborov-Smolensky low-degree approximators for ACC^0[p], force the top spectrum of an ACC^0 function to live in a low-Hamming-weight, hence low-Freiman-dimensional, region of F_2^n, which implies a small Plunnecke doubling on its XOR-sumset. Plunnecke-Ruzsa doubling of the spectrum is a Freiman-Ruzsa invariant essentially absent from the ACC^0 lower-bound literature (<5 papers cross additive-combinatorics doubling with ACC^0), giving a fresh, locally-checkable natural property aligned with Williams' algorithmic-method circuit-SAT framework. Because pseudorandom function candidates with a designed sparse heavy-spectrum will have |S|<=2 and trivially small pi, R is not useful against PRGs, providing a concrete attempt at minimal-violation natural property.

Novelty

- Judge: " over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by NATURAL_PROOFS barrier

Matroid Rank Lower-Bounds Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Randomized Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-08 06:23 UTC
- **Entry ID**: `13efcfb2aa35`

Statement

For any matroid M representable over GF(2), the communication complexity $R_{1/3}$(DISJ) of the characteristic function of M's independent sets satisfies $R_{1/3}$(DISJ) ≥ rank(M) · $\log_2$(n).

Rationale

Matroid rank captures combinatorial structure of independence, while DISJOINTNESS requires coordination across subsets. Linking them via GF(2) representability exposes algebraic constraints on communication protocols.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1083e7e.py", line 76, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1083e7e.py", line 57, in run_trial
    A = generate_random_matroid(n, p)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1083e7e.py", line 19, in generate_rand
    while not is_independent_set(A, n, p):
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1083e7e.py", line 25, in is_independen
    if any(all(B[j][k] == A[i][k] for k in range(i)) for j in range(i)):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1083e7e.py", line 25, in <genexpr>
    if any(all(B[j][k] == A[i][k] for k in range(i)) for j in range(i)):
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e1083e7e.py", line 25, in <genexpr>
    if any(all(B[j][k] == A[i][k] for k in range(i)) for j in range(i)):
               ^
NameError: name 'B' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'B' in is_independent_set, preventing any data collection | next: Fix the NameError in is_independent_set by properly initializing matrix B

Tree-to-Dimension Tightness for MAJ_3: Optimal KW Protocols Yield Multiplicity-4 Covers of X_{MAJ_3}

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse-Geometric KW Bounds (CG-KW) framework; coarse geometry and asymptotic dimension (Yu/Nowak-Yu) applied to Karchmer-Wigderson games × De Morgan formula depth lower bounds in the Karchmer-Wigderson communication model, specifically for the monotone threshold function MAJ_3 whose known optimal depth D(MAJ_3) = 3
- **Recorded**: 2026-05-08 06:34 UTC
- **Entry ID**: `792c33fb69ec`

Statement

Let X_{MAJ_3} = MAJ_3^{-1}(0) x MAJ_3^{-1}(1) be the KW_metric_space of the 3-bit majority function under the disagreement-Hamming metric d((x,y),(x',y')) = Hamming(x,x') + Hamming(y,y'). Under axiom A1 (Tree-to-Dimension), the optimal depth-3 KW Coarse_partition_tree for MAJ_3 induces, via tree_to_cover, an AsymDim_cover {U_a} of X_{MAJ_3} whose r-multiplicity is at most 4 = D(MAJ_3)+1 for every r >= 1, with each U_a having diameter <= 4. Equivalently, asdim_at_scale(X_{MAJ_3}) <= D(MAJ_3), and conversely no cover of multiplicity <= 3 achieves diameter <= 2, witnessing the depth lower bound asdim(X_{MAJ_3}) >= 2 predicted by the framework's target invariant formula_depth_D(f).

Rationale

This sub-conjecture stress-tests axiom A1 on the smallest non-trivial monotone threshold function whose optimal formula depth is exactly known (D(MAJ_3)=3, attained by the standard balanced AC^0 formula). If A1 is correct, an explicit depth-3 KW protocol must yield a cover with multiplicity bounded by 4 and small diameter; conversely the converse one-sided gap (no multiplicity-3 cover at small scale) certifies that the framework's coarse invariant captures genuine depth lower bounds rather than collapsing trivially. MAJ_3 also serves as a non-PARITY witness: PARITY's KW geometry is highly symmetric (a metric quotient of the cube), whereas MAJ_3 has nontrivial threshold structure, so confirming A1 here generalizes the existing PARITY sub-conjecture to a genuinely different combinatorial regime.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8f5e247f.py", line 110, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8f5e247f.py", line 58, in run_trial
    x = X[leaf]
        ~^^^^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError before producing any RESULT line, so neither the support nor the falsification condition was evaluated. Per rule 2, a crashed test yields INCONCLUSIVE. | next: Fix the leaf-indexing bug in run_trial (ensure the KW protocol's leaf enumeration matches the X array length) and rerun the 5-seed protocol plus brute-force multiplicity check.

Asymptotic Dimension and Resolution Length for Tseitin Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric Group Theory (Asymptotic Dimension) × Resolution Complexity of Tseitin Formulas
- **Recorded**: 2026-05-08 06:34 UTC
- **Entry ID**: `e708abcb82c7`

Statement

For a graph G, let ν(G) be its asymptotic dimension. The Tseitin formula on G requires Resolution length ≥ 2^Ω(ν(G)).

Rationale

Higher asymptotic dimension indicates more complex geometric structure, which increases the difficulty of resolving the formula, leading to exponential growth in proof length.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.09s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with extended computation limits

---

Matroid Spread Bounded by Monotone DNF Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory (polymatroid spread) × Monotone DNF Size for k-CLIQUE
- **Recorded**: 2026-05-08 06:40 UTC
- **Entry ID**: `27f88a70aaf4`

Statement

For any monotone DNF formula Φ computing the k-CLIQUE function on n variables, the spread of the associated polymatroid (defined by Φ's clauses) satisfies spread(Φ) = Ω(k^{1/4} log n), and the DNF size is $n^{\{\Omega(k}${1/4})}.

Rationale

The spread of the polymatroid captures the variability in the rank function, reflecting the structural complexity of the DNF. A higher spread indicates a more intricate dependency structure, necessitating exponential growth in the DNF size to represent the k-CLIQUE function.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.87s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_04ff47ab.py", line 99, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_04ff47ab.py", line 82, in run_trial
    spread_value = spread(dnf)
                   ^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_04ff47ab.py", line 66, in spread
    rank_function = [rank([clause[:i] + clause[i+1:] for clause in dnf]) for i in range(n)]
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_04ff47ab.py", line 43, in rank
    reduced_matrix = gaussian_elimination(matrix)
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_04ff47ab.py", line 28, in gaussian_elim
    factor = Fraction(A[j][i], A[i][i])
             ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/fractions.py", line 281, in __new__
    raise ZeroDivisionError('Fraction(%s, 0)' % numerator)
ZeroDivisionError: Fraction(0, 0)
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing evaluation of the conjecture's validity. | next: Fix the Gaussian elimination implementation to handle zero denominators and re-run tests with diverse seeds

---

Plethysm Coefficient Gap in Symmetric Powers of Permanent Polynomials

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Boolean Circuit Complexity of Permanent
- **Recorded**: 2026-05-08 06:49 UTC
- **Entry ID**: `66a7f7df2a8b`

Statement

For any n ≥ 2, the plethysm coefficient c_{λ}$^{\{Sym}$k(Sym$^{n(\mathbb{C}}$m))} of the permanent polynomial Perm_n satisfies c_λ ≥ 2^{Ω(n)} when λ is a hook partition with arm length ≥ k, while for the determinant Det_m, c_λ ≤ poly(m) for all k ≤ m^{1/2} and m < n^{1.5}.

Rationale

Schur-Weyl duality decomposes tensor powers into irreducible representations, and plethysm coefficients measure how symmetric powers interact with these decompositions. Permanent polynomials exhibit exponential growth in specific hook partitions, while determinants remain polynomially bounded, reflecting their distinct orbit closure structures under GL(n).

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_432a7123.py", line 103, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_432a7123.py", line 60, in run_trial
    m = math.isqrt(n**1.5)
        ^^^^^^^^^^^^^^^^^^
TypeError: 'float' object cannot be interpreted as an integer
```

Judge reasoning

Test crashed due to type error; unable to evaluate conjecture validity | next: Fix integer conversion in test code (replace math.isqrt(n**1.5) with math.isqrt(int(n**1.5)))

Sandpile-Group 2-Rank Lower-Bounds Tseitin DPLL Tree Size

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Chip-firing / abelian sandpile combinatorics: the critical (sandpile) group K(G) of a connected graph, recovered as Z^{V-1}/im(L′(G)) from the reduced graph Laplacian L′(G), with its invariant-factor decomposition obtained by Smith normal form over Z; the 2-rank k_2(G) := dim_{F_2}(K(G)/2K(G)) is a polynomial-time graph invariant essentially absent from the proof-complexity literature × Resolution / DPLL refutation length of Tseitin formulas Tseitin(G,σ) on 3-regular graphs (the Urquhart-style canonical sub-Frege gateway target whose Frege proof complexity remains open and is the natural launchpad for any EF lower-bound program in the spirit of Krajicek-Buss-Razborov-Impagliazzo)
- **Recorded**: 2026-05-08 06:55 UTC
- **Entry ID**: `d46d829f1f79`

Statement

For every connected 3-regular graph G on V vertices and every charge σ ∈ F_2^V with parity sum 1, let k_2(G) := dim_{F_2}(K(G)/2K(G)) be the 2-rank of the sandpile group (equivalently V-1-rank_{F_2}(L′ mod 2)) and let T(G,σ) be the number of decision nodes in the DPLL search tree for Tseitin(G,σ) using fixed-lex variable order with unit propagation. Then log_2 T(G,σ) ≥ k_2(G); a single (G,σ) with T(G,σ) < 2^{k_2(G)} refutes the conjecture.

Rationale

Tseitin(G,σ) encodes solvability of an F_2-linear system whose obstruction is precisely the image of σ under the F_2-augmentation of the integral cycle/cut sequence Z^E → Z^V → K(G); the 2-rank k_2(G) measures how much of that obstruction survives integrally, so a refutation must implicitly certify each independent F_2-cocycle generator in K(G). This connects an integral-lattice invariant of expanders to proof length without invoking algebraic geometry or representation theory. The bridge from chip-firing combinatorics to Frege/EF proof-complexity targets is, to our knowledge, untouched, and gives a falsifiable shadow on the resolution sub-system whose extension to bounded-depth Frege depth is the natural follow-up.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_899352df.py", line 276, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_899352df.py", line 171, in run_trial
    while not is_connected(G):
              ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_899352df.py", line 24, in is_connected
    for v in G[u]:
             ~^^^
KeyError: 0
```

Judge reasoning

The test crashed with a KeyError in is_connected before any instances were evaluated, so no data on log_2 T(G,σ) vs k_2(G) was produced. Per Rule 2, a crash yields INCONCLUSIVE. | next: Fix the graph construction/representation bug (ensure G is a dict keyed by all V vertices before calling is_connected) and rerun the 150-instance sweep.

Secant Variety Dimension Lower-Bounds Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry of Secant Varieties × Randomized Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-08 06:56 UTC
- **Entry ID**: `49cc5859d824`

Statement

For a DISJOINTNESS matrix M ∈ {0,1}^{n×n}, the dimension of its secant variety σ(M) satisfies dim(σ(M)) ≥ c·n for an absolute constant c > 0. This implies R_{1/3}(DISJ_n) ≥ Ω(n) via the dimension-based lower bound for tensor rank.

Rationale

Secant variety dimensions capture the intrinsic complexity of tensor rank, which is directly tied to communication complexity via the tensor rank-communication complexity duality. The DISJOINTNESS matrix's secant dimension scales with n due to its combinatorial structure, offering a novel algebraic geometry-based lower bound.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.05s`


```
: 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'secant_dimension', 'metric_value': 15, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'secant_dimension', 'metric_value': 40, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'secant_dimension', 'metric_value': 40, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'secant_dimension', 'metric_value': 10, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'secant_dimension', 'metric_value': 10, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'secant_dimension', 'metric_value': 5, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'secant_dimension', 'metric_value': 15, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'secant_dimension', 'metric_value': 15, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'secant_dimension', 'metric_value': 40, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'secant_dimension', 'metric_value': 30, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'secant_dimension', 'metric_value': 10, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'secant_dimension', 'metric_value': 30, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'secant_dimension', 'metric_value': 10, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'secant_dimension', 'metric_value': 10, 'instances_tested': 1, 'conjecture_ho
RESULT: SUPPORTED mean=20.0 std=12.649110640673518 support_fraction=1.0
```


Judge reasoning

All trials used n=1, making the metric scale trivially with n=1. The conjecture requires verifying linear growth with n, which was not tested. | next: Test with varying n ≥ 2 to observe metric scaling and validate linear lower bound

Free Cumulant Gap in Read-Twice BPs Bounds Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Disjointness Communication Complexity
- **Recorded**: 2026-05-08 07:12 UTC
- **Entry ID**: `6004bf0dc5da`

Statement

For a random 3-CNF formula Φ on n variables, let μ(Φ) denote the free cumulant gap between its clause indicator variables. Then the randomized communication complexity $R_{1/3}$(DISJ_n) ≥ Ω(μ(Φ) / log n).

Rationale

Free cumulants capture non-crossing partition dependencies in clause variables, which may reveal structural asymmetries in disjointness protocols. The gap between cumulant growth rates in read-twice BPs and the exponential separation in communication complexity could expose hidden algebraic constraints in parity-like functions.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1506b8e3.py", line 92, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1506b8e3.py", line 72, in run_trial
    A = gaussian_elimination(A)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1506b8e3.py", line 27, in gaussian_elim
    factor = A[j][i] / A[i][i]
             ~~~~~~~~^~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing data collection. The pre-registered support condition cannot be evaluated without successful trials. | next: Debug gaussian_elimination() to handle zero denominators, then re-run trials with diverse seeds

---

Gowers Uniformity Norm of Boolean Functions Bounds ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Gowers Uniformity Norms in Additive Combinatorics × ACC^0 Circuit Size
- **Recorded**: 2026-05-08 07:21 UTC
- **Entry ID**: `b61cbcb3e1c4`

Statement

For a Boolean function f on n variables, if the Gowers uniformity norm of order 3 is at least Ω(n^ε), then the minimal ACC^0 circuit size for f is Ω(n^{2-ε})

Rationale

High Gowers uniformity indicates the function is not well-approximated by low-degree polynomials, which are efficiently computable by ACC^0 circuits. Thus, such functions require larger ACC^0 circuits.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.02s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with identical parameters

---

Low-Degree Fourier Mass of Cut Indicator Bounds Tseitin Resolution

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier-analytic / hypercontractivity for boolean functions (Bonami–Beckner low-degree concentration of cut indicators on the boolean cube {0,1}^V, in the spirit of Friedgut-Kalai–Linial junta theorems) × Resolution refutation length L_R(Tseitin(G,σ)) of 3-regular Tseitin formulas with odd vertex charge σ (the Urquhart / Ben-Sasson–Wigderson canonical sub-Frege target)
- **Recorded**: 2026-05-08 07:33 UTC
- **Entry ID**: `ff19cf66644d`

Statement

Let G be a connected 3-regular graph on n vertices and m=3n/2 edges, and let σ∈F_2^V have odd total parity. Define the cut-threshold indicator f_G:{0,1}^V→{0,1} by f_G(x)=1 iff |E(S_x,V§x)|≤m/3, where S_x={v:x_v=1}, and let its level-≤k Fourier mass be W{≤k}[f_G]=Σ_{|S|≤k} f̂G(S)^2. Define ν(G):=−log_2 W{≤⌊n/8⌋}[f_G]−log_2(1/E[f_G]+1). The conjecture: log_2 L_R(Tseitin(G,σ))≥c·ν(G) for an absolute constant c≥1/12, while ν(G)=O(1) whenever G admits a vertex separator of size O(1) with O(1) edge boundary; a single 3-regular instance on n≤40 violating log_2 L_R≥(1/12)·ν(G) refutes the bound.

Rationale

On expander G, cuts are sharply concentrated near m/2 by Bonami–Beckner hypercontractivity applied to the (n-junta) cut function, so f_G is a near-zero, low-influence boolean function and its low-degree mass W_{≤n/8} is exponentially small, forcing ν(G)=Ω(n) and matching the Urquhart 2^{Ω(n)} resolution lower bound. On non-expanders the small-cut event is realized by a constant-size junta of separator vertices, so f_G is essentially a low-degree function and W_{≤n/8} is Ω(1), driving ν(G)=O(1). This bridges the Friedgut–Bourgain low-degree-mass / junta machinery (rarely applied to proof complexity) directly to Tseitin resolution via the cut indicator, without invoking spectral λ_2 or vertex expansion explicitly.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s without producing any RESULT line, so the pre-registered support condition (Spearman ρ ≥ 0.6 plus per-instance bound) could not be evaluated. Per the rules, a crash/timeout yields INCONCLUSIVE. | next: Reduce the instance pool (e.g., n ∈ {10,12,14}, 10 seeds × 3 families) and cap the SAT solver's decision budget per instance so the 450-instance sweep finishes within the timeout.

Monomial Ideal Generators and Frege Proof Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial Commutative Algebra × Frege Proof Complexity
- **Recorded**: 2026-05-08 07:38 UTC
- **Entry ID**: `89c741c0e36f`

Statement

For a CNF formula Φ, the minimal number of generators of the monomial ideal I(Φ) generated by the clauses of Φ is polynomially bounded if and only if Φ has a Frege proof of polynomial size.

Rationale

The structure of the monomial ideal reflects the logical dependencies in the clauses, and the minimal generators correspond to the essential clauses needed for a proof. The complexity of generating these ideals mirrors the inherent difficulty of deriving contradictions in Frege systems.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`


```
metric_value': None, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'negative s
TRIAL: {'metric_name': 'Generators', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Generators', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Generators', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Generators', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Generators', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Generators', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Generators', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Generators', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Generators', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Generators', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Generators', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds'
```

```
TRIAL: {'metric_name': 'Generators', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds'
RESULT: FALSIFIED counterexample="mapping_undefined" first_failing_seed=11
```

Judge reasoning

The test only used 1 instance (n=1), violating the pre-registered support condition. The 'Generators' metric is undefined (None), and the counterexample 'negative shift count' suggests potential metric definition bugs. | next: Test with ≥1000 instances and validate metric definitions before re-evaluation

---

Positivstellensatz Rank Lower-Bounds SOS Max-CUT Degree

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × Sum-of-Squares Degree
- **Recorded**: 2026-05-08 07:43 UTC
- **Entry ID**: `8f0ca0b1270f`

Statement

For a max-CUT instance with n variables, if its corresponding polynomial $f \in \mathbb{R}[x_1,\ldots,x_n]$ is non-negative but not a sum of squares, then the SOS degree required to approximate f within 1-ε is $\Omega(\log n)$.

Rationale

Positivstellensatz characterizes non-negativity via algebraic certificates, while SOS hierarchies approximate this via bounded-degree sums of squares. Non-SOS-non-negative polynomials force higher-degree approximations, revealing structural gaps between algebraic geometry and convex optimization.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_992de759.py", line 149, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_992de759.py", line 112, in run_trial
    M = moment_matrix(poly, n)
        ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_992de759.py", line 83, in moment_matrix
    m = len(poly)
        ^^^^^^^^^
TypeError: object of type 'int' has no len()
```

Judge reasoning

Test crashed due to TypeError; 'poly' was an integer instead of a polynomial object | next: Debug moment_matrix() implementation to ensure 'poly' is a sequence-type object

---

Real Stable Rank Lower-Bounds AC⁰ PARITY Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × AC^0 Circuit Size for PARITY
- **Recorded**: 2026-05-08 07:54 UTC
- **Entry ID**: `3dac7381047e`

Statement

For any AC^0 circuit C computing PARITY on n bits, the real stable rank of the matrix M_C representing C's linear transformation satisfies rank_stable(M_C) ≥ Ω(log n).

Rationale

Real stable rank captures the minimal number of rank-1 matrices needed to approximate a linear transformation. PARITY's linearity imposes structural constraints on M_C, which may translate to lower bounds on its stable rank, reflecting inherent complexity in AC^0 representations.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `2.12s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5dcbe19c.py", line 96, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5dcbe19c.py", line 76, in run_trial
    M_C = walsh_hadamard_transform(truth_table)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5dcbe19c.py", line 31, in walsh_hadamar
    transform[j] = (a - b) * math.sqrt(2)
                    ~~^~~
TypeError: unsupported operand type(s) for -: 'list' and 'list'
```

Judge reasoning

Test crashed due to TypeError in Walsh-Hadamard transform calculation, preventing data collection | next: Fix the type error in the Walsh-Hadamard transform implementation by ensuring numerical operations on scalar values

---

Real Variety Connectedness Lower Bound on SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × Communication Complexity
- **Recorded**: 2026-05-08 08:00 UTC
- **Entry ID**: `8b3175ffc0f7`

Statement

For any Boolean function f with communication matrix M, the SOS degree required to approximate the max-CUT of f is at least the number of connected components of the real algebraic set defined by the equations of M's entries.

Rationale

The structure of the real algebraic set defined by the communication matrix's equations imposes constraints on the polynomial representations needed for the SOS relaxation. The number of connected components reflects the complexity of the variety, thereby affecting the minimum SOS degree required.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `1.46s`

```
'counterexample': ''}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 10, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 10, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 10, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 10, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 10, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 10, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 10, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 10, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 10, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 10, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 10, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 10, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 10, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 10, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 10, 'instances_tested': 1, 'conjecture_holds': T
RESULT: SUPPORTED mean=10.0 std=0.0 support_fraction=1.0
```

Judge reasoning

While 100% support is observed, the critic highlights insufficient sample size (1 instance per trial) and metric saturation that render the bound vacuous. The pre-registered criterion requires ≥80% support, which is met, but the challenge invalidates metric scaling analysis. | next: Test adversarial cases with varying connected components and larger instance counts to validate the conjecture's robustness.

Legendre-Symbol Partial-Sum Anomaly Excludes ACC^0

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Analytic number theory: Polya-Vinogradov / Burgess partial sums of the Legendre symbol over Z_p × ACC^0[m] circuit complexity for explicit functions, framed via communication-matrix discrepancy and the Razborov-Rudich non-largeness escape
- **Recorded**: 2026-05-08 08:01 UTC
- **Entry ID**: `bff6969c1719`

Statement

For f:{0,1}^n -> {-1,+1} index inputs as integers k in {0,...,2^n-1}, let p_n be the largest prime below 2^n, and let chi_n be the Legendre symbol mod p_n. Define Q(f) = (1/sqrt(p_n)) * max_{1 <= M < p_n} | sum_{k=1}^M f(bin_n(k)) * chi_n(k) |. Conjecture: (i) for uniform random f, Pr[Q(f) >= n^{1/4}] = o(1) (non-large, dodging Razborov-Rudich); (ii) every f computable by an ACC^0[m] circuit of size at most $2^{\{n}${1/3}} satisfies Q(f) <= (log n)^{O(1)}; (iii) the depth-3 Sipser function f_S^{(3)} with alternating OR/AND/OR of bottom fan-in ceil(n^{1/3}) satisfies Q(f_S^{(3)}) >= c * n^{1/8} for an absolute c>0. Therefore R(f) := [Q(f) >= n^{1/4}] is a non-large, polynomial-time-on-truth-table natural property excluding poly-size ACC^0[m].

Rationale

ACC^0[m] functions are Z_2^n-Fourier-structured (Razborov-Smolensky, Bonami-Beckner) and possess approximate low-degree symmetric polynomial representations, which align with the additive group of indices but carry no multiplicative-character (Z_{p_n}^*) structure. Polya-Vinogradov gives sqrt(p) log p partial-sum cancellation for chi alone; twisting by an ACC^0 truth-table should preserve square-root cancellation since the two algebraic regimes (Z_2^n vs Z_p) interact pseudorandomly. Sipser's deeply-alternating gates create long monochromatic runs in the integer-indexed truth-table that resonate with the slow sign-change pattern of chi partial sums, breaking cancellation. Because random f also achieves cancellation (LIL gives Q ~ sqrt(log p_n)), the property [Q >= n^{1/4}] is satisfied by a vanishing fraction of random functions, sidestepping the natural-proofs largeness clause.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5579605f.py", line 116, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5579605f.py", line 62, in run_trial
    chi_n = [legendre_symbol(k, p_n) for k in range(2**n)]
             ^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5579605f.py", line 38, in legendre_symb
    return legendre_symbol(2, p) * legendre_symbol(a // 2, p)
           ^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5579605f.py", line 38, in legendre_symb
    return legendre_symbol(2, p) * legendre_symbol(a // 2, p)
           ^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5579605f.py", line 38, in legendre_symb
    return legendre_symbol(2, p) * legendre_symbol(a // 2, p)
           ^^^^^^^^^^^^^^^^^^^^^

  [Previous line repeated 995 more times]
RecursionError: maximum recursion depth exceeded
```

Judge reasoning

The test crashed with a RecursionError in the legendre_symbol implementation before any data was produced, so neither the support nor falsification criteria could be evaluated. | next: Rewrite legendre_symbol iteratively (or use Euler's criterion via pow(a,(p-1)//2,p)) to avoid recursion depth issues, then rerun the pre-registered protocol for n in {10,12,14,16}.

Möbius Coinvariant of Minterm Intersection Lattice Bounds k-CLIQUE Monotone DNF

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Rota–Whitney Möbius function theory of intersection-closure lattices (the alternating-zeta-inversion calculus of finite meet-semilattices, classically used in subspace-arrangement combinatorics and almost never applied to monotone DNF analysis) × Monotone DNF size for the k-CLIQUE indicator (Razborov 1985 sunflower regime; the gateway to feasible-interpolation lower bounds for Extended Frege via monotone-circuit-to-EF transfer)
- **Recorded**: 2026-05-08 08:12 UTC
- **Entry ID**: `29dff0c510e9`

Statement

For a monotone DNF F on N variables with minimal-minterm family M(F)={M_1,…,M_m}⊆2[1], let L(F) be the meet-semilattice generated by {▯,[N]}∪M(F) under intersection, ordered by inclusion, with Möbius function μ_L. Define Λ(F)=log_2(1+Σ_{x∈L(F)}|μ_L(0̂,x)|). We conjecture: (i) for every monotone DNF F of size s on N=C(n,2) graph variables, Λ(F)≤4·log_2(s+1)·log_2(n+1); (ii) for the k-CLIQUE indicator with k=3 on n vertices, every monotone DNF F representing it satisfies Λ(F)≥(1/8)·n·log_2(n). A single (s,F) pair with Λ(F)>4·log_2(s+1)·log_2(n+1), or a single triangle-DNF F with Λ(F)<(1/8)·n·log_2(n), refutes the conjecture.

Rationale

Razborov's sunflower argument shows triangle minterms have rigid pairwise-intersection patterns; Möbius mass on the intersection lattice is exactly the alternating overlap signature that random/structured small DNFs (whose intersection lattices are nearly Boolean and hence have μ-values cancelling) cannot replicate. Möbius coinvariants are subadditive under join (Whitney 1935; Rota 1964), giving a natural conjunction-submodularity hook, and their values are computable in O(|L(F)|^2) time by zeta inversion — a concrete computational shortcut that avoids any reliance on uncomputable algebraic invariants.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_872b909c.py", line 115, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_872b909c.py", line 81, in run_trial
    mu_triangle = mobius_function(L_triangle)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_872b909c.py", line 29, in mobius_functi
    if lattice[j].issubset(lattice[i]):
```

[1] N

```
       ^^^^^^^^^^^^^^^^^^^
AttributeError: 'tuple' object has no attribute 'issubset'
```

Judge reasoning

The test crashed with an AttributeError ('tuple' object has no attribute 'issubset') in the Möbius function computation before any data was produced, so neither support nor falsification conditions could be evaluated. | next: Fix the lattice representation by converting elements to frozensets (or apply issubset to the underlying set) before calling mobius_function, then rerun the pre-registered sweep.

Network Reliability Deficit Lower-Bounds Tseitin Resolution

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Network reliability theory / Tutte polynomial percolation evaluations (Whitney–Tutte–Welsh–Colbourn–Sokal): the all-terminal reliability polynomial R_G(p) = Pr[bond percolation at retention p leaves G connected], a specialization of the Tutte polynomial T_G at (1/(1-p), 1/p) up to prefactors × Resolution refutation length L_R(Tseitin(G,σ)) for 3-regular graphs G with odd-charge σ:V→F_2 — the canonical Urquhart / Ben-Sasson–Wigderson sub-Frege gateway target
- **Recorded**: 2026-05-08 08:22 UTC
- **Entry ID**: `8ddbca827c85`

Statement

For every 3-regular connected graph G on |V|≤40 with odd-charge σ, define the reliability deficit ν(G) := -log_2(1 - R_G(1/2)), where R_G(1/2) is the probability that bond-percolation with edge-retention 1/2 keeps G connected (computable as a Tutte-polynomial value, estimable in O(|V|+|E|) per Monte-Carlo trial). We conjecture there exist absolute constants c_lo ≥ 1/16 and C_hi ≤ 16 such that c_lo · ν(G) ≤ log_2 L_R(Tseitin(G,σ)) ≤ C_hi · ν(G) · log_2|V|, in particular ν(G)=O(1) whenever G has a bridge or sparse cut (so the bound is vacuous for non-expanders) while ν(G)=Θ(|V|) for spectral expanders.

Rationale

Disconnection probability under p=1/2 percolation collapses to ≥1/2 the moment G admits a small edge cut (so ν(G)=O(1) for any non-expander), but decays as e^{-Θ(|V|)} for graphs whose every cut has width Θ(|V|) — matching exactly the edge-isoperimetric premise that Ben-Sasson–Wigderson width-size arguments need. Reliability polynomial values are a Tutte-polynomial specialisation that has, to my knowledge, never been used as the certificate ν in a Tseitin lower-bound program, and unlike spectral or Hashimoto invariants it is genuinely combinatorial-probabilistic (a sum over the lattice of edge subsets) rather than eigenvalue-based, so it bypasses the natural-proofs criticism that has dogged purely spectral surrogates.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_170c0feb.py", line 166, in <module>
```

```
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_170c0feb.py", line 137, in run_trial
    ν_G = reliability_deficit(G)
          ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_170c0feb.py", line 120, in reliability_
    R_hat = sum(percolation(G, p) for _ in range(5000)) / 5000
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_170c0feb.py", line 120, in <genexpr>
    R_hat = sum(percolation(G, p) for _ in range(5000)) / 5000
                ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_170c0feb.py", line 59, in percolation
    uf(u, v)
TypeError: run_trial.<locals>.union_find.<locals>.find() takes 1 positional argument but 2 were given
```

Judge reasoning

The test crashed with a TypeError in the union-find implementation (find() called with wrong arity) before any data was produced, so neither the Spearman correlation nor the cohort-specific bounds could be evaluated. | next: Fix the union-find closure (use proper find(x)/union(u,v) signatures) and rerun the 300-instance protocol to obtain ν and log L_R measurements.

Three-Distance Beatty Ordering Matches Avg DPLL on Random 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Diophantine approximation: Steinhaus' Three Distance Theorem and Beatty/Sturmian sequences $\lfloor i\varphi \rfloor$ for the golden ratio $\varphi=(\sqrt{5}-1)/2$; the gap-set of $\{i\varphi \bmod 1\}_{i=1..n}$ has at most three distinct values, giving a number-theoretic 'low-discrepancy' substitute for uniform randomness — essentially absent from the DPLL / proof-complexity literature × Average-case DPLL search tree size $T(\Phi,\sigma)$ on uniform random 3-SAT above threshold (m=8n clauses), as a derandomization-style avg-to-worst transfer in the Impagliazzo–Wigderson tradition: replace a random variable order by a single deterministic Beatty order
- **Recorded**: 2026-05-08 08:28 UTC
- **Entry ID**: `ce444850cb4a`

Statement

Let Φ be a uniform random 3-SAT instance on n variables with m=8n clauses (UNSAT w.h.p.). Let σ_φ be the permutation of [n] obtained by sorting the fractional parts $\{i\varphi\}_{i=1..n}$ $(\varphi=(\sqrt{5}-1)/2)$, *and let* σ_R *denote a uniformly random permutation. Conjecture: for every* $n\in[12,24]$ *and every instance seed,* $\log_2 T_{DPLL}(\Phi,\sigma_\varphi) \le E_{\sigma_R}[\log_2 T_{DPLL}(\Phi,\sigma_R)] + 4\cdot\log_2 n$, i.e. the single Three-Distance Beatty order matches the typical random order up to a logarithmic slack on every instance, not merely on average over instances.

Rationale

The Three Distance Theorem guarantees that a Beatty rotation produces a maximally regular partition of [0,1), making σ_φ a structured surrogate for a random branching order. If the conjecture holds, a purely number-theoretic ordering rule simulates the average behavior of random orderings instance-by-instance, which is exactly the kind of cheap derandomization (worst-case-instance ≤ avg-over-randomness) that Impagliazzo–Wigderson-style transfers predict; if it fails, a single instance pinpoints obstacle to such transfers for DPLL.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c88f08b6.py", line 157, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c88f08b6.py", line 127, in run_trial
    T_phi = dpll(instance, order_phi)
            ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c88f08b6.py", line 101, in dpll
    if not any(lit in assignment and not assignment[lit] for lit in instance):
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c88f08b6.py", line 101, in <genexpr>
    if not any(lit in assignment and not assignment[lit] for lit in instance):
               ^^^^^^^^^^^^^^^^^
TypeError: unhashable type: 'list'
```

Judge reasoning

The test crashed with a TypeError ('unhashable type: list') in the DPLL routine before producing any results, so neither the per-instance slack bound nor the Spearman correlation could be evaluated. | next: Fix the DPLL implementation to represent literals as hashable integers (or check membership by iterating the clause) and rerun the 120 (n,seed) trials.

Lie Nilpotency Class of Layer Algebra Distinguishes Read-Twice BPs from IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Engel–Lie nilpotency of finitely-generated matrix Lie subalgebras (descending central series and Lazard-style commutator filtrations from pro-p / Magnus-algebra theory), a classical tool in nilpotent group theory but essentially absent from branching-program lower-bound literature × Read-twice branching program (BP) size for IP_2_n = ⊕_{i=1}^n x_i·y_i, in the Borodin–Razborov–Smolensky 1993 model
- **Recorded**: 2026-05-08 08:32 UTC
- **Entry ID**: `11647c329d11`

Statement

For a layered BP P with state set S of size s and layers k=1..L, let M_k = M_k$^{\{0\}+M_k}${1} be the (variable-marginalized) integer transition matrix of layer k, set M̄_k = M_k − (tr M_k / s)·I_s, and let g(P) ⊂ M_s(Q) be the Lie subalgebra generated under [A,B]=AB−BA by {M̄_1,…,M̄_L}. Define the distinguishing tensor ρ(P) := nilpotency class of g(P) (smallest c such that every (c+1)-fold nested commutator of generators is the zero matrix; ρ(P)=∞ if g(P) is not nilpotent). Conjecture: (A) for every read-twice BP P, ρ(P) ≤ 4·log_2(size(P)) + O(1); (B) for the canonical unrestricted product BP for IP_2_n, ρ ≥ ⌈n/2⌉; together (A)+(B) imply any read-twice BP for IP_2_n has size ≥ 2^{Ω(n)}.

Rationale

Reading a variable twice forces the same layer matrix to appear at two filtration depths, which should collapse iterated commutators (cap nilpotency class by log-size), whereas the inner-product BP encodes a non-degenerate symplectic form whose product representation interleaves Heisenberg-type commutators of depth $\Theta(n)$. This is a basis-free combinatorial invariant orthogonal to spectral / free-cumulant approaches that have repeatedly crashed in the blacklist, and it does not rely on any natural-property test on truth-tables, dodging the natural-proofs barrier.

Novelty

- Judge: `NOVEL` over 3 arXiv hits

Top hits consulted: - [1709.08890v2] Partial matching width and its application to lower bounds for branching programs - [2001.07570v2] Hom 3-Lie-Rinehart Algebras - [1807.11011v2] The capability and certain functors of some nilpotent Lie algebras of class two

Empirical Test

- exit code: `124`, elapsed: `240.11s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s without producing a RESULT line, so neither the support nor falsification condition could be evaluated. Per the rules, a timeout yields INCONCLUSIVE. | next: Reduce the computational load (e.g., cap n at 8, reduce seeds to 10, and use sparse/structured commutator computations or early-exit when nilpotency class exceeds the threshold) and rerun with a longer timeout.

SVD Participation Dispersion of Mid-Layer Lower-Bounds Read-Twice BPs for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Spectral discrepancy / dispersion theory: Lindsey-style $L^{2\text{-vs-}L}\infty$ transference and the Bourgain–Kashin–Chazelle 'effective rank' (participation ratio) of 0/1 incidence matrices, used as a coordinate-free dispersion proxy that is essentially absent from the branching-program literature × Leveled read-twice branching programs (BPs) computing IP_2_n = ⊕_{i=1}^{n/2} x_i·y_i; the Borodin–Razborov–Smolensky 1993 read-twice barrier where the read-once vs unrestricted gap is exponential but the read-twice vs unrestricted gap is wide open
- **Recorded**: 2026-05-08 08:40 UTC
- **Entry ID**: `97b82c049269`

Statement

For a leveled BP P of size s on N=2n variables, let B_mid ∈ {0,1}^{s × 2^N} be the state-input incidence matrix at the middle level (the level after exactly ⌊L(P)/2⌋ queries on every source-to-sink path), and define the dispersion DISP(P) := log_2 ((Σ_i σ_i(B_mid)$^{2)}$2 / Σ_i σ_i(B_mid)^4); equivalently the log-participation-ratio of the singular spectrum, which always satisfies DISP(P) ≤ log_2 s. CONJECTURE: every leveled read-twice BP P computing IP_2_n satisfies DISP(P) ≥ n/2 − c·log_2 n for an absolute constant c ≤ 4, so that size(P) ≥ 2^{n/2 − c·log n}. A single read-twice BP P that correctly computes IP_2_n yet has DISP(P) < n/2 − 4·log_2 n falsifies the conjecture.

Rationale

Read-twice structure forces every state's input-fiber to be expressible as the intersection of a 'first-pass' subcube with a 'second-pass' coset, so the rows of B_mid lie in a structured low-discrepancy family — exactly the regime where Chazelle–Bourgain–Kashin participation-ratio dispersion is informative but ordinary rank or σ_2/σ_1 spectral gaps are degenerate (the canonical IP_2 BP has all middle singular values equal, killing naive gap-based bounds). The participation ratio is precisely the right coordinate-free dispersion measure that matches the canonical 2^{n/2} read-once IP_2 BP and we conjecture it transfers to read-twice. If true, this gives a 'distinguishing tensor' invariant ρ=DISP that is (i) trivially ≤ log s and (ii) ≥ n/2 − O(log n) on IP_2 BPs, yielding 2^{n/2 − O(log n)} read-twice lower bounds — the first matching read-once for read-twice.

Novelty

- Judge: `NOVEL` over 7 arXiv hits

Top hits consulted: - [1709.08890v2] Partial matching width and its application to lower bounds for branching programs - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [0906.0693v3] An improved lower bound on the counterfeit coins problem - [2210.01601v2] Quantum communication complexity of linear regression - [2401.14623v1] Structure in Communication Complexity and Constant-Cost Complexity Classes

Empirical Test

- exit code: `0`, elapsed: `30.26s`

```
_name': 'DISP', 'metric_value': 17.74664271579066, 'instances_tested': 6, 'conjecture_holds': True,
TRIAL: {'metric_name': 'DISP', 'metric_value': 17.817241456842243, 'instances_tested': 6, 'conjectur
TRIAL: {'metric_name': 'DISP', 'metric_value': 17.772055556368766, 'instances_tested': 6, 'conjectur
TRIAL: {'metric_name': 'DISP', 'metric_value': 17.813045718248656, 'instances_tested': 6, 'conjectur
TRIAL: {'metric_name': 'DISP', 'metric_value': 17.751510874544746, 'instances_tested': 6, 'conjectur
TRIAL: {'metric_name': 'DISP', 'metric_value': 17.858137668491107, 'instances_tested': 6, 'conjectur
TRIAL: {'metric_name': 'DISP', 'metric_value': 17.750584718409037, 'instances_tested': 6, 'conjectur
TRIAL: {'metric_name': 'DISP', 'metric_value': 17.78967489000406, 'instances_tested': 6, 'conjecture
TRIAL: {'metric_name': 'DISP', 'metric_value': 17.805817289705633, 'instances_tested': 6, 'conjectur
TRIAL: {'metric_name': 'DISP', 'metric_value': 17.748623796070323, 'instances_tested': 6, 'conjectur
TRIAL: {'metric_name': 'DISP', 'metric_value': 17.786551890716403, 'instances_tested': 6, 'conjectur
TRIAL: {'metric_name': 'DISP', 'metric_value': 17.793781724798567, 'instances_tested': 6, 'conjectur
TRIAL: {'metric_name': 'DISP', 'metric_value': 17.82067653221687, 'instances_tested': 6, 'conjecture
TRIAL: {'metric_name': 'DISP', 'metric_value': 17.764702392643578, 'instances_tested': 6, 'conjectur
RESULT: SUPPORTED mean=17.79099415384149 std=0.03165067694000282 support_fraction=1.0
```

Judge reasoning

The test reports SUPPORTED with DISP ≈ 17.79 across all trials, but this near-constant value across only 6 instances suggests metric saturation on a canonical symmetric BP rather than a genuine search over adversarial read-twice BPs; the pre-registered criterion required testing across N∈{4,6,8,10,12,14} with 30 random seeds plus controls, and the output shows no N-variation or control results, so the support condition is not unambiguously verified. | next: Re-run with explicit per-N breakdowns

---

Hypercontractive L4/L2 Flatness of Clause-Count Lower-Bounds DPLL Tree

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Fourier-analytic / hypercontractivity for Boolean functions — the (2→4) Bonami–Beckner hypercontractive ratio R(g)=‖g‖_4/‖g‖_2 applied to the centered clause-satisfaction-count function of a CNF, a KKL/Friedgut–Kalai–Linial/Mossel-style spectral-flatness invariant essentially absent from average-case DPLL analysis × DPLL search-tree size T_DPLL(Φ) on unsatisfiable random 3-CNFs Φ above the SAT threshold, under unit propagation and a fixed lexicographic branching order — the canonical sub-resolution average-case proof-search target
- **Recorded**: 2026-05-08 08:45 UTC
- **Entry ID**: `91f33a861a00`

Statement

For every unsatisfiable 3-CNF Φ on n variables with m clauses, define f_Φ:{0,1}^n→[0,1] by f_Φ(x)=(1/m)·#{C∈Φ : x⊨C}, let g_Φ := f_Φ − E_x[f_Φ], and define the hypercontractive flatness exponent E(Φ) := ‖g_Φ‖_4^4 / ‖g_Φ‖_2^4 (with conventions E=1 when g_Φ≡0; by Cauchy–Schwarz E≥1). We conjecture log_2 T_DPLL(Φ) ≥ (n − log_2 m) / E(Φ) − 5 for every such Φ; a single instance violating this bound falsifies the conjecture.

Rationale

Bonami–Beckner hypercontractivity tells us that small E(g)=‖g‖_4^4/‖g‖_2^4 forces the Walsh spectrum of g to be 'flat' (close to white noise), so a flat clause-count function lacks the low-degree structure that DPLL's unit propagation can exploit — predicting near-2^n search trees — while a peaked spectrum (large E) signals a low-degree certificate that DPLL can find quickly. The bridge from the (2,4)-hypercontractive ratio to DPLL search-tree size via the Walsh transform of the per-clause satisfaction count appears unexplored, despite being a single Walsh-Hadamard transform away from textbook KKL machinery.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2602.17248v2] The optimal hypercontractive constants for $\mathbb{Z}_3$ and biased Bernoulli random variables - [2511.07739v3] A Lower Bound for the Fourier Entropy of Boolean Functions on the Biased Hypercube - [1908.05361v2] Placing quantified variants of 3-SAT and Not-All-Equal 3-SAT in the polynomial hierarchy - [1804.02478v5] Complexity of the CNF-satisfiability problem - [2003.09703v1] Variance function of boolean additive convolution

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_81dccc3a.py", line 116, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_81dccc3a.py", line 85, in run_trial
    if is_unsatisfiable(phi):
       ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_81dccc3a.py", line 66, in is_unsatisfia
    if all(phi[i][j] == (assignment[j] ^ phi[i][j]) for i in range(len(phi))):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_81dccc3a.py", line 66, in <genexpr>
    if all(phi[i][j] == (assignment[j] ^ phi[i][j]) for i in range(len(phi))):
                  ^
NameError: cannot access free variable 'j' where it is not associated with a value in enclosing scope
```

Judge reasoning

The test crashed with a NameError in is_unsatisfiable before any data was produced, so neither the per-instance bound nor the Spearman criterion could be evaluated. | next: Fix the scoping bug in is_unsatisfiable (properly bind j in the generator/loop) and rerun the 120-instance trial.

Gauss-Kuzmin Deviation Bounds Truth-Table Lempel-Ziv Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Diophantine approximation / Gauss-Kuzmin distributional theory of continued fractions (the log_2(4/3) frequency of partial-quotient 1 and Khinchin-Levy concentration), classically applied to metric number theory and almost never to descriptive complexity × Time-bounded Kolmogorov complexity K^poly of Boolean truth tables — the meta-complexity object underlying MCSP, accessed via the polynomial-time-computable LZ77 phrase-count proxy LZ(T) which sandwiches K^poly up to logarithmic factors
- **Recorded**: 2026-05-08 09:02 UTC
- **Entry ID**: `b81e6538851e`

Statement

For a Boolean function f:{0,1}^n -> {0,1} let T_f in {0,1}$^{\{2}$n} be its truth table, r_f = sum_{i<2^n} T_f[i] * 2^{-i-1} in [0,1], and r_f = [0; a_1,...,a_K] its finite continued fraction expansion. Define freq_1(f) = (1/min(K,4n)) * #{i <= min(K,4n) : a_i = 1}, the Gauss-Kuzmin deviation GKD(f) = |freq_1(f) - log_2(4/3)|, and LZ(f) = LZ77 phrase count of T_f. Conjecture: for every n >= 6 and every f, GKD(f) * log_2(LZ(f) + 2) <= 4 * sqrt(n), so any single (f,n) with product exceeding 4*sqrt(n) refutes it.

Rationale

Gauss-Kuzmin governs partial-quotient statistics of almost every irrational, but a truth table is a specific dyadic rational T_f / 2$^{\{2}$n} whose CF expansion is a number-theoretic fingerprint of its descriptive structure. Highly compressible T_f (small LZ ~ K^poly) tend to have arithmetic structure that biases partial-quotient frequencies away from log_2(4/3), while incompressible T_f obey Khinchin-Levy concentration with GKD ~ O(1/sqrt(K)); the conjectured product bound asserts a uniform sqrt(n) trade-off bridging these regimes. The statement is a UNIVERSAL inequality on a product of two computable invariants — neither a largeness nor a usefulness separator — so it sidesteps the natural-proofs barrier that killed the Mahler-measure attempt in this approach.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a0501520.py", line 177, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a0501520.py", line 112, in run_trial
    truth_table = evaluate_k_DNF(generate_truth_table(n), generate_random_k_DNF(n, k))
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a0501520.py", line 74, in evaluate_k_DN
```

```
    if all(truth_table[i + var] == 1 for var in term):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a0501520.py", line 74, in <genexpr>
    if all(truth_table[i + var] == 1 for var in term):
           ~~~~~~~~~~~^^^^^^^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in evaluate_k_DNF before producing any RESULT line, so neither the support nor falsification criteria could be evaluated. | next: Fix the indexing bug in evaluate_k_DNF (likely the i + var truth-table lookup) and rerun the 450-instance pre-registered protocol.

SOS Moment Matrix Rank Bounded by Square Root of Variables for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded**: 2026-05-08 09:28 UTC
- **Entry ID**: `00198069e342`

Statement

For any max-CUT instance on n variables, the rank of the degree-d SOS moment matrix is at most $O(\sqrt{n})$ if $d \leq 2$, and strictly increases with d. Any 0.879-approximator for max-CUT must have a moment matrix violating this rank bound for d=2.

Rationale

The SOS hierarchy's moment matrix encodes polynomial identities over the max-CUT feasible region. Real algebraic geometry provides tools to analyze its rank, which reflects the complexity of the relaxation. A low-rank bound for d=2 implies structural limitations, while higher d requires increased rank to capture the problem's geometry.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_980de3dd.py", line 105, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_980de3dd.py", line 65, in run_trial
    rank2 = sdp_rank(P2)
            ^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_980de3dd.py", line 46, in sdp_rank
    return gaussian_elimination(matrix)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_980de3dd.py", line 41, in gaussian_elim
```

```
    factor = A[row][col] / A[pivot_row][col]
             ~~~~~~~~~~~~^~~~~~~~~~~~~~~~~~~~
ZeroDivisionError: division by zero
```

Judge reasoning

The test crashed with a ZeroDivisionError in the gaussian_elimination routine before any rank data was produced, so the pre-registered support condition could not be evaluated. | next: Replace the hand-rolled Gaussian elimination with numpy.linalg.matrix_rank (or SVD-based numerical rank with the specified 1e-6 tolerance) and rerun the 60 seeds.

Non-backtracking Spectral Gap Lower-Bounds Tseitin DPLL Trees

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Ihara zeta / non-backtracking (Hashimoto) spectrum of regular graphs (Ihara 1966, Bass 1992, Friedman 2008, Alon-Boppana for B(G)) — an expansion proxy strictly sharper than the ordinary Laplacian gap and essentially absent from the proof-complexity literature × Resolution / DPLL refutation length L_R(Tseitin(G,σ)) of 3-regular Tseitin formulas with odd vertex charge σ:V→F_2 (the Urquhart / Ben-Sasson-Wigderson canonical sub-Frege gateway target)
- **Recorded**: 2026-05-08 09:37 UTC
- **Entry ID**: `d1d1e2ec5d2a`

Statement

Let G be a connected 3-regular graph on n vertices and let B(G) ∈ {0,1}^{2|E|×2|E|} be the non-backtracking (Hashimoto) matrix defined by B[(u,v),(v,w)] = 1 iff w ≠ u, with second-largest absolute eigenvalue $\lambda_2(B)$. Define the poly-time graph invariant $\nu(G) := n \cdot \max(0, \sqrt{2} - (\lambda_2(B) - \sqrt{2}))/\sqrt{2}$, which is O(1) for any G with $\lambda_2(B) \geq 2\sqrt{2}$ (the non-expander regime, including disconnected and tree-like G) and Θ(n) for Ramanujan-quality 3-regular expanders where $\lambda_2(B) \approx \sqrt{2}$. Conjecture: for every odd σ, $\log_2$ T_DPLL(Tseitin(G,σ)) ≥ (1/16)·ν(G), where T_DPLL is the leaf count of unit-propagation DPLL with fixed lexicographic branching; one trial with $\log_2 T < \nu/16 - 2$ falsifies.

Rationale

Friedman's theorem identifies √(d−1) = √2 as the true Alon-Boppana threshold for 3-regular non-backtracking spectra, and exactly this gap controls the mixing-time of cycle covers that Ben-Sasson-Wigderson width arguments require to lift from vertex-expansion to Resolution-length lower bounds. Standard $\lambda_2(L)$-based invariants (Algebraic Connectivity, Sparsest-Cut LP) blur small-set expansion with bottleneck cuts, whereas $\lambda_2(B)$ is sensitive only to genuine Ramanujan-style mixing — which is precisely the regime Tseitin is hard. The Hashimoto spectrum is computable in $O(|E|^3)$ and has never been used as a proof-complexity invariant, yet it is the natural quantitative refinement of 'expander' that the open characterisation problem demands.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.02s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode 124) before producing any RESULT line, so neither the support nor falsification condition can be evaluated. | next: Reduce the largest size tier and/or cap DPLL leaf counts more aggressively, then rerun with a longer timeout to obtain the 150 trials required by the pre-registered criterion.

Symmetrized Univariate Approximate Degree Bounds AC0 PARITY

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Univariate L^∞-approximation theory on the integer lattice (Chebyshev-Bernstein-Markov inequalities for polynomials on $\{0,1,\ldots,n\}$); the symmetrization-projection trick of Minsky-Papert combined with a poly-time LP-dual extraction of the approximate degree, applied as an S_n-natural circuit invariant rarely used outside the symmetric-function literature × AC^0 depth-d circuit size for PARITY (and any S_n-symmetric Boolean function); Hastad-style switching-lemma target with the Bun-Thaler approximate-degree perspective
- **Recorded**: 2026-05-08 09:52 UTC
- **Entry ID**: `e67822267f11`

Statement

For a Boolean function $f:\{0,1\}^n\to\{-1,+1\}$ computed by an AC^0 circuit C, define its S_n-symmetrization $g_f:\{0,\ldots,n\}\to[-1,1]$ by $g_f(k)=E_{|x|=k}[f(x)]$, and let $\Psi_{sym}(C):=\Psi_{sym}(f)$ be the smallest k for which there exists a univariate real polynomial q of degree $\leq k$ with $\max_{j\in\{0,\ldots,n\}} |q(j)-g_f(j)| \leq 1/3$ (computable in poly(n) time via a (k+1)-variable, (n+1)-constraint LP). We conjecture that for every depth-d AC^0 circuit C of size s on n inputs, $\Psi_{sym}(C) \leq (c\cdot\log s)^{d-1}$ for an absolute constant $c\leq 8$, while $\Psi_{sym}(\text{PARITY})=\Theta(n)$ by Paturi's symmetric-approximate-degree bound, yielding $\log s \geq \Omega(n^{1/(d-1)})$ for any AC^0 circuit computing PARITY. The conjecture is falsified by exhibiting a single AC^0 circuit C with $\Psi_{sym}(C) > (8\cdot\log_2 \text{size}(C))^{\text{depth}(C)-1}$.

Rationale

Under the symmetrization projection, the entire hardness budget of an AC^0 circuit collapses onto its hamming-weight profile, where univariate Chebyshev/Markov machinery gives a clean LP-dual certificate of approximate degree; this turns the elusive multivariate approximate-degree calculation into a tractable n+1-point Remez problem, whose value is provably S_n-natural and is conjecturally controlled by the $(\log s)^{d-1}$ switching-lemma scale. The bridge is unusual because univariate Markov-Bernstein theory has been used inside Bun-Thaler-style lower-bound proofs only as a tool, never as a poly-time circuit invariant attached to individual AC^0 circuits, and the symmetrization step gives a bona-fide group-action naturality (under S_n) that survives composition with random restrictions. If sharp, the conjecture replaces the random-restriction argument by a single LP and pins down the optimal Hastad constant c.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
(no output)
```

Judge reasoning

The stdout contains no RESULT line or numerical evidence — only the pre-registered criteria are echoed. Combined with the critic's CHALLENGE flagging vacuous-bound/metric-saturation concerns at small n and possible selection bias, the support condition is not unambiguously demonstrated. | next: Re-run with explicit reporting of per-cell Ψ_sym values, Spearman ρ, and Ψ_sym(PARITY_n), and include adversarial symmetric circuits (e.g., approximate threshold/MAJORITY-like AC^0 constructions) plus lar

---

Bivariate Mahler Measure of Sign Polynomial Lower-Bounds R(DISJ)

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Mahler measure / heights of multivariate polynomials in algebraic number theory (Smyth's L'(χ_{-3},-1) closed form for m(1+s+t); Boyd–Deninger zeta-value formulae; Lehmer's problem). The logarithmic Mahler measure m(P) = ∫*{T^k} log|P| dμ is a classical height invariant essentially absent from the communication-complexity literature. × Randomized two-party communication complexity R*{1/3}(f) of sign matrices M_f ∈ {-1,+1}^{2^n×2^n}, with DISJ_n as the canonical Ω(n) target (Razborov 1992) accessed through the discrepancy-style proxy R_{1/3}(f) ≥ (1/2) log_2(1/disc(M_f)) − O(1).
- **Recorded**: 2026-05-08 09:57 UTC
- **Entry ID**: `ddc77206bc09`

Statement

For f:{0,1}^n × {0,1}^n → {-1,+1} with sign matrix M_f, define the multilinear matrix polynomial F_f(s_1,…,s_n,t_1,…,t_n) = Σ_{x,y∈{0,1}^n} M_f(x,y) · ∏*{i=1}^n s_i^{x_i} t_i^{y_i} ∈ Z[s,t], and its 2n-torus logarithmic Mahler measure m(F_f) = ∫*{[0,1]^{2n}} log|F_f(e^{2πiθ_1},…,e^{2πiφ_n})| dθ dφ. Conjecture: (a) m(F_{DISJ_n}) − m(F_{const}) ≥ n · m(1+s+t) − o(n) = (0.32307…)·n − o(n), and (b) for every f, R_{1/3}(f) ≥ c · (m(F_f) − n·log 2) − O(log n) for an absolute c > 0. A single instance with R_{1/3} = O(log n) but m(F_f) − n·log 2 = ω(log n) refutes (b); an instance of DISJ with sublinear m refutes (a).

Rationale

The multilinear polynomial F_f is the natural Fourier-side / generating-function avatar of M_f under the {0,1}^n × {0,1}^n product structure; for DISJ the polynomial factors as 2∏(1+s_i+t_i) − ∏(1+s_i)(1+t_i), whose Mahler measure inherits Smyth's exact L-value per coordinate and gives a clean Ω(n) signature, while easy functions (constant, equality up to a row/column permutation, low-rank patterns) collapse F_f into a product of small-height factors with bounded m. Heights/Mahler measure are coordinate-respectful (multiplicative under tensor product, monotone under specialization), so they should track the multiplicative structure that makes DISJ hard, in a way orthogonal to spectral or discrepancy invariants.

Novelty

- Judge: " over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by NATURAL_PROOFS barrier

Hochster Betti Sum of Monotone f Lower-Bounds Log-Rank of IND-Lifted Matrix

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial commutative algebra — specifically Hochster's formula for the multi-graded Betti numbers of the Stanley-Reisner ring k[Δ] of a simplicial complex via reduced simplicial homology of vertex-restrictions Δ_σ over F_2 (Hochster 1977; Bruns-Herzog Ch. 5); the total Hochster Betti sum β(Δ) := ∑*{σ ⊆ V}* ∑{i ≥ -1} dim_{F_2} H̃*i(Δ_σ; F_2) is poly-time computable from the face list and is essentially absent from communication-complexity and lifting literature. × Deterministic two-party communication complexity D(f ∘ IND_2^k) of Raz-McKenzie style lifted Boolean functions, accessed through the F_2 log-rank lower bound D(M) ≥ log_2 rank*{F_2}(M); f is monotone on k variables and IND_2 is the 2-bit Index gadget (Alice holds 1 pointer per coordinate, Bob holds the 2-bit array), the canonical Göös–Pitassi–Watson / Raz–McKenzie lifting setup whose monotone-circuit and proof-complexity consequences are open above the certificate-complexity barrier.
- **Recorded**: 2026-05-08 10:04 UTC
- **Entry ID**: `06221fec8ebe`

Statement

Let f: {0,1}^k → {0,1} be monotone (k ≤ 6), let Δ_f := {S ⊆ [k] : f(1_S) = 0} be its (downward-closed) Stanley-Reisner simplicial complex, and let β(Δ_f) := ∑*{σ ⊆ [k]}* ∑{i = -1}^{|σ|-1} dim_{F_2} H̃*i(Δ*{f,σ}; F_2) be the total Hochster Betti sum, where Δ_{f,σ} = {S ∈ Δ_f : S ⊆ σ}. Then the F_2 log-rank of the IND_2-lifted communication matrix satisfies log_2 rank_{F_2}(M_{f ∘ IND_2^k}) ≥ ⌈log_2(1 + β(Δ_f))⌉ + 1; a single monotone f for which log_2 rank_{F_2}(M_{f ∘ IND_2^k}) < ⌈log_2(1+β(Δ_f))⌉ + 1 refutes the conjecture.

Rationale

Hochster's formula expresses Betti numbers of k[Δ_f] as homological obstructions to contractibility of restrictions of Δ_f, which encode the hierarchy of minimal certificates of f at every coordinate-subset scale; lifting theorems pump exactly such 'multiscale certificate structure' from f into the communication matrix of f ∘ IND_2^k via the gadget. Because F_2 log-rank is an oblivious lower bound on D, and Hochster sums add up obstructions across all scales rather than just the worst minterm/maxterm (which is ordinary certificate complexity), the conjecture refines the standard certificate-lifting bound by the fine homological structure of the false-set complex. The bridge is genuinely new: Stanley-Reisner Betti theory has been used in algebraic statistics and toric geometry but, to our knowledge, never against IND-lifted matrices.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3d932602.py", line 99, in <module>
    result = run_trial(seed)
```

```
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3d932602.py", line 63, in run_trial
    Δ_f = [S for S in range(1 << k) if all(f(bin(S)[2:].zfill(k)[:i+1]) == 0 for i in range(len(bin(S)[2
1))]
                          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3d932602.py", line 63, in <genexpr>
    Δ_f = [S for S in range(1 << k) if all(f(bin(S)[2:].zfill(k)[:i+1]) == 0 for i in range(len(bin(S)[2
1))]
                                           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3d932602.py", line 60, in <lambda>
      f = lambda x: all(x[i] <= x[j] for i, j in combinations(range(k), 2))
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3d932602.py", line 60, in <genexpr>
      f = lambda x: all(x[i] <= x[j] for i, j in combinations(range(k), 2))
                              ~^^^
IndexError: string index out of range
```

Judge reasoning

The test crashed with an IndexError before producing any data, due to a bug in constructing Δ_f (slicing a bit-string of variable length rather than the zero-padded representation). No instances were evaluated, so neither support_fraction nor Spearman ρ could be computed. | next: Fix the Δ_f construction by iterating S over subsets of [k] using bit-indexing on the zero-padded length-k string (or use bitmask membership directly), then rerun the 120-instance protocol.

Cut-Polynomial Kurtosis Excess Bounds GW SDP Integrality Gap

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier-analytic hypercontractivity on the Boolean cube (Bonami-Beckner (4,2)-norm comparison and excess-kurtosis of low-degree polynomials in the KKL/Friedgut/Mossel tradition) × Goemans-Williamson / SOS-degree-2 (Delorme-Poljak) SDP integrality gap for unweighted Max-Cut on 3-regular graphs
- **Recorded**: 2026-05-08 10:10 UTC
- **Entry ID**: `485f006f57f6`

Statement

For every connected 3-regular graph G on n vertices with adjacency matrix A, define the cut polynomial $g_G(x) := x^T A x$ for $x \in \{\pm1\}^n$ and the Bonami-Beckner kurtosis excess $\kappa(G) := E_x[g_G(x)^4] / E_x[g_G(x)^2]^2 - 3$, where E_x is uniform over $\{\pm1\}^n$. Then the Goemans-Williamson / SOS-2 integrality gap obeys $(DP(G) - MaxCut(G)) / |E(G)| \leq 0.5 \cdot \max(\kappa(G), 1/n)^{1/4}$, where $DP(G) = |E|/2 + n \cdot |\lambda_{min}(A)|/4$ is the Delorme-Poljak eigenvalue SDP bound. A single 3-regular G with $n \leq 20$ violating this inequality falsifies the conjecture.

Rationale

Goemans-Williamson rounding succeeds exactly when the SDP-induced quadratic form g_G behaves Gaussianly under random ±1 inputs; the Bonami-Beckner (4,2)-hypercontractive excess kurtosis κ is the canonical degree-2 deviation-from-Gaussianity invariant and is therefore a natural Fourier-analytic certificate for SDP tightness. Since g_G is degree-2, its full Fourier mass sits at level ≤ 2, so κ measures purely the 4th-cumulant cross-edge structure that GW rounding fails to capture. If true, this gives a polynomial-time computable moment-matrix-style witness $P(M) = \kappa$ that any 0.879-approximator must respect, transferable under polynomial reductions that preserve quadratic structure.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dcfa6a46.py", line 96, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dcfa6a46.py", line 74, in run_trial
    max_cut_value = max_cut(G)
                    ^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dcfa6a46.py", line 44, in max_cut
   cut_value = sum(G[u][v] if (u, v) in G else G[v][u] for u in cut for v in cut if (u, v) not in G and (v
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dcfa6a46.py", line 44, in <genexpr>
   cut_value = sum(G[u][v] if (u, v) in G else G[v][u] for u in cut for v in cut if (u, v) not in G and (v
                                                   ~^^^
KeyError: 0
```

Judge reasoning

The test crashed with a KeyError in the max_cut routine before any instances were evaluated, so neither the per-instance bound nor the Spearman correlation criterion was assessed. | next: Fix the max_cut implementation (use networkx edge iteration or a correct cut-edge sum over G.edges) and rerun the 120-instance protocol.

Golden-Ratio Rotation Discrepancy of Truth Tables Excludes ACC^0[m]

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Diophantine equidistribution: Erdős–Turán / Koksma rotation discrepancy of the golden-ratio Kronecker sequence (Weyl 1916, Erdős–Turán 1948, Schoissengeier bounds) applied as a P-uniform statistic on Boolean truth tables — essentially absent from circuit complexity × ACC^0[m] circuit size lower bounds for explicit P-time functions, in the Williams 2014 NEXP⊄ACC^0 / Nisan–Wigderson hardness-vs-randomness lineage
- **Recorded**: 2026-05-08 10:17 UTC
- **Entry ID**: `d329802a46f1`

Statement

For f:{0,1}^n→{0,1} with truth table T_f∈{0,1}$^{\{2}$n} and φ=(√5−1)/2, define the golden-ratio rotation discrepancy D_φ(f) := max_{1≤N≤2^n} |∑_{i<N} (−1)^{T_f[i]} · e^{2πi φ i}| / √N. We conjecture (a) any f computed by depth-d size-s ACC^0[m] satisfies D_φ(f) ≤ (s·log m)^{c·d} for an absolute constant c, while (b) the explicit Beatty function f_φ(x):=⌊φ·int(x)⌋ mod 2 (computable in P) has D_φ(f_φ) ≥ 2^{n/4}/poly(n); together these imply f_φ ∉ ACC^0[m] of polynomial size and yield the natural property R(f):=[D_φ(f) ≥ (log 2$^{n)}${ω(1)}] which is poly(2^n)-constructive, useful, but non-large (random f has D_φ=O(√n) by Khintchine), threading Razborov–Rudich at the largeness axis only.

Rationale

An irrational rotation by φ is spectrally orthogonal to every rational period dividing m, so any cyclotomic structure inherited by an ACC^0[m] truth table from Razborov–Smolensky low-degree GF(m)-polynomial approximators contributes only polylog to D_φ, whereas a Beatty word saturates D_φ by Weyl's theorem — exactly the asymmetry NW designs exploit when an irrational-rotation source fools mod-counting tests. Picking the largeness axis as the natural-proofs 'minimal violation' is principled because subexponential-time computability and ACC^0-uselessness are both desired by the Williams program, but largeness is the only Razborov–Rudich axiom not entangled with the Williams algorithmic-method machinery. No prior work links Erdős–Turán Beatty discrepancy of truth tables to ACC^0 size.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [2002.10956v2] Upper bounds on Kronecker coefficients with few rows - [1410.7087v3] Bounds on Kronecker and q-binomial coefficients - [0805.1385v3] Almost-natural proofs - [0302155v3] Generalized additive bases, Konig's lemma, and the Erdos-Turan conjecture

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f98e3adc.py", line 115, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f98e3adc.py", line 73, in run_trial
    f_phi_truth_table = beatty_function(n)
                                        ^
UnboundLocalError: cannot access local variable 'n' where it is not associated with a value
```

Judge reasoning

The test crashed with an UnboundLocalError before any data was produced, so neither the support nor falsification condition can be evaluated. | next: Fix the scoping bug in beatty_function (pass n as a parameter or define it in scope) and rerun the 30-seed n=20 trial.

Galois Concept Lattice Height Lower-Bounds Monotone KW Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Formal Concept Analysis (Wille 1982; Ganter-Wille 1999): Galois concept lattices L(K) of binary contexts K=(G,M,I), accessed via the NextClosure algorithm and the chain-height invariant h(L(K)) := length of the longest concept chain — a poly-time order-theoretic feature of bipartite incidences essentially absent from circuit/proof complexity. × Karchmer-Wigderson monotone formula depth D_+(f) of monotone Boolean f:{0,1}^n→{0,1}, equivalently the optimal monochromatic-rectangle depth of the KW relation on Min(f) × Max(f) with answers in [n] (Karchmer-Wigderson 1990; Raz-Wigderson; Tzameret).
- **Recorded**: 2026-05-08 10:25 UTC
- **Entry ID**: `bba09a0a4ff8`

Statement

For every monotone Boolean f:{0,1}^n→{0,1} with |Min(f)|≥2 and |Max(f)|≥2, define the FCA context K_f := (Min(f), Max(f)×[n], I_f) with I_f(x,(y,i))=1 iff x_i=1 and y_i=0; let L(K_f) be its Galois concept lattice and h(K_f) the length of its longest concept chain. Then the monotone Karchmer-Wigderson formula depth obeys D_+(f) ≥ ⌈log_2 h(K_f)⌉. A single monotone f exhibiting h(K_f) > 2^{D_+(f)} falsifies the conjecture.

Rationale

Each formal concept is a maximal KW-consistent rectangle of (true-input, false-input×answer) pairs; a chain of concepts is a strict refinement sequence of such rectangles, so any KW protocol must spend at least ⌈log_2 h(K_f)⌉ rounds to resolve the chain. FCA chain-height is unaffected by lifting tricks that defeat rank/discrepancy bounds, so it bypasses algebrization while retaining a poly-time proxy. The bound is tight for AND_n and OR_n (h=1) and grows with implicant-clause incidence richness, where existing rank/log-rank certificates plateau.

Novelty

- Judge: `NOVEL` over 5 arXiv hits

Top hits consulted: - [0710.4339v1] Heavy-Quark Masses from the Fermilab Method in Three-Flavor Lattice QCD - [1111.0981v1] Form factors for B to Kll semileptonic decay from three-flavor lattice QCD - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [1210.2401v1] Distributed Formal Concept Analysis Algorithms Based on an Iterative MapReduce Framework - [2311.04204v3] Sharp Thresholds Imply Circuit Lower Bounds: from random 2-SAT to Planted Clique

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_47b39da1.py", line 147, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_47b39da1.py", line 122, in run_trial
    new_closure = NextClosure(K_f, (Min_f, Max_f))
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_47b39da1.py", line 98, in NextClosure
    if all(x[i] == (y[0][i], y[1][i]) for i in range(n)):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_47b39da1.py", line 98, in <genexpr>
    if all(x[i] == (y[0][i], y[1][i]) for i in range(n)):
                    ~~~~^^^
TypeError: 'set' object is not subscriptable
```

Judge reasoning

The test crashed with a TypeError (‘set’ object is not subscriptable) inside NextClosure before producing any RESULT line, so no seeds were evaluated against the pre-registered criterion. | next: Fix the NextClosure implementation to represent attribute sets as indexable/sorted structures (e.g., frozenset converted to sorted tuple, or bitmask) and re-run the 30-seed trial.

Lyndon Factor Count Separates MOD_3 from Depth-2 ACC^0[2]

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorics on words (Chen-Fox-Lyndon factorization, Duval's linear-time algorithm) × ACC^0[2] circuit complexity, specifically depth-2 XOR-of-AND circuits (the canonical Razborov-Smolensky target where MOD_p vs MOD_q lower bounds are proven)
- **Recorded**: 2026-05-08 10:31 UTC
- **Entry ID**: `2243c6c24ab3`

Statement

Let L(T) denote the length of the unique Chen-Fox-Lyndon factorization of a binary string T, computable in linear time via Duval's algorithm. For every depth-2 ACC^0[2] circuit C on n variables of size at most n^2 (top XOR gate over AND gates of width at most ceil(log_2 n)+2), the truth table TT(C) in {0,1}$^{\{2}$n} read in lexicographic order satisfies L(TT(C)) <= (2^n)/sqrt(n). In contrast, the explicit family MOD_3_n(x) = [popcount(x) mod 3 == 0] satisfies L(TT(MOD_3_n)) >= (2^n)/8 for all n >= 6, and the gap (2^n)/8 vs (2^n)/sqrt(n) constitutes a Lyndon-witness of MOD_3 not in depth-2-ACC^0[2]_{n^2}.

Rationale

Depth-2 XOR-of-AND circuits compute F_2-affine-piecewise functions whose truth tables decompose along F_2-cosets, producing strongly self-similar binary strings that Lyndon factorization compresses into few large runs (Thue-Morse-like behavior dominates). MOD_3 has inherent period-3 ternary structure incompatible with any small F_2-affine cover, so its truth table fragments into many short incomparable Lyndon factors. The Chen-Fox-Lyndon decomposition has not, to our knowledge, been used as a quantitative ACC^0 invariant, even though it is one of the cheapest non-trivial word-combinatorial statistics available.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [1610.03808v2] Lyndon word decompositions and pseudo orbits on q-nary graphs - [1602.01530v8] Randomness Extraction in AC0 and with Small Locality

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_54f36a74.py", line 110, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_54f36a74.py", line 79, in run_trial
    tt_circuit = [circuit[i] for i in range(2**n)]
                  ~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError before producing any data, so neither the support nor the falsification condition was evaluated. Per the rules, a crashed test yields INCONCLUSIVE. | next:

Fix the circuit truth-table construction (ensure the circuit object is indexable over the full 2^n domain) and rerun the pre-registered protocol for n in {8,10,12}.

KKL Influence-Spread of Edge-Product Threshold Lower-Bounds Tseitin Resolution

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier analysis on the Boolean cube — Kahn-Kalai-Linial 1988 max-influence theorem and Bonami-Beckner (2->4) hypercontractivity, specifically the influence-spread quotient nu(f) := TI(f)/MI(f) = (sum_v Inf_v f)/(max_v Inf_v f) of a graph-derived ferromagnetic 2-spin threshold function — an invariant essentially absent from proof complexity but central to Friedgut-Bourgain junta theorems. × Resolution refutation length L_R(Tseitin(G,sigma)) of 3-regular graphs G with odd vertex charge sigma:V->F_2 (the Urquhart / Ben-Sasson-Wigderson canonical sub-Frege gateway target).
- **Recorded**: 2026-05-08 10:39 UTC
- **Entry ID**: `8d6bdbe44c56`

Statement

For a connected 3-regular graph G on n vertices and odd sigma, let f_G:{-1,+1}^V -> {-1,+1} be the edge-product threshold f_G(x) := sign(sum_{(u,v) in E} x_u x_v) with sign(0):=+1, and define nu_KKL(G) := (sum_{v in V} Inf_v(f_G)) / (max_{v in V} Inf_v(f_G)) under uniform measure on {-1,+1}^V, with Inf_v(f_G) = Pr_x[f_G(x) != f_G(x^{(v)})]. Conjecture: there is an absolute constant c > 0 such that for every 3-regular G with odd sigma, L_R(Tseitin(G,sigma)) >= 2^{c * nu_KKL(G)}. A single instance with nu_KKL(G) > 12*log_2(L_R(Tseitin(G,sigma))+2) refutes the conjecture.

Rationale

On expanders the edge-product Hamiltonian f_G is a ‘spread’ threshold function and KKL forces TI/MI to be Theta(n/log n) — influences are smeared across all vertices because no local minority can flip the global ferromagnetic sign cheaply. On non-expanders with a bottleneck, a single bridge vertex carries macroscopic influence, collapsing TI/MI to O(1); this matches the Ben-Sasson-Wigderson dichotomy that Tseitin is hard exactly when expansion holds, but routes it through a Fourier-influence proxy that has never been computed for Tseitin instances.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_872cbc6e.py", line 147, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_872cbc6e.py", line 123, in run_trial
    G = family(n)
        ^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_872cbc6e.py", line 24, in generate_rand
    neighbors = random.sample(range(v + 1, n), 2)
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 430, in sample
```

```
    raise ValueError("Sample larger than population or is negative")
ValueError: Sample larger than population or is negative
```

Judge reasoning

The test crashed in the random 3-regular graph generator (ValueError: Sample larger than population) before any instances were evaluated, so neither the Spearman correlation nor the per-instance bound could be assessed. | next: Replace the ad-hoc 3-regular generator with a robust construction (e.g., networkx.random_regular_graph(3, n) with even n) and rerun the 360-instance protocol.

Schur-Horn Diagonal-Spectrum Defect Bounds SOS-2 Max-Cut Gap

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Horn diagonal-spectrum majorization theory in matrix analysis (Schur 1923, Horn 1954, Klyachko 1998 Horn cone): the Schur-Horn defect SH(M) := ||diag(M) − sort_desc(λ(M))||_1 of a Hermitian matrix M, an L^1 measure of how far the diagonal is from the eigenvalue vector in majorization order — classical in matrix analysis but essentially absent from SOS / CSP integrality-gap literature × Sum-of-Squares degree-2 (Goemans-Williamson / Delorme-Poljak basic SDP) integrality gap for Max-Cut on 3-regular graphs, the canonical SoS hierarchy CSP target
- **Recorded**: 2026-05-08 10:49 UTC
- **Entry ID**: `77f332fbbf06`

Statement

Let G be a 3-regular graph on n vertices with 0/1 adjacency matrix A. Define SH(G) := ||diag(A^2) − sort_desc(eig(A^2))||_1 = Σ_i |3 − λ_i(A)^2| (a poly-time computable invariant). Let SOS2(G) := (n/4)(3 − λ_min(A)) be the Delorme-Poljak SOS-2 Max-Cut value, and let MaxCut(G) be the exact maximum cut. Then SOS2(G) − MaxCut(G) ≤ SH(G)/8 for every 3-regular G; a single 3-regular graph with SOS2(G) − MaxCut(G) > SH(G)/8 refutes the conjecture.

Rationale

The Delorme-Poljak SOS-2 bound is purely spectral, while MaxCut depends on diagonal-aligned 0/±1 cut indicators, so the integrality gap should be controlled by how far the Gram structure of A^2 deviates from being diagonalizable on the standard basis — exactly what the Schur-Horn defect quantifies. Majorization theory hands us a coordinate-free measure of this 'eigenvector misalignment' that is poly-time computable and respects the symmetry of the SDP relaxation. If true, SH(G) becomes a sharp diagnostic for which sparse instances exhibit SOS-2 integrality gaps without invoking pseudo-distribution refutations.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out at 240s (returncode 124) before emitting any RESULT line, so neither the support fraction nor the Spearman correlation could be evaluated. With no data and the critic challenging, the pre-registered support condition cannot be confirmed. | next: Reduce the workload (e.g., fewer/smaller instances such as n=8,10 only, or replace exact MaxCut with a faster solver) and rerun with a higher timeout to obtain the support_fraction and Spearman ρ.

Bregman-Minc Permanent Defect Lower-Bounds SOS-2 Max-Cut Gap on Triangle-Free Cubic Graphs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Permanental graph theory and the Bregman-Minc / Egorychev-Falikman permanent inequalities (combinatorial matrix theory: matrix permanents of (0,1)-matrices, with Schrijver-type tightness on bipartite regular blocks). This corner of the permanent calculus, classically tied to van der Waerden / matching-counting problems, is essentially absent from CSP and SOS integrality-gap literature. × Sum-of-squares degree-2 (Delorme-Poljak / Goemans-Williamson basic SDP) integrality gap for unweighted Max-Cut on 3-regular triangle-free graphs, the canonical SOS hierarchy CSP target.
- **Recorded**: 2026-05-08 11:00 UTC
- **Entry ID**: `139cdbb34775`

Statement

For every triangle-free 3-regular graph G on n vertices with adjacency matrix A_G, define the Bregman-Minc defect pi(G) := (n/3) * log_2(6) - log_2(perm(A_G)), which is nonnegative by Bregman-Minc and zero iff G is a disjoint union of K_{3,3}'s (Egorychev-Falikman tightness for r-regular (0,1)-matrices). Conjecture: there exists an absolute constant c >= 1/100 such that SDP_2(G) - MaxCut(G) >= c * pi(G) * sqrt(n) for all such G, where SDP_2(G) = m/2 + (n/4)*|lambda_min(A_G)| is the Delorme-Poljak basic-SDP value. Equivalently, any triangle-free cubic G whose permanent strictly undershoots the Bregman-Minc baseline must exhibit a quantitatively positive degree-2 SOS Max-Cut gap.

Rationale

The Bregman-Minc baseline 6^{n/3} is saturated exactly by bipartite K_{3,3}-block cubic graphs - the same family on which the GW SDP is integral (gap = 0); deviation pi(G) > 0 detects non-bipartite-cubic 'permanental dispersion' that obstructs any bipartite-style rounding. Because the permanent encodes matching-mass spread across all sign patterns simultaneously, a deficit pi(G) plausibly forces spectral asymmetry between A_G and its bipartite envelope, which is exactly what feeds the |lambda_min| term in Delorme-Poljak. This would import a permanent-inequality / matching-polynomial machinery completely outside the current SOS / pseudoexpectation toolbox, yielding a poly-time-computable structural lower bound on the SOS-2 gap.

Novelty

- Judge: `NOVEL` over 3 arXiv hits

Top hits consulted: - [1901.00043v1] On the structure of (claw,bull)-free graphs - [2109.02238v1] Optimal solutions and ranks in the max-cut SDP - [1403.5248v3] NP-hardness results for partitioning graphs into disjoint cliques and a triangle-free subgraph

Empirical Test

- exit code: `124`, elapsed: `240.07s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out after 240s without producing a RESULT line, so the pre-registered Spearman/support-fraction criteria could not be evaluated. Per rules, a crashed/timed-out test yields INCONCLUSIVE. | next: Reduce the instance set (e.g., $n \le 14$ or 10 seeds) or replace exact permanent computation with Ryser's formula plus a per-instance timeout, then rerun to obtain the pre-registered statistics.

Selberg Log-Gas Rigidity Defect Bounds SOS-2 Max-CUT Gap

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Random matrix theory: the β=2 Selberg-integral / log-gas free energy $F[\lambda] = \Sigma_{i<j} \log|\lambda_i - \lambda_j|$ measuring spectral rigidity relative to the Wigner-semicircle / Kesten-McKay quantile law (Mehta, Forrester, Wigner-Dyson 'rigidity' tradition), used in β-ensemble fluctuation theory but essentially absent from SOS / CSP integrality-gap literature and provably distinct from free-probability cumulant calculus × Sum-of-squares degree-2 (Lasserre level-2 / Goemans-Williamson basic SDP) integrality gap for unweighted max-CUT on 3-regular graphs, the canonical SOS hierarchy CSP target
- **Recorded**: 2026-05-08 11:07 UTC
- **Entry ID**: `14fdafc31b7e`

Statement

For a 3-regular graph G on n vertices, let $\lambda_1 \ge \ldots \ge \lambda_n$ be the eigenvalues of A_G and let $\mu_i := 2\sqrt{2}\cdot\cos((i-1/2)\pi/n)$ be the Kesten-McKay quantile points (the d=3 Wigner-semicircle inverse-CDF on n equispaced probabilities). Define the Selberg log-defect $SLD(G) := (2/n^2)\cdot[\Sigma_{i<j} \log|\lambda_i - \lambda_j| - \Sigma_{i<j} \log|\mu_i - \mu_j|]$, an unsigned spectral-rigidity score where $SLD(G)<0$ means strictly more rigid than the semicircle. Conjecture: there exists an absolute constant $c \ge 1/8$ such that for every 3-regular graph G, $SDP_2(G)/MC(G) \ge 1 + c\cdot\max(0, -SLD(G))^2$, where $SDP_2(G)$ is the GW basic-SDP value $\max_{\{X\succeq 0,\mathrm{diag}(X)=1\}} (|E|-\langle A_G,X\rangle/2)/2$ and MC(G) is the exact max-cut; equivalently any 0.879-approximating moment matrix has $SLD(G) \ge -\sqrt{(0.121/c)}$.

Rationale

Random 3-regular graphs are Ramanujan-like and $SLD(G)\approx 0$, the regime where SDP is tightest; off-spectrum graphs with crystalline (rigid, sub-Wigner) bulk are exactly those whose few outlier eigenvalues drive a slack $\langle A_G,X\rangle$ in the GW relaxation. Log-gas rigidity is a coordinate-free, S_n-natural invariant — orthogonal in technique to Fourier/hypercontractivity, kurtosis, and Positivstellensatz arguments that have already failed on this target — so the bridge probes new structure: rigidity-induced over-relaxation. The mapping is GCT-flavored in that the β=2 Selberg integral is the partition function of the U(n)-invariant Haar measure on Hermitian conjugacy classes, the same representation-theoretic object Mulmuley invokes in plethystic asymptotics.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.29s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_832d5e7a.py", line 109, in <module>
```

```
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_832d5e7a.py", line 89, in run_trial
    lambda_ = eigenvalues(adjacency_matrix)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_832d5e7a.py", line 47, in eigenvalues
    v /= math.sqrt(sum(x * x for x in v))
TypeError: unsupported operand type(s) for /=: 'list' and 'float'
```

Judge reasoning

The test crashed with a TypeError in the eigenvalues routine (list /= float), so no data was produced and the pre-registered Spearman/instance criteria could not be evaluated. | next: Fix the eigenvector normalization bug (use a numpy array or divide elementwise) and rerun the 150-instance trial.

Lasserre Moment-Rank Gap of Accept Set Bounds AC0 PARITY Size

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Sum-of-squares / Lasserre hierarchy moment-matrix theory (Lasserre 2001, Parrilo 2003, Grigoriev 2001) — specifically the empirical degree-k pseudo-moment Gram matrix M_k(f) := (1/|f^{-1}(1)|) sum_{x in f^{-1}(1)} v_k(x) v_k(x)^T over the multilinear monomial vector v_k(x) of degree <= k on the boolean cube, used here as an S_n-equivariant non-uniform circuit invariant rather than as a CSP relaxation; its R-rank is poly-time via SVD and is essentially absent from the AC^0 / switching-lemma literature. × AC^0 depth-d formula size for PARITY_n in the Hastad / Razborov-Smolensky lineage; the canonical NEXP-vs-AC^0 gateway target where tight size constants for d>=3 remain open.
- **Recorded**: 2026-05-08 11:12 UTC
- **Entry ID**: `ad24d125d8b7`

Statement

For every boolean f:{0,1}^n -> {0,1} and integer k <= n/2, let rk_k(f) := rank_R(M_k(f)) <= N_k where N_k = sum_{j<=k} C(n,j). Conjecture (A): rk_k(PARITY_n) = N_k for every k <= floor(n/2) (full Lasserre rank of the even-parity accept set up to half-degree). Conjecture (B): for every AC^0 depth-d formula F of size s computing g on n variables, rk_k(g) <= s * k * C(n, k-1) for every 2 <= k <= n/2. Combining (A) and (B) at k = ceil(n^{1/(d-1)}) forces any AC^0 depth-d formula for PARITY_n to have size >= N_k / (k * C(n,k-1)) = $2^{\{\mathrm{Omega}(n}$ {1/(d-1)} / log n)}; one counterexample to (A) at any (n,k <= n/2) or to (B) at any (F,k,g) refutes the conjecture.

Rationale

PARITY’s accept set is the unique S_n-invariant antipodal half-cube and so should saturate every Lasserre moment-matrix dimension below half-degree, while AC^0 formulas of size s decompose their accept set as a union of s subcubes whose moment-matrix support sits inside an O(s)-dimensional (k*C(n,k-1))-spanned subspace of the multilinear monomial basis (one degree-(k-1) factor per gate fan-in). Re-expressing the Hastad/Razborov barrier as a moment-matrix rank deficit makes the lower bound a single linear-algebraic invariant computable in O((2^n + N_k) N_k^2) time. The bridge sits squarely in the SOS-hierarchy framework — Lasserre level-k rank for empirical pseudo-distributions on f^{-1}(1) — while being a non-uniform circuit lower bound rather than a CSP integrality gap, occupying a corner with <5 papers.

Novelty

- Judge: “ over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by NATURAL_PROOFS barrier

Forman-Ricci Negative-Edge Count Lower-Bounds Tseitin DPLL Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Discrete Ricci curvature à la Forman (Forman 1998, 'Bochner's method for cell complexes'): the augmented combinatorial Ricci curvature F(e) of an edge in a graph, computed in closed form from triangle and 4-cycle counts at e, distinct from Ollivier-Ricci (no transport LP) and essentially absent from proof complexity (≈0 papers, vs. Skenderi 2025 for Ollivier-Ricci on SAT) × DPLL / tree-resolution refutation size T_DPLL(Tseitin(G,σ)) of 3-regular Tseitin formulas with odd vertex charge σ:V→F_2 (the Urquhart / Ben-Sasson–Wigderson canonical sub-Frege gateway target)
- **Recorded**: 2026-05-08 11:22 UTC
- **Entry ID**: `a687b7685342`

Statement

For a 3-regular graph G=(V,E) and odd charge σ:V→F_2, define the augmented Forman-Ricci curvature on edge e=(u,v) by F(e) := 2·t(e) + 2·q(e) − (deg(u)+deg(v)−2), where t(e) is the number of triangles through e and q(e) is the number of induced 4-cycles through e. Let N_−(G) := #{e ∈ E : F(e) < 0}. We conjecture absolute constants c ≥ 0.4 and C ≤ 5 such that log_2 T_DPLL(Tseitin(G,σ)) ≥ c · N_−(G) − C for every odd-charged 3-regular Tseitin instance with |V| ≥ 12, where T_DPLL counts recursive calls under unit propagation and a fixed lex variable order; one instance with log_2 T_DPLL < c·N_−(G) − C falsifies it.

Rationale

Negative Forman-Ricci on an edge encodes local 'tree-like' branching geometry — neither triangles nor short squares can short-circuit width-building inferences across e — so the count N_−(G) is a poly-time, S_n-natural proxy for the 'expander mass' that powers Ben-Sasson–Wigderson width lower bounds, but read off from purely local edge statistics rather than λ_2 or Cheeger flow. Because Forman curvature decouples cleanly from spectral and metric-embedding invariants already attempted on Tseitin (Laplacian gap, Ollivier-Ricci, Ihara zeta, Leighton-Rao), an empirical correlation would expose a new local-to-global handle on the Urquhart target. The bridge is barrier-safe: the invariant is graph-only (no truth-table access), so no natural-proofs largeness, and it relativizes/algebrizes trivially.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results; pre-registered support condition cannot be verified | next: Increase timeout duration and re-run with stricter resource limits

---

Matroid Rank Gap in Monotone DNF Depth for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: matroid theory × monotone circuit lower bounds for k-CLIQUE
- **Recorded**: 2026-05-08 11:42 UTC
- **Entry ID**: `bfe3aa3f84d1`

Statement

Let μ(f) denote the maximal number of disjoint terms in a DNF representation of a Boolean function f. For any monotone DNF formula of size poly(n), μ(f) = O(log n). For the k-CLIQUE indicator function, μ(f) = Ω(n^{1/2})

Rationale

Matroid rank functions are inherently submodular, and their growth rate under conjunction directly relates to the structural complexity of DNF representations. The k-CLIQUE function's exponential DNF complexity forces μ(f) to scale with √n, while polynomial-size DNFs can be compressed into logarithmic rank via matroid decomposition techniques.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_81a60df6.py", line 73, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_81a60df6.py", line 52, in run_trial
    mu_f = compute_mu(dnf_formula)
           ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_81a60df6.py", line 42, in compute_mu
    if is_independent_set(S, dnf_formula):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_81a60df6.py", line 33, in is_independen
    if any(S[i] == S[j] == 1 for term in dnf_formula if term[u] == term[v]):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_81a60df6.py", line 33, in <genexpr>
    if any(S[i] == S[j] == 1 for term in dnf_formula if term[u] == term[v]):
             ^
NameError: name 'i' is not defined. Did you mean: 'id'?
```

Judge reasoning

Test crashed due to NameError, preventing data collection. Pre-registered support condition cannot be evaluated without successful runs. | next: Fix the NameError in the test script (undefined variable 'i' in line 33) and re-run experiments

---

Fourier Coefficient Flatness Implies Worst-Case Hardness for Boolean Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: FOURIER_ANALYSIS × CIRCUIT_COMPLEXITY
- **Recorded**: 2026-05-08 11:59 UTC
- **Entry ID**: `8f9c6a311a28`

Statement

For a Boolean function f on n variables, if the average of the absolute values of its Fourier coefficients is less than 1/√n, then f is hard to compute in the worst case (i.e., requires exponential time).

Rationale

Functions with small Fourier coefficients are 'flat' and have limited correlation with parity functions, suggesting they are hard to compute in the worst case. This connects average-case properties (small average Fourier coefficients) to worst-case hardness.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.41s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and higher resource allocation

---

Fourier Coefficient Product Lower Bound on Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: FOURIER_ANALYSIS × COMMUNICATION_COMPLEXITY
- **Recorded**: 2026-05-08 12:13 UTC
- **Entry ID**: `cd6772561f42`

Statement

For a Boolean matrix M representing the disjointness function on n elements, the product of the minimal and maximal absolute Fourier coefficients, τ(M) = min_k |F(k)| * max_k |F(k)|, satisfies τ(M) ≥ c * n for some absolute constant c > 0.

Rationale

The product of the minimal and maximal Fourier coefficients captures the spread of the function's energy in the Fourier domain. A large product implies significant variation in the Fourier coefficients, which would require more communication to distinguish the inputs.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `-9`, elapsed: `79.78s`

```
(no output)
```

Judge reasoning

Test crashed before producing data, preventing evaluation of support fraction or counterexamples. | next: Re-run test with increased time/memory limits and deterministic seed configuration

---

Fourier Coefficient Decay Implies SOS Degree Lower Bound for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: FOURIER_ANALYSIS × SOS_DEGREE
- **Recorded**: 2026-05-08 12:27 UTC
- **Entry ID**: `7d81194403f3`

Statement

For any Boolean function f: {0,1}^n → {0,1}, let μ_k(f) denote the k-th level Fourier coefficient. If μ_k(f) decays faster than 1/k^α for α > 2, then any degree-d SOS relaxation achieving a 0.878-approximation for max-CUT on f must satisfy d ≥ Ω(n^{1/α}).

Rationale

The decay rate of Fourier coefficients governs the spectral structure of Boolean functions, while SOS degree captures the algebraic complexity of approximating max-CUT. Rapid decay implies low-frequency dominance, requiring higher SOS degrees to resolve high-frequency components critical for tight approximations.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.34s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing validation of support fraction or counterexample detection. | next: Run test with extended timeout and higher resource allocation to complete execution

---

Communication Matrix Discrepancy Lower Bound for AC⁰ PARITY Circuits

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Communication Complexity × AC⁰ Circuit Complexity
- **Recorded**: 2026-05-08 12:50 UTC
- **Entry ID**: `dd369e80c960`

Statement

For any AC⁰ circuit C computing PARITY on n inputs, the discrepancy of its communication matrix M_C satisfies disc(M_C) ≥ c · log(size(C)) for some absolute constant c > 0.

Rationale

The discrepancy of the communication matrix for PARITY is tightly linked to its communication complexity, which is known to be Ω(log n) for AC⁰ circuits. By leveraging the invariance of discrepancy under variable permutations (a non-trivial group action), this invariant provides a natural measure of circuit complexity that replaces the random restriction argument in switching lemmas.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ac6b0fa4.py", line 60, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ac6b0fa4.py", line 42, in run_trial
    circuit = generate_ac0_circuit(n, depth)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ac6b0fa4.py", line 20, in generate_ac0_
    subcircuits = [generate_ac0_circuit(n // 2, depth - 1) for _ in range(2)]
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ac6b0fa4.py", line 20, in generate_ac0_
    subcircuits = [generate_ac0_circuit(n // 2, depth - 1) for _ in range(2)]
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ac6b0fa4.py", line 20, in generate_ac0_
    subcircuits = [generate_ac0_circuit(n // 2, depth - 1) for _ in range(2)]
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
   [Previous line repeated 1 more time]
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ac6b0fa4.py", line 21, in generate_ac0_
    return [subcircuits[0][i] ^ subcircuits[1][i] for i in range(n)]
            ~~~~~~~~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError during recursion, preventing data collection. Pre-registered support condition cannot be evaluated. | next: Fix the recursive circuit generation to handle base cases properly and re-run tests with increased seed coverage

---

Seed Length of Nisan-Wigderson PRG Bounded by Minimal Linear Approximation Count

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: LINEAR_ALGEBRA × PSEUDORANDOM_GENERATORS
- **Recorded**: 2026-05-08 13:02 UTC
- **Entry ID**: `141266783ffe`

Statement

For a hard function f against small-depth circuits, let L(f) denote the minimal number of linear functions needed to approximate f within ε = 1/n. The seed length of the Nisan-Wigderson PRG fooling f is at most log(L(f)) + O(log n).

Rationale

The minimal number of linear functions required to approximate f reflects its structural simplicity. If f can be approximated by few linear functions, it implies lower computational complexity, enabling a shorter seed length for the PRG. This connects algebraic approximation properties to pseudorandomness efficiency.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cc733f77.py", line 80, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cc733f77.py", line 50, in run_trial
    L_f = rank(A)
          ^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cc733f77.py", line 42, in rank
    A, _ = gaussian_elimination(A, [0]*len(A))
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cc733f77.py", line 36, in gaussian_elim
    factor = A[j][i] / A[i][i]
             ~~~~~~~~^~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with division by zero error during Gaussian elimination, preventing validation of the conjecture | next: Fix code to handle singular matrices and re-run experiments with diverse test cases

---

Operator Norm Separation for Read-Twice Branching Programs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: OPERATOR_THEORY × BRANCHING_PROGRAMS
- **Recorded**: 2026-05-08 13:14 UTC
- **Entry ID**: `bac847b867fa`

Statement

For any read-twice branching program P computing a function f, the spectral norm of the associated transition matrix M_P is O(log size(P)), whereas for the trivial read-twice BP computing the inner product mod 2 (IP_2), the spectral norm is Ω(n).

Rationale

The spectral norm of the transition matrix captures the 'spread' of the BP's transitions. Read-twice BPs, being more constrained, have matrices with smaller spectral norms, while the IP_2 function, which requires exponential size, has a much larger norm. This separation could lead to lower bounds on the size of read-twice BPs for certain functions.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `1.68s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5b091259.py", line 78, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5b091259.py", line 50, in run_trial
    M_P = gaussian_elimination(M_P)
          ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5b091259.py", line 24, in gaussian_elim
    factor = A[j][i] / A[i][i]
             ~~~~~~~~^~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing result collection. No data to assess conjecture validity. | next: Debug gaussian_elimination() to handle singular matrices, then re-run with robust error checking

---

Plethysm Multiplicity Gap in Symmetric Powers of Permanent vs Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl duality, plethysm, algebraic combinatorics × Algebraic complexity theory (Extended Frege / Frege lower bounds)

- **Recorded**: 2026-05-08 13:26 UTC
- **Entry ID**: `ee4cd4f186fc`

Statement

For all n ≥ 2 and m < n^{1.5}, the multiplicity of the trivial representation in the decomposition of S^m(perm_n) under GL(n) exceeds that in $S^{m(det\_m}O(1))$ by a factor of Ω(n^{1/3})

Rationale

Plethysm multiplicities capture orbit closure invariants under linear substitutions. If perm_n's decomposition contains exponentially more trivial components than det_m^O(1) for m < n^{1.5}, this would separate their orbit closures and imply super-polynomial EF lower bounds via Landsberg's tensor rank framework.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38989d51.py", line 87, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38989d51.py", line 59, in run_trial
   perm_multiplicities, det_multiplicities = compute_multiplicities(max(n_values), m_values)
                                             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38989d51.py", line 52, in compute_multi
    perm_multiplicities = [trivial_multiplicity(m, n) for m in m_values]
                           ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38989d51.py", line 41, in trivial_multi
    return plethysm(m, n)
           ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38989d51.py", line 37, in plethysm
    result += binomial_coefficient(m, k) * plethysm(k, n)
                                           ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38989d51.py", line 37, in plethysm
    result += binomial_coefficient(m, k) * plethysm(k, n)
                                           ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38989d51.py", line 37, in plethysm
    result += binomial_coefficient(m, k) * plethysm(k, n)
                                           ^^^^^^^^^^^^^^
   [Previous line repeated 992 more times]
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38989d51.py", line 30, in binomial_coef
    return factorial(n) // (factorial(k) * factorial(n - k))
           ^^^^^^^^^^^^
RecursionError: maximum recursion depth exceeded
```

Judge reasoning

Test crashed due to recursion depth limits, preventing empirical validation of the conjecture | next: Optimize plethysm computation with memoization and iterative approaches to handle large n/m values

Moment-Matrix Rank Deficit in Max-CUT Approximation

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: REAL_ALGEBRAIC_GEOMETRY × SOS_DEGREE
- **Recorded**: 2026-05-08 13:36 UTC
- **Entry ID**: `1c394bcaff4a`

Statement

For any graph G with maximum cut α(G), any degree-d Sum-of-Squares (SOS) relaxation achieving approximation ratio 0.879 + ε must have a moment matrix M_d with rank ≥ Ω(√n). This rank threshold is preserved under polynomial reductions between Max-CUT instances.

Rationale

The moment matrix rank captures the 'complexity' of the SOS relaxation. By linking it to the graph's structure, we isolate a geometric obstruction to improving the Goemans-Williamson bound. This connects spectral concentration in random matrices (real algebraic geometry) to SOS hierarchy limitations.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run with stricter resource limits to complete the experiment

---

Secant Variety Dimension Lower Bounds Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: ALGEBRAIC_GEOMETRY_OF_SECANT_VARIETIES × COMMUNICATION_COMPLEXITY
- **Recorded**: 2026-05-08 13:42 UTC
- **Entry ID**: `b35b32763c26`

Statement

For a Boolean matrix M representing the disjointness function on n-bit inputs, let τ(M) be the dimension of the secant variety of the set of points { (x, y) | M[x][y] = 1 } in the projective space P^{n-1} × P^{n-1}. Then, the randomized communication complexity R(M) satisfies R(M) ≥ Ω(τ(M)) with τ(DISJ_n) = Ω(n).

Rationale

Secant varieties capture the geometric complexity of point configurations, and their dimension reflects the minimal number of rank-1 tensors needed to span the matrix. By linking this algebraic invariant to communication complexity, we isolate a novel geometric obstruction to efficient protocols for disjointness.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c34eb15.py", line 91, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c34eb15.py", line 72, in run_trial
    secant_dim = secant_dimension(M)
                 ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c34eb15.py", line 60, in secant_dimens
    secant_ideal.append(gaussian_elimination(A, b))
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c34eb15.py", line 33, in gaussian_elim
    A[j][k] += factor * A[i][k]
    ~~~~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed during gaussian_elimination with index error, preventing dimension calculation | next: Debug gaussian_elimination implementation to handle matrix indexing properly

---

Matroid Rank Deficit in Monotone DNF Depth for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: MATROID_THEORY × MONOTONE_CIRCUIT_COMPLEXITY
- **Recorded**: 2026-05-08 13:48 UTC
- **Entry ID**: `86fc235d2aba`

Statement

Let M be a matroid of rank r on n elements, and let μ(M) = r - rank(groundset - closure(DNF_terms)). For any monotone DNF formula with size S(n), μ(M) ≤ log n. For the k-CLIQUE indicator function, μ(M) ≥ Ω(n^{1 - 1/(k-1)}).

Rationale

Matroid rank functions are submodular and capture combinatorial dependencies. By linking matroid rank deficits to DNF size, we create a structural invariant that could amplify communication complexity lower bounds for k-CLIQUE via combinatorial duality. This avoids algebraic barriers by staying in discrete matroid theory.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b53c5454.py", line 126, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b53c5454.py", line 101, in run_trial
    DNF_terms = generate_k_clique_dnf(n, k=3)
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b53c5454.py", line 85, in generate_k_cl
    term.add(clique_edges[i*k + j - (i+1)*(i//2)][0])
             ~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError during DNF term generation for k=3, preventing result collection. | next: Debug generate_k_clique_dnf() implementation for k=3 edge indexing

---

Monochromatic Rectangle Density Lower Bounds for PARITY Communication Matrices

- **Verdict**: INCONCLUSIVE
- **Bridge**: COMMUNICATION_COMPLEXITY × AC0_PARITY
- **Recorded**: 2026-05-08 14:08 UTC
- **Entry ID**: 604222a146e1

Statement

For any monotone AC^0 formula C computing PARITY on n variables, the number of monochromatic rectangles in its communication matrix is at least $\Omega(n)$, implying that the formula size is exponential in n.

Rationale

Monochromatic rectangle density in the communication matrix of PARITY captures structural constraints on monotone AC^0 formulas. By linking this combinatorial invariant to the KW game's protocol cost, we derive a lower bound on formula size, bypassing random restriction arguments.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 266.78s

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out without producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with extended computation time

---

Spectral Gap Lower Bound on Tseitin Resolution Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: SPECTRAL_GRAPH_THEORY × RESOLUTION_LENGTH
- **Recorded**: 2026-05-08 14:26 UTC
- **Entry ID**: `14b9250ca476`

Statement

For a d-regular graph G with algebraic connectivity λ_2, the resolution length of its Tseitin formula is at least `2^{Ω(λ₂)}`. Non-expanders (with $\lambda_2 = o(1)$) have resolution length subexponential in λ_2.

Rationale

Spectral gap (λ_2) quantifies expansion properties of graphs. Expanders (large λ_2) are known to require exponential resolution for Tseitin formulas. This conjecture links λ_2 directly to resolution complexity, offering a new graph invariant for hardness characterization.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and higher resource allocation

---

Hilbert Series Complexity of Orbit Closures for Permanent vs Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: ALGEBRAIC_GEOMETRY_OF_ORBIT_CLOSURES × GEOMETRIC_COMPLEXITY_THEORY
- **Recorded**: 2026-05-08 14:45 UTC
- **Entry ID**: `4cd8cbfd5e94`

Statement

For all n ≥ 2 and m < `n^{1.5}`, the Hilbert series of the coordinate ring of the orbit closure of the permanent polynomial perm_n under GL(n) has a strictly higher degree than the Hilbert series of the orbit closure of the determinant polynomial det_m under GL(m), and this invariant is preserved under linear substitutions.

Rationale

The Hilbert series encodes the growth of the dimensions of graded components in the coordinate ring of an orbit closure. If perm_n's orbit closure has a higher-degree Hilbert series than det_m's, it suggests structural complexity differences that could contribute to separating their complexity classes. This ties to GCT's goal of using algebraic geometry to distinguish det and perm.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a3564bca.py", line 124, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a3564bca.py", line 93, in run_trial
    m = random.randint(1, math.isqrt(n**1.5))
                          ^^^^^^^^^^^^^^^^^^
TypeError: 'float' object cannot be interpreted as an integer
```

Judge reasoning

Test crashed due to type error in calculating m; insufficient data to assess conjecture | next: Fix integer conversion in test code to handle n**1.5 properly

---

Secant Variety Dimension Lower Bounds for Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: ALGEBRAIC_GEOMETRY_OF_SECANT_VARIETIES × COMMUNICATION_COMPLEXITY
- **Recorded**: 2026-05-08 14:52 UTC
- **Entry ID**: `adedf0b3ac89`

Statement

Let M be the communication matrix of the disjointness function DISJ_n. Define τ(M) as the dimension of the secant variety of the variety V(M) spanned by the rank-1 tensors of M. Then τ(M) ≥ Ω(n), and equality holds for all n ≤ 40.

Rationale

Secant variety dimensions capture the minimal number of rank-1 tensors needed to approximate a matrix, which directly relates to communication complexity. For DISJ_n, the secant variety's dimension grows linearly with n, matching the Ω(n) lower bound. This connects algebraic geometry to communication complexity via tensor rank theory.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.14s`

```
ic_name": "secant_variety_dimension", "metric_value": 0, "instances_tested": 6, "conjecture_holds":
TRIAL: {"seed": 547, "metric_name": "secant_variety_dimension", "metric_value": 0, "instances_tested
TRIAL: {"seed": 593, "metric_name": "secant_variety_dimension", "metric_value": 0, "instances_tested
TRIAL: {"seed": 631, "metric_name": "secant_variety_dimension", "metric_value": 0, "instances_tested
TRIAL: {"seed": 677, "metric_name": "secant_variety_dimension", "metric_value": 0, "instances_tested
TRIAL: {"seed": 727, "metric_name": "secant_variety_dimension", "metric_value": 0, "instances_tested
TRIAL: {"seed": 773, "metric_name": "secant_variety_dimension", "metric_value": 0, "instances_tested
TRIAL: {"seed": 821, "metric_name": "secant_variety_dimension", "metric_value": 0, "instances_tested
TRIAL: {"seed": 877, "metric_name": "secant_variety_dimension", "metric_value": 0, "instances_tested
TRIAL: {"seed": 929, "metric_name": "secant_variety_dimension", "metric_value": 0, "instances_tested

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_204982c5.py", line 130, in <module>
    mean = sum(result['metric_value'] for result in results) / len(results)
                                                    ^^^^^^^
NameError: name 'results' is not defined. Did you mean: 'result'?
```

Judge reasoning

Test crashed due to undefined variable 'results', preventing reliable evaluation of conjecture support. | next: Fix the test code's results handling and re-run experiments with proper validation

---

Free Entropy Lower Bound on Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: FREE_PROBABILITY × COMMUNICATION_COMPLEXITY
- **Recorded**: 2026-05-08 15:00 UTC
- **Entry ID**: `cd1afb38ca62`

Statement

For any n ≥ 1, let M_n be the communication matrix of the DISJOINTNESS function on n-bit inputs. Define τ(M_n) as the free entropy of the non-commutative random variables derived from M_n's spectral decomposition. Then τ(M_n) ≥ Ω(n) and τ(DISJ_n) = Ω(n) for all n.

Rationale

Free entropy quantifies the 'complexity' of non-commutative distributions, capturing dependencies in high-dimensional matrices. By linking M_n's spectral structure to free probability invariants, we derive a lower bound on communication complexity via the non-commutative nature of DISJOINTNESS's matrix representation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3435fa78.py", line 75, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3435fa78.py", line 62, in run_trial
    tau_M_n = free_entropy(M_n)
              ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3435fa78.py", line 53, in free_entropy
    A = gaussian_elimination(M_n)
        ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3435fa78.py", line 26, in gaussian_elim
    factor = Fraction(A[j][i], A[i][i])
             ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/fractions.py", line 277, in __new__
    raise TypeError("both arguments should be "
TypeError: both arguments should be Rational instances
```

Judge reasoning

Test crashed during fraction conversion, preventing data collection. The error suggests a type mismatch in the Gaussian elimination implementation. | next: Debug the Fraction type conversion in gaussian_elimination() by ensuring all inputs are Rational instances before division

---

Invariant Degree Gap in Read-Twice Branching Programs for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: INVARIANT_THEORY × BRANCHING_PROGRAMS
- **Recorded**: 2026-05-08 15:06 UTC
- **Entry ID**: `5d978c228eaa`

Statement

For a read-twice branching program P computing a function f: {0,1}^n → {0,1}, let deg_inv(f) denote the minimal degree of a non-trivial polynomial invariant under the action of the general linear group GL(n,2) that vanishes on f. Then deg_inv(f) = O(log size(P)) for all f computable by read-twice BPs, but deg_inv(IP_2) = Ω(n) where IP_2 is the inner product mod 2 function.

Rationale

Invariant theory provides polynomial functions invariant under group actions, which can distinguish functions with different symmetry properties. Read-twice BPs for IP_2 must encode high-degree invariants due to the function's non-linearity, while general read-twice functions have low-degree invariants. This conjecture bridges algebraic geometry and circuit complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_573776bd.py", line 84, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_573776bd.py", line 61, in run_trial
    min_invariant_degree = find_min_invariant_degree(bp, ip2_value)
                           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_573776bd.py", line 50, in find_min_inva
    if is_invariant(poly, bp) and polynomial_degree(poly) > 0:
                                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: '>' not supported between instances of 'tuple' and 'int'
```

Judge reasoning

Test crashed due to type error in degree comparison; unable to evaluate conjecture's validity | next: Fix polynomial_degree return type to integer and revalidate invariant degree calculations

---

Bounded Arithmetic Complexity of Algebraic Closure in Finite Fields

- **Verdict**: `BARRIER_HIT`
- **Bridge**: BOUNDED_ARITHMETIC × ALGEBRAIC_GEOMETRY
- **Recorded**: 2026-05-08 15:19 UTC
- **Entry ID**: `9863b4502e8f`

Statement

For a system of polynomial equations over GF(p^n), the minimal proof size in bounded arithmetic (PSPACE) required to certify the existence of a solution is polynomially bounded by the number of variables and the degree of the polynomials. Conversely, if the system has no solutions, the proof size is exponential in the number of variables.

Rationale

Bounded arithmetic theories (e.g., S^1_2) are central to understanding proof complexity and separation of computational classes. Linking their proof size to algebraic geometry's solvability conditions (e.g., Hilbert's Nullstellensatz) provides a bridge between logical complexity and algebraic structure. This conjecture tests whether algebraic closure properties inherently constrain or expand proof complexity in bounded arithmetic.

Novelty

- Judge: " over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by ALGEBRIZATION barrier

Graph Energy Lower Bound on Tseitin Resolution Length

- **Verdict**: INCONCLUSIVE
- **Bridge**: SPECTRAL_GRAPH_THEORY × RESOLUTION_LENGTH
- **Recorded**: 2026-05-08 15:39 UTC
- **Entry ID**: e90418e8fa5a

Statement

For a graph G with adjacency matrix A, let Energy(G) = Σ_{i=1}^n |λ_i| where λ_i are the eigenvalues of A. The Tseitin formula on G requires Resolution length ≥ 2^{Ω(Energy(G))}.

Rationale

Graph energy, a spectral invariant, captures structural complexity. High energy implies non-trivial eigenvalue distribution, which may correlate with the exponential growth of resolution refutations needed to handle the formula's dependencies.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_00bb6b0c.py", line 112, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_00bb6b0c.py", line 93, in run_trial
    energy_value = energy(G)
                   ^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_00bb6b0c.py", line 75, in energy
    eigenvals = eigenvalues(A)
                ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_00bb6b0c.py", line 47, in eigenvalues
    Q, R = qr_decomposition(A)
           ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_00bb6b0c.py", line 58, in qr_decomposit
    v[k] += norm_v
    ~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError before producing results | next: Debug qr_decomposition implementation for out-of-range indexing

---

Multiplicity Gap in Symmetric Powers of Permanent vs Determinant Representations

- **Verdict**: INCONCLUSIVE
- **Bridge**: SCHUR_WYEL_DUALLY × PERMANENT_DETERMINANT_COMPLEXITY

- **Recorded**: 2026-05-08 15:45 UTC
- **Entry ID**: `2746a7cbee6d`

Statement

For all integers m < n^{1.5}, the multiplicity of the irreducible representation indexed by the partition (n - m, m) in the symmetric power S^m(perm_n) exceeds its multiplicity in S^m(det_n), where perm_n and det_n denote the permanent and determinant polynomials of size n.

Rationale

This conjecture leverages Schur-Weyl duality to identify a representation-theoretic invariant that distinguishes the symmetric structures of permanent and determinant polynomials. The multiplicity gap in symmetric powers could provide a geometric obstruction to embedding the permanent's orbit within the determinant's orbit closure, aligning with GCT's separation strategy.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fa1ea576.py", line 115, in <module>
    results = [run_trial(seed) for seed in seeds]
               ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fa1ea576.py", line 94, in run_trial
    perm_multiplicity = multiplicity((n - m, m), [[random.randint(-10, 10) for _ in range(n)] for _ in r
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fa1ea576.py", line 73, in multiplicity
    perm = young_tableau_to_permutation(yt)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fa1ea576.py", line 34, in young_tableau
    perm[yt[i][j]] = i * len(yt[i]) + j
    ~~~~^^^^^^^^^^
IndexError: list assignment index out of range
```

Judge reasoning

Test crashed before collecting data; cannot evaluate support fraction or counterexamples | next: Fix the IndexError in young_tableau_to_permutation by validating tableau dimensions

---

Fourier Coefficient Sum Separation for Read-Twice Branching Programs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: FOURIER_ANALYSIS × BRANCHING_PROGRAMS
- **Recorded**: 2026-05-08 15:54 UTC
- **Entry ID**: `e0efe9716b37`

Statement

For any read-twice branching program P computing the inner product mod 2 function IP_2 on n variables, the sum of the absolute values of the Fourier coefficients of the function computed by P is $\Omega(\sqrt{n})$, whereas for any function that can be computed by a read-once branching program, this sum is $O(\log n)$.

Rationale

The Fourier coefficients of IP_2 exhibit a structure that requires a larger read-twice branching program to capture, as reflected by the sum of their absolute values, which is significantly larger than for functions with simpler Fourier structures computable by read-once programs.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.48s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9cf3c361.py", line 76, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9cf3c361.py", line 50, in run_trial
    ip2_once_sum = fourier_coefficient_sum(ip2_once, n)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9cf3c361.py", line 38, in fourier_coeff
    for state, (a, b) in program:
               ^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

Judge reasoning

Test crashed due to unpacking error, preventing data collection. Support condition cannot be evaluated without successful trials. | next: Fix the unpacking error in the fourier_coefficient_sum function by verifying the program's state structure

---

Invariant Polynomial Generation Complexity in Determinant vs Permanent Orbits

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Algebraic Groups × Complexity of Polynomial Functions
- **Recorded**: 2026-05-08 16:16 UTC
- **Entry ID**: `934a387d12d9`

Statement

The minimal number of algebraically independent invariant polynomials required to generate the coordinate ring of the determinant's orbit under the action of $GL(n)\times GL(n)\times GL(n)$ is strictly less than that required for the permanent's orbit for all $n \geq 2$.

Rationale

The determinant's orbit is known to have a simpler structure in terms of invariant polynomials, which could reflect its lower computational complexity. The permanent's orbit may require more invariant polynomials due to its higher complexity, suggesting a deeper algebraic structure that is harder to compute.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
_name': 'Invariant Polynomial Count', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': F
TRIAL: {'metric_name': 'Invariant Polynomial Count', 'metric_value': 0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'Invariant Polynomial Count', 'metric_value': 1, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'Invariant Polynomial Count', 'metric_value': 1, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'Invariant Polynomial Count', 'metric_value': 1, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'Invariant Polynomial Count', 'metric_value': 0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'Invariant Polynomial Count', 'metric_value': 0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'Invariant Polynomial Count', 'metric_value': 1, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'Invariant Polynomial Count', 'metric_value': 1, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'Invariant Polynomial Count', 'metric_value': 1, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'Invariant Polynomial Count', 'metric_value': 0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'Invariant Polynomial Count', 'metric_value': 1, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'Invariant Polynomial Count', 'metric_value': 1, 'instances_tested': 1, 'conje
RESULT: FALSIFIED counterexample="" first_failing_seed=11
```

Judge reasoning

parse_fail | next: NONE

Covering Radius Bound on Set System Discrepancy

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometry of Numbers × Communication Complexity
- **Recorded**: 2026-05-08 17:02 UTC
- **Entry ID**: `4ee94497fe16`

Statement

For any set system S on n elements, the discrepancy disc(S) is bounded by the covering radius of the integer lattice Z^n scaled by the volume of the convex hull of the characteristic vectors of S.

Rationale

Geometry of numbers provides tools to analyze lattice coverings, which can quantify how well points in space are approximated by lattice points. This connects to communication complexity via discrepancy, where the covering radius reflects the 'spread' of set system representations, influencing lower bounds on communication protocols.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.17s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8c996cbf.py", line 124, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8c996cbf.py", line 108, in run_trial
    Z_n_covering_radius = covering_radius(n)
                          ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8c996cbf.py", line 88, in covering_radi
    distances = [distance(point, p) for p in points]
                 ^^^^^^^^
NameError: name 'distance' is not defined. Did you mean: 'distances'?
```

Judge reasoning

Test crashed due to undefined 'distance' function, preventing data collection. | next: Implement missing distance metric and revalidate conjecture with corrected code

---

Algebraic Geometry Solution Count Lower Bound on ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry over Finite Fields × ACC^0 Circuit Size
- **Recorded**: 2026-05-08 17:17 UTC
- **Entry ID**: `2419600249d3`

Statement

For any system of linear equations over a finite field GF(q), the number of solutions S is exponentially related to the size of the minimal ACC^0 circuit required to compute it. Specifically, if a system has S solutions, then the minimal ACC^0 circuit size is Ω(log S).

Rationale

The structure of the solution space in algebraic geometry imposes constraints on the computational resources needed to compute it, leading to lower bounds on circuit complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `17.74s`

```
tested': 30, 'conjecture_holds': False, 'counterexample': 'mapping_undefined'}
TRIAL: {'metric_name': 'Number of Solutions', 'metric_value': 0.8666666666666667, 'instances_tested'
TRIAL: {'metric_name': 'Number of Solutions', 'metric_value': 0.8333333333333334, 'instances_tested'
TRIAL: {'metric_name': 'Number of Solutions', 'metric_value': 0.8666666666666667, 'instances_tested'
TRIAL: {'metric_name': 'Number of Solutions', 'metric_value': 0.6666666666666666, 'instances_tested'
TRIAL: {'metric_name': 'Number of Solutions', 'metric_value': 0.8333333333333334, 'instances_tested'
TRIAL: {'metric_name': 'Number of Solutions', 'metric_value': 0.9, 'instances_tested': 30, 'conjectur
TRIAL: {'metric_name': 'Number of Solutions', 'metric_value': 0.8333333333333334, 'instances_tested'
```

```
TRIAL: {'metric_name': 'Number of Solutions', 'metric_value': 0.8333333333333334, 'instances_tested'

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a21dcae7.py", line 100, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a21dcae7.py", line 100, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                              ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed before completing evaluation; support fraction cannot be validated. | next: Re-run test with seed tracking enabled and error handling for undefined mappings

---

Matroid Rank Deficit in Monotone DNF Depth for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Monotone Circuit Depth
- **Recorded**: 2026-05-08 17:23 UTC
- **Entry ID**: `9d5d3a5ea1bf`

Statement

For any monotone DNF formula F with n variables, let M_F be the matroid whose circuits are the minimal non-empty terms of F. Define rank_deficit(F) = rank(M_F) - log_2(size(F)). Then rank_deficit(F) ≤ 2 log n for all F with size(F) ≤ n^3, and rank_deficit(k-CLIQUE) ≥ Ω(n^{1/2}) for k ≥ 3.

Rationale

Matroid rank captures structural dependencies in DNF terms, which may expose limitations in representing k-CLIQUE. The deficit metric isolates the 'wastefulness' of DNF representations, linking matroid invariants to circuit complexity through combinatorial optimization.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `2.78s`

```
: ''}
TRIAL: {'metric_name': 'rank_deficit', 'metric_value': -13.200208961930477, 'instances_tested': 1, '
TRIAL: {'metric_name': 'rank_deficit', 'metric_value': -11.982637133669424, 'instances_tested': 1, '
TRIAL: {'metric_name': 'rank_deficit', 'metric_value': -10.915505962231931, 'instances_tested': 1, '
TRIAL: {'metric_name': 'rank_deficit', 'metric_value': -8.280770770130603, 'instances_tested': 1, 'c
TRIAL: {'metric_name': 'rank_deficit', 'metric_value': -4.285402218862249, 'instances_tested': 1, 'c
TRIAL: {'metric_name': 'rank_deficit', 'metric_value': -11.553869243196068, 'instances_tested': 1, '
TRIAL: {'metric_name': 'rank_deficit', 'metric_value': -11.440350119200563, 'instances_tested': 1, '
TRIAL: {'metric_name': 'rank_deficit', 'metric_value': -8.361943773735241, 'instances_tested': 1, 'c
TRIAL: {'metric_name': 'rank_deficit', 'metric_value': -12.444108996147115, 'instances_tested': 1, '
TRIAL: {'metric_name': 'rank_deficit', 'metric_value': -10.98903963696239, 'instances_tested': 1, 'c
```

```
TRIAL: {'metric_name': 'rank_deficit', 'metric_value': -13.283667167491501, 'instances_tested': 1, '
TRIAL: {'metric_name': 'rank_deficit', 'metric_value': -11.659995892429977, 'instances_tested': 1, '
TRIAL: {'metric_name': 'rank_deficit', 'metric_value': -11.086136225027309, 'instances_tested': 1, '
RESULT: SUPPORTED mean=-10.735960113292682 std=2.3443356317573154 support_fraction=1.0
```

Judge reasoning

parse_fail | next: NONE

Betti Number Exponential Lower Bound on Tseitin Resolution Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic topology (Betti numbers of clique complexes) × Resolution length of Tseitin formulas
- **Recorded**: 2026-05-08 17:36 UTC
- **Entry ID**: `9c6ddd093b59`

Statement

For a graph G, the resolution length of its Tseitin formula is at least 2^{β_1(G)}, where β_1(G) is the first Betti number of the clique complex of G. This holds for all graphs with $\beta_1(G) \geq 1$ and $n \leq 40$.

Rationale

Higher Betti numbers indicate nontrivial topological structure in the clique complex, which may necessitate exponentially longer resolution proofs due to the increased combinatorial complexity of the Tseitin formula's dependencies.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
ecture_holds': False, 'counterexample': 'beta_1(G) < 1'}
TRIAL: {'metric_name': 'resolution_length', 'metric_value': None, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'resolution_length', 'metric_value': None, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'resolution_length', 'metric_value': None, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'resolution_length', 'metric_value': 0, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'resolution_length', 'metric_value': None, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'resolution_length', 'metric_value': None, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'resolution_length', 'metric_value': 0, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'resolution_length', 'metric_value': None, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'resolution_length', 'metric_value': 0, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'resolution_length', 'metric_value': None, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'resolution_length', 'metric_value': None, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'resolution_length', 'metric_value': None, 'instances_tested': 1, 'conjecture
RESULT: FALSIFIED counterexample="beta_1(G) < 1" first_failing_seed=0
```

Judge reasoning

parse_fail | next: NONE

Tensor Rank Lower Bound on SOS Refutation Size for Symmetric CSPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry (Tensor Rank of Symmetric Tensors) × Sum-of-Squares Hierarchy (CSP Refutation Size)
- **Recorded**: 2026-05-08 17:49 UTC
- **Entry ID**: `fcc13ae0d3f9`

Statement

For a symmetric polynomial f(x) = det(A) where A is an n×n matrix of variables, the SOS refutation size required to prove f ≡ 0 is at least Ω(n^{1.5} / log n), while the tensor rank of f is Θ(n^{1.5}). This implies that the SOS rank of f is asymptotically tied to its tensor rank, and the refutation size is governed by the same growth rate.

Rationale

The tensor rank of symmetric polynomials (e.g., determinants) captures their intrinsic complexity, while SOS refutation size measures the computational effort to prove unsatisfiability. Linking these two invariants bridges algebraic geometry and CSP complexity, leveraging the symmetry of the problem to ensure the conjecture is non-trivial and testable.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.07s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing conclusive data; pre-registered support condition cannot be evaluated. | next: Increase timeout duration and re-run test with stricter resource limits

---

Newton Polytope Vertex Count Bounds SOS Degree for Max-CUT Approximation

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry (Newton Polytopes) × SOS Degree for Max-CUT Approximation
- **Recorded**: 2026-05-08 18:02 UTC
- **Entry ID**: `03c7a9ecad95`

Statement

For any graph G with m edges, the SOS degree d required to approximate Max-CUT with a ratio better than 0.878 must satisfy d ≥ log(m), where the SOS degree is the minimal degree of a sum-of-squares relaxation achieving the approximation ratio.

Rationale

The Newton polytope's vertex count reflects the polynomial's structural complexity. A higher vertex count implies a more intricate polynomial, necessitating a higher SOS degree to capture its properties, thereby linking combinatorial graph structure to algebraic complexity.

Novelty

- Judge: `INCONCLUSIVE` over 10 arXiv hits

Top hits consulted: - [1512.01594v2] Pruning Algorithms for Pretropisms of Newton Polytopes - [1604.01183v2] Approximate Polytope Membership Queries - [2107.00574v1] Trigonometric approximation of the Max-Cut Polytope is star-like - [2409.14294v2] A lower bound theorem for d-polytopes with $2d + 2$ vertices - [0305179v3] Polynomial Degree and Lower Bounds in Quantum Complexity: Collision and Element Distinctness with Small Range

Empirical Test

- exit code: `124`, elapsed: `240.08s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results; insufficient data to evaluate support fraction or counterexamples | next: Increase timeout duration and re-run test with extended computation resources

---

Noncommutative Quantum Dimension Lower Bound on Disjointness

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative Geometry × Communication Complexity (Disjointness)
- **Recorded**: 2026-05-08 18:22 UTC
- **Entry ID**: `82033f938942`

Statement

For the communication matrix M of the Disjointness problem on n bits, the minimal quantum dimension of the noncommutative algebra generated by M's entries is Ω(n), implying randomized communication complexity Ω(n).

Rationale

Noncommutative geometry provides tools to analyze the structure of algebras generated by matrices, which can capture the complexity of communication protocols. The quantum dimension, a measure of the algebra's structure, relates to the minimal resources needed for communication, offering a lower bound.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.13s`

```
ion', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'noncommutative_dimension', 'metric_value': 40, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'noncommutative_dimension', 'metric_value': 40, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'noncommutative_dimension', 'metric_value': 40, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'noncommutative_dimension', 'metric_value': 40, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'noncommutative_dimension', 'metric_value': 40, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'noncommutative_dimension', 'metric_value': 40, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'noncommutative_dimension', 'metric_value': 40, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'noncommutative_dimension', 'metric_value': 40, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'noncommutative_dimension', 'metric_value': 40, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'noncommutative_dimension', 'metric_value': 40, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'noncommutative_dimension', 'metric_value': 40, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'noncommutative_dimension', 'metric_value': 40, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'noncommutative_dimension', 'metric_value': 40, 'instances_tested': 1, 'conje
RESULT: SUPPORTED mean=40.0 std=0.0 support_fraction=1.0
```

Judge reasoning

The test used n=1 instances, which cannot validate Ω(n) scaling. Metric saturation suggests the bound may be vacuous for Disjointness. | next: Test with n ≥ 100 to verify scaling behavior and metric variability

---

Noncommutative Norm Gap in Read-Twice Branching Programs for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative Geometry × Read-twice Branching Programs (BP) for IP_2
- **Recorded**: 2026-05-08 18:36 UTC
- **Entry ID**: `aed3c06488a0`

Statement

For a read-twice BP P computing IP_2 on n variables, the noncommutative operator norm ||T_P||∞ *of its transition matrix T_P satisfies* ||*T_P*||∞ = O(log n), whereas for a general BP computing IP_2, ||T_P||_∞ = Ω(n).

Rationale

Noncommutative operator norms capture the structure of the transition matrix in a way that reflects the read-twice constraint, potentially exposing a gap between structured and unrestricted BPs. This connects algebraic topology (via operator algebras) to branching program complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_53b95e16.py", line 65, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_53b95e16.py", line 45, in run_trial
```

```
    read_twice_norm = spectral_norm(read_twice_matrix)
                      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_53b95e16.py", line 34, in spectral_norm
    v = matrix_multiply(matrix, v)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_53b95e16.py", line 27, in matrix_multip
    C[i][j] += A[i][k] * B[k][j]
                         ~~~~^^^
TypeError: 'float' object is not subscriptable
```

Judge reasoning

Test crashed with TypeError, preventing data collection. The error suggests a code issue rather than a mathematical failure. | next: Fix the TypeError in matrix_multiply by ensuring proper handling of matrix and vector types

---

Laplacian Eigenvalue Deficit Bounds SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded**: 2026-05-08 18:47 UTC
- **Entry ID**: `1947d5fb4445`

Statement

For a graph G with Laplacian eigenvalues $\lambda_1 \leq \ldots \leq \lambda\_n$, the SOS degree d required to approximate Max-CUT(G) with ratio 0.878 - ε satisfies $d \geq \Omega(\sqrt{(n / (\lambda\_n - \lambda_1))})$ for $\varepsilon \leq 1/n^2$.

Rationale

The SOS hierarchy's ability to capture spectral gaps via polynomial identities ties to real algebraic geometry's Positivstellensatz. Eigenvalue deficits ($\lambda\_n - \lambda_1$) encode structural constraints on polynomial certificates, creating a bridge between algebraic geometry and SOS complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results; insufficient data to evaluate support fraction or counterexamples. | next: Increase timeout duration and re-run test with deterministic seed configuration

---

Diophantine Approximation Bound on Resolution Proof Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Diophantine Approximation × Resolution Proof Complexity

- **Recorded**: 2026-05-08 18:56 UTC
- **Entry ID**: `729b19bbcdf7`

Statement

For a CNF formula Φ with n variables, the size of the shortest resolution proof is at least Ω((log n)^{1/2}) if the discrepancy of Φ's clauses is bounded by a function related to the approximation of irrational numbers by rationals.

Rationale

Diophantine approximation provides tools to analyze how well real numbers can be approximated by rationals, which may relate to the structure of clauses in a CNF formula and the complexity of resolving them. This conjecture links the discrepancy of the formula's clauses to the required proof size, leveraging the interplay between number theory and proof complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`


```
ion_proof_size', 'metric_value': 46, 'instances_tested': 1, 'conjecture_holds': True, 'counterexampl
TRIAL: {'metric_name': 'resolution_proof_size', 'metric_value': 45, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'resolution_proof_size', 'metric_value': 36, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'resolution_proof_size', 'metric_value': 33, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'resolution_proof_size', 'metric_value': 19, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'resolution_proof_size', 'metric_value': 8, 'instances_tested': 1, 'conjectur
TRIAL: {'metric_name': 'resolution_proof_size', 'metric_value': 36, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'resolution_proof_size', 'metric_value': 33, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'resolution_proof_size', 'metric_value': 10, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'resolution_proof_size', 'metric_value': 48, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'resolution_proof_size', 'metric_value': 31, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'resolution_proof_size', 'metric_value': 55, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'resolution_proof_size', 'metric_value': 32, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'resolution_proof_size', 'metric_value': 33, 'instances_tested': 1, 'conjectu
RESULT: SUPPORTED mean=32.766666666666666 std=15.938806591321558 support_fraction=1.0
```


Judge reasoning

The test used insufficiently large n values and lacked adversarial cases with required Diophantine properties, violating the conjecture's conditions despite 100% support fraction. | next: Test with larger n and explicitly constructed Diophantine discrepancy instances

---

Submodular Coverage Deficit in Monotone DNF for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorics (Submodular Functions) × Monotone Circuit Complexity
- **Recorded**: 2026-05-08 19:10 UTC
- **Entry ID**: `d4c1f94494d9`

Statement

Let $\mu(D) = \Sigma|T_i| - \Sigma|T_i \cap T_j|$ for a DNF D with terms T_i. For any DNF of size poly(n), $\mu(D) \leq O(\log n)$. For the k-CLIQUE indicator function, $\mu(D) \geq \Omega(n^{1/2})$ when $k \geq 3$.

Rationale

Submodular coverage deficit captures term overlaps in DNFs. For k-CLIQUE, the exponential number of terms creates large overlaps, making $\mu(D)$ scale with $n^{1/2}$. This links combinatorial submodularity to monotone circuit lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.41s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out without producing results; insufficient data to evaluate conjecture. | next: Run test with extended timeout and debug crash logs

---

Quandle Isomorphism Complexity and ACC^0 Circuit Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Quandle Theory × ACC^0 Circuit Complexity
- **Recorded**: 2026-05-08 19:22 UTC
- **Entry ID**: `c39cd0b7c42e`

Statement

For any finite quandle Q of size n, the minimum depth of an ACC^0 circuit computing the quandle isomorphism problem is $\Omega(\log n)$. Equivalently, the quandle isomorphism problem cannot be solved by constant-depth ACC^0 circuits for any $n \geq 2$.

Rationale

Quandles, algebraic structures encoding knot-theoretic symmetries, exhibit complex combinatorial properties. Their isomorphism problem, though seemingly abstract, may mirror the inherent difficulty of circuit lower bounds. By analyzing the algebraic constraints of quandles, we hypothesize that their structural richness necessitates super-constant depth for efficient computation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d648f52d.py", line 76, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d648f52d.py", line 48, in run_trial
    while are_isomorphic(q1, q2):
          ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d648f52d.py", line 34, in are_isomorphi
    if q1[(i, j)] != q2[perm[i] - 1][perm[j] - 1]:
                     ~~^^^^^^^^^^^^^
KeyError: 0
```

Judge reasoning

Test crashed with KeyError before producing results, preventing evaluation of conjecture | next: Debug test code's permutation handling and re-run with validated inputs

---

Schur-Weyl Decomposition Rank Lower Bounds Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl duality × Disjointness communication complexity
- **Recorded**: 2026-05-08 19:28 UTC
- **Entry ID**: `c2c1aaf5edd5`

Statement

For any bipartite graph G = (A ∪ B, E) with |A| = |B| = n, let λ(G) denote the minimal number of complete bipartite subgraphs needed to cover E. The Schur-Weyl decomposition rank R(G) of the adjacency matrix equals λ(G). For all G with λ(G) ≥ 2, R(G) ≥ Ω(n^{1/2})

Rationale

Schur-Weyl duality provides a decomposition of tensor products into irreducible representations, which can quantify symmetry in bipartite graphs. This decomposition rank captures structural properties of edge covers, directly linking to disjointness protocols' inherent communication requirements via the covering number's combinatorial lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_646169d4.py", line 104, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_646169d4.py", line 81, in run_trial
    lambda_G = max_matching(G)
               ^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_646169d4.py", line 44, in max_matching
    dfs(u)
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_646169d4.py", line 35, in dfs
    if not visited[v]:
           ~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError, preventing validation of conjecture. The error suggests a bug in graph processing code that needs fixing before results can be assessed. | next: Fix the index out of range error in max_matching function by adding bounds checking for node indices

---

Persistent Homology Barcode Length Inversely Proportional to DPLL Tree Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Persistent Homology × DPLL Search Tree
- **Recorded**: 2026-05-08 19:41 UTC
- **Entry ID**: `42bdd22c09b4`

Statement

For 3-SAT instances with m clauses and n variables, the total barcode length L of the persistent homology of the clique complex built from clause-sharing edges satisfies L = Θ(1 / T_DPLL), where T_DPLL is the DPLL tree size. For all instances with n ≤ 40, |L * T_DPLL| ≤ 1 + ε for ε < 0.1.

Rationale

Persistent homology captures topological structure of clause interactions, which may correlate with search space complexity. Higher connectivity (persistent features) could imply more complex landscapes, inversely affecting DPLL efficiency via branching factors.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_61711205.py", line 79, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_61711205.py", line 57, in run_trial
    adjacency_matrix = build_clique_complex(clauses)
                       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_61711205.py", line 31, in build_clique_
    n = max(abs(l) for l in set.union(*clauses))
                            ^^^^^^^^^^^^^^^^^^^
TypeError: descriptor 'union' for 'set' objects doesn't apply to a 'list' object
```

Judge reasoning

Test crashed due to TypeError in set.union operation on list objects, preventing data collection | next: Fix the build_clique_complex function to properly handle list inputs (e.g., convert to sets first)

Additive Energy Threshold Exceeds ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive Combinatorics × ACC^0 Circuit Size
- **Recorded**: 2026-05-08 20:09 UTC
- **Entry ID**: `916f234e492e`

Statement

For a Boolean function f: {0,1}^n → {0,1}, let E(f) be its additive energy: E(f) = |{(a,b,c,d) ∈ {0,1}^{4n} : f(a) + f(b) = f(c) + f(d)}|. Then, for all f with E(f) ≥ n^{2.5}, the minimal ACC^0 circuit size computing f is ≥ n^{1.5} log n.

Rationale

Additive energy captures structural regularity in Boolean functions, while ACC^0 circuits lack depth to exploit such regularity. High additive energy implies complex interaction patterns that require more gates to simulate, exposing a separation between pseudorandom-like functions and shallow circuit classes.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.27s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing validation of support fraction or counterexample detection. | next: Increase timeout duration and re-run test with extended computation limits

---

Zarankiewicz-Free Communication Matrices Imply Monotone Circuit Lower Bounds

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Extremal Graph Theory × Monotone Circuit Complexity
- **Recorded**: 2026-05-08 20:19 UTC
- **Entry ID**: `8086a3a03a65`

Statement

Let f: {0,1}^n × {0,1}^n → {0,1} be a Boolean function with communication matrix M_f. If M_f is K_{s,t}-free (no complete bipartite subgraph K_{s,t}), then the monotone circuit size of f is at least $z(n,n,s,t)^2$, where z(n,n,s,t) is the Zarankiewicz number for K_{s,t}-free bipartite graphs.

Rationale

Extremal graph theory provides sharp bounds on dense K_{s,t}-free graphs, which may constrain the structure of communication matrices. Monotone circuit complexity is tightly linked to communication complexity via the Karchmer-Wigderson game, suggesting that extremal graph-theoretic obstructions could translate into circuit size lower bounds.

Novelty

- Judge: `INCONCLUSIVE` over 11 arXiv hits

Top hits consulted: - [1102.2932v2] Applications of Monotone Rank to Complexity Theory - [1709.09876v2] Communication Complexity of Cake Cutting - [2210.06739v1] Computing the Best Case Energy Complexity of Satisfying Assignments in Monotone Circuits - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [1604.03423v5] Graph Matrices: Norm Bounds and Applications

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6d543051.py", line 31
    def z(n, n, s, t):
             ^
SyntaxError: duplicate argument 'n' in function definition
```

Judge reasoning

Test crashed due to syntax error in function definition | next: Fix the duplicate parameter error in the z(n,n,s,t) function definition and rerun experiments

---

Kronecker Coefficient Growth Separates Permanent vs Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Permanent vs Determinant Complexity
- **Recorded**: 2026-05-08 20:40 UTC
- **Entry ID**: `488f1af8a053`

Statement

For all n ≥ 2, the maximum Kronecker coefficient in the decomposition of $\mathrm{Sym}^{n(\mathrm{Sym}}k(V))$ grows polynomially in n for determinants but exponentially in n for permanents, where V is the standard n-dimensional representation of GL_n(C).

Rationale

Schur-Weyl duality decomposes tensor products into irreducible representations, and Kronecker coefficients count multiplicities. The permanent's asymmetric decomposition (via symmetric powers) suggests exponential complexity, while determinants' symmetric structure implies polynomial bounds. This bridges representation theory to algebraic complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
TRIAL: {"seed": 11, "metric_name": "Max Kronecker Coefficient", "metric_value": 0.000000, "instances_
TRIAL: {"seed": 23, "metric_name": "Max Kronecker Coefficient", "metric_value": 0.000000, "instances_

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6f418874.py", line 74, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6f418874.py", line 50, in run_trial
   kronecker_coeffs = [kronecker_coefficient(shape[:i+1], shape[i+1:]) for i in range(len(shape))]
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6f418874.py", line 34, in kronecker_coe
    numerator *= hook_length_formula(lam[:i+1]) * hook_length_formula(mu[:j+1])
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6f418874.py", line 22, in hook_length_f
    for j in range(len(shape[i])):
                   ^^^^^^^^^^^^^
TypeError: object of type 'int' has no len()
```

Judge reasoning

Test crashed due to a TypeError in hook_length_formula, preventing reliable evaluation of the conjecture | next: Fix the shape handling in hook_length_formula to properly process integer inputs as list-like objects

---

Real Radical Dimension and SOS Degree for Max-CUT

- **Verdict**: INCONCLUSIVE
- **Bridge**: Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded**: 2026-05-08 20:58 UTC
- **Entry ID**: 5a78c0e11f98

Statement

For a Max-CUT instance on n variables, the SOS degree required to approximate the Max-CUT value is at least Ω(log d), where d is the dimension of the real radical of the system of polynomials derived from the instance.

Rationale

The real radical captures the real solutions of the polynomial system, and higher-dimensional radicals may require higher SOS degrees to approximate the Max-CUT value, as they encode more complex real constraints.

Novelty

- Judge: INCONCLUSIVE over 9 arXiv hits

Top hits consulted: - [2512.09960v1] Numerical Behavior of the Riemann Zeta Function Using Real-to-Complex Conversion Without Gram Points or Bracketing - [0401058v1] Multimer Radical Ions and Electron/Hole Localization in Polyatomic Molecular Liquids: A critical review - [1806.09998v1] Real time state monitoring and fault diagnosis system for motor based on LabVIEW - [1507.06191v1]

Convexifying positive polynomials and sums of squares approximation - [2203.06291v2] Moments, Sums of Squares, and Tropicalization

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e0272f63.py", line 111, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e0272f63.py", line 101, in run_trial
    "conjecture_holds": sos_degree >= math.log(d),
                                      ^^^^^^^^^^^
ValueError: math domain error
```

Judge reasoning

Test crashed with math domain error, preventing evaluation of conjecture validity | next: Fix logarithm argument validation to handle d=0/negative values

---

Permutation Polynomial Width Conjecture

- **Verdict**: INCONCLUSIVE
- **Bridge**: Permutation polynomials over finite fields × Branching program width
- **Recorded**: 2026-05-08 21:07 UTC
- **Entry ID**: ce6a6c40a567

Statement

The minimal width of a branching program computing a permutation polynomial of degree d over GF(q) is Θ(log d).

Rationale

Permutation polynomials encode combinatorial symmetries that influence branching program width via their algebraic structure, linking finite field arithmetic to NC^1 complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s


```
olds': False, 'counterexample': 'not_a_permutation'}
TRIAL: {'metric_name': 'branching_program_width', 'metric_value': None, 'instances_tested': 1, 'conj
TRIAL: {'metric_name': 'branching_program_width', 'metric_value': None, 'instances_tested': 1, 'conj
TRIAL: {'metric_name': 'branching_program_width', 'metric_value': None, 'instances_tested': 1, 'conj
TRIAL: {'metric_name': 'branching_program_width', 'metric_value': 1, 'instances_tested': 1, 'conject
TRIAL: {'metric_name': 'branching_program_width', 'metric_value': None, 'instances_tested': 1, 'conj
TRIAL: {'metric_name': 'branching_program_width', 'metric_value': None, 'instances_tested': 1, 'conj
TRIAL: {'metric_name': 'branching_program_width', 'metric_value': None, 'instances_tested': 1, 'conj
TRIAL: {'metric_name': 'branching_program_width', 'metric_value': None, 'instances_tested': 1, 'conj
```

```
TRIAL: {'metric_name': 'branching_program_width', 'metric_value': 1, 'instances_tested': 1, 'conject
TRIAL: {'metric_name': 'branching_program_width', 'metric_value': 1, 'instances_tested': 1, 'conject
TRIAL: {'metric_name': 'branching_program_width', 'metric_value': None, 'instances_tested': 1, 'conj
TRIAL: {'metric_name': 'branching_program_width', 'metric_value': None, 'instances_tested': 1, 'conj
RESULT: FALSIFIED counterexample="not_a_permutation" first_failing_seed=11
```

Judge reasoning

The test failed to meet the pre-registered support fraction threshold (80%) and the critic challenges the validity of the counterexample due to insufficient instance testing (all trials tested 1 instance). The 'not_a_permutation' counterexample may indicate invalid test cases rather than a true falsification. | next: Run tests with ≥100 instances across varying d and q to validate scaling behavior and confirm permutation polynomial validity

---

Plethysm Coefficient Gap in Symmetric Powers of Permanent vs Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Monotone Circuit Complexity
- **Recorded**: 2026-05-08 21:14 UTC
- **Entry ID**: `3424afae2a44`

Statement

For all n ≥ 2, the plethysm coefficient of Sym^k(perm_n) exceeds that of Sym^k(det_m) by a factor of Ω(2^{n/2}) for m < n^{1.5} and k = ⌊n/2⌋.

Rationale

Schur-Weyl duality decomposes tensor powers into irreducible representations; plethysm coefficients measure how these decompose under symmetric power operations. Permanent and determinant have distinct orbit closures under GL(n), suggesting their symmetric power decompositions should exhibit exponential separation in specific coefficients.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7f38910c.py", line 76, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7f38910c.py", line 54, in run_trial
    perm_coeff = plethysm_coefficient([n - k + 1] + [1] * (n - k), k)
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7f38910c.py", line 47, in plethysm_coef
    coeff += hook_length_formula([t[i] + 1 for i in range(k)])
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7f38910c.py", line 36, in hook_length_f
    for i, cell in enumerate(row):
```

```
                    ^^^^^^^^^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

Test crashed with TypeError before producing results; cannot evaluate support fraction or counterexamples. | next: Fix the hook_length_formula parameter type error in the test code to enable proper execution

---

Cluster Algebra Mutation Distance Bounds ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Cluster Algebras × ACC^0 Circuit Size
- **Recorded**: 2026-05-08 21:20 UTC
- **Entry ID**: `56bd1aa631e5`

Statement

For any CNF formula Φ with n variables, the mutation distance between its initial and final cluster variables in the cluster algebra generated by Φ's clauses is Θ(log n) if and only if the minimal ACC^0 circuit size for Φ is Θ(log n).

Rationale

Cluster algebras encode combinatorial mutations that mirror circuit transformations. By mapping clauses to cluster variables and interactions to exchange matrices, mutation distance could quantify structural complexity, offering a novel algebraic lens on circuit size lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ab58fe47.py", line 85, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ab58fe47.py", line 55, in run_trial
    clauses = generate_cnf(n, m)
              ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ab58fe47.py", line 25, in generate_cnf
    clause = [random.choice(variables), -random.choice(variables)]
              ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 348, in choice
    return seq[self._randbelow(len(seq))]
           ~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: 'set' object is not subscriptable
```

Judge reasoning

Test crashed due to TypeError in generate_cnf function. No data collected to evaluate support fraction or counterexamples. | next: Fix the generate_cnf function to use a subscriptable data structure (e.g., list) instead of a set for variable selection.

Moment Matrix Eigenvalue Decay and SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded**: 2026-05-08 21:37 UTC
- **Entry ID**: `1ca152a211e8`

Statement

For a random Max-CUT instance on n variables, the minimal eigenvalue of the degree-d moment matrix of its SOS relaxation decays as Θ(1/d) with constant depending on n. For all n ≤ 40 and d ≥ 2, the absolute value of the minimal eigenvalue satisfies |λ_min| ≥ c(n)/d where c(n) = Θ(n^α) for α ∈ (0, 1).

Rationale

The eigenvalues of the moment matrix encode the geometry of the feasible region for the SOS relaxation. Their decay rate with degree d reveals how the relaxation's tightness scales, linking algebraic structure to optimization complexity. This connects real algebraic geometry's spectral properties to SOS hierarchy behavior.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c2f9359.py", line 130, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c2f9359.py", line 69, in run_trial
    eigenvalues = solve_linear_system(M, [0] * (n + 1))
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c2f9359.py", line 55, in solve_linear_
    x[i] = A_b[i][-1]
           ~~~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError before producing results, preventing validation of the conjecture. | next: Debug the IndexError in solve_linear_system to ensure proper matrix dimensions for all n ≤ 40 and d ≥ 2

Matroid Representation Complexity of Monotone DNF for k-CLIQUE

- **Verdict**: INCONCLUSIVE
- **Bridge**: Matroid Representability over Finite Fields × Monotone DNF Formulas
- **Recorded**: 2026-05-08 21:44 UTC
- **Entry ID**: 499c5d46d38e

Statement

For a monotone DNF formula Φ over n variables, define μ(Φ) as the minimal size of a finite field representation of its term lattice. Then, μ(Φ) ≤ O(log n) for any Φ with O(poly(n)) terms, and μ(k-CLIQUE) ≥ Ω(n^{1-1/k}) for k ≥ 3.

Rationale

Matroid representability over finite fields captures combinatorial structure of DNF terms. The conjecture links this to k-CLIQUE's inherent complexity, suggesting that k-CLIQUE's term lattice requires exponentially larger representations than general DNFs.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

```
ize', 'metric_value': 58, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': ''}
TRIAL: {'metric_name': 'matroid_representation_size', 'metric_value': 28, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'matroid_representation_size', 'metric_value': 112, 'instances_tested': 1, 'c
TRIAL: {'metric_name': 'matroid_representation_size', 'metric_value': 106, 'instances_tested': 1, 'c
TRIAL: {'metric_name': 'matroid_representation_size', 'metric_value': 36, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'matroid_representation_size', 'metric_value': 62, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'matroid_representation_size', 'metric_value': 146, 'instances_tested': 1, 'c
TRIAL: {'metric_name': 'matroid_representation_size', 'metric_value': 96, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'matroid_representation_size', 'metric_value': 130, 'instances_tested': 1, 'c
TRIAL: {'metric_name': 'matroid_representation_size', 'metric_value': 154, 'instances_tested': 1, 'c

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_62da5526.py", line 77, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_62da5526.py", line 77, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                               ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed before producing valid results; no counterexample found but conjecture_holds is False for multiple trials | next: Implement error handling for 'seed' field and re-run tests with sufficient sample size

Plethysm Coefficient Exponential Gap in Symmetric Squares of Permutation Representations for Random CNFs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality, Plethysm, Algebraic Combinatorics × ACC^0 Circuit Size
- **Recorded**: 2026-05-08 22:01 UTC
- **Entry ID**: `2515221a101f`

Statement

For a random CNF formula with n variables, the plethysm coefficient of the symmetric square of the permutation representation decomposed into irreducibles is Ω(2^n), which implies that ACC^0 circuits cannot compute the permanent of such formulas.

Rationale

The exponential growth of plethysm coefficients for symmetric squares of permutation representations suggests deep structural differences between permanent and determinant, potentially exposing lower bounds on ACC^0 circuit size.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `1.80s`

```
tested": 30, "conjecture_holds": False, "counterexample": ""}
TRIAL: {"seed": 593, "metric_name": "plethysm_coefficient", "metric_value": 5.052894465957788e-12, "instances_tested": 30, "conjecture_holds": False, "counterexample": ""}
TRIAL: {"seed": 631, "metric_name": "plethysm_coefficient", "metric_value": 8.018485485069505e-12, "instances_tested": 30, "conjecture_holds": False, "counterexample": ""}
TRIAL: {"seed": 677, "metric_name": "plethysm_coefficient", "metric_value": 7.608903062601938e-12, "instances_tested": 30, "conjecture_holds": False, "counterexample": ""}
TRIAL: {"seed": 727, "metric_name": "plethysm_coefficient", "metric_value": 3.770662207648113e-12, "instances_tested": 30, "conjecture_holds": False, "counterexample": ""}
TRIAL: {"seed": 773, "metric_name": "plethysm_coefficient", "metric_value": 7.053303052209947e-12, "instances_tested": 30, "conjecture_holds": False, "counterexample": ""}
TRIAL: {"seed": 821, "metric_name": "plethysm_coefficient", "metric_value": 5.589756678349182e-12, "instances_tested": 30, "conjecture_holds": False, "counterexample": ""}
TRIAL: {"seed": 877, "metric_name": "plethysm_coefficient", "metric_value": 6.7381142507277544e-12, "instances_tested": 30, "conjecture_holds": False, "counterexample": ""}
TRIAL: {"seed": 929, "metric_name": "plethysm_coefficient", "metric_value": 2.9968433257691583e-12, "instances_tested": 30, "conjecture_holds": False, "counterexample": ""}

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92e1a0ef.py", line 111, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92e1a0ef.py", line 111, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                              ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

parse_fail | next: NONE

Symmetric Group Multiplicity Sum Lower Bound for Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups × Communication Complexity of Disjointness
- **Recorded**: 2026-05-08 22:16 UTC
- **Entry ID**: `0062aadf70ee`

Statement

For every n ≥ 1, the sum of the squares of the multiplicities in the decomposition of the communication matrix M of the DISJOINTNESS problem into irreducible representations of the symmetric group S_n is Ω(n).

Rationale

The decomposition into symmetric group representations captures the combinatorial structure of the communication matrix, and the sum of squares of multiplicities reflects the complexity of the interaction between the two parties' inputs. This invariant is expected to scale with n, providing a lower bound on communication complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.10s`


```
iplicities', 'metric_value': 40.0, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample
TRIAL: {'metric_name': 'Sum of Squares of Multiplicities', 'metric_value': 40.0, 'instances_tested':
TRIAL: {'metric_name': 'Sum of Squares of Multiplicities', 'metric_value': 40.0, 'instances_tested':
TRIAL: {'metric_name': 'Sum of Squares of Multiplicities', 'metric_value': 40.0, 'instances_tested':
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TRIAL: {'metric_name': 'Sum of Squares of Multiplicities', 'metric_value': 40.0, 'instances_tested':
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TRIAL: {'metric_name': 'Sum of Squares of Multiplicities', 'metric_value': 40.0, 'instances_tested':
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TRIAL: {'metric_name': 'Sum of Squares of Multiplicities', 'metric_value': 40.0, 'instances_tested':
TRIAL: {'metric_name': 'Sum of Squares of Multiplicities', 'metric_value': 40.0, 'instances_tested':
TRIAL: {'metric_name': 'Sum of Squares of Multiplicities', 'metric_value': 40.0, 'instances_tested':
RESULT: SUPPORTED mean=40.0 std=0.0 support_fraction=1.0
```


Judge reasoning

The test only verified n=30 with perfect support, but the conjecture requires validation for all n≥1. The pre-registered criterion was met for this single case, but the conjecture's universal quantifier remains unproven. | next: Test the conjecture for multiple values of n (e.g., n=10, 20, 50) to establish general validity

Polymatroid Rank Deficit in Monotone DNF for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Polymatroid Theory × Monotone DNF Formulas
- **Recorded**: 2026-05-08 22:24 UTC
- **Entry ID**: `1562169c18c2`

Statement

For any k-CLIQUE instance with n variables, the polymatroid rank of its clause hypergraph satisfies rank(H) ≥ Ω(n/k). For any monotone DNF formula with size S(n) = poly(n), the rank(H) is bounded by O(log n).

Rationale

Polymatroid rank captures combinatorial structure of clause interactions, which may expose inherent limitations in monotone DNF representations. The conjecture links hypergraph rank to circuit size, leveraging submodularity to isolate k-CLIQUE's complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8e369c5b.py", line 87, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8e369c5b.py", line 67, in run_trial
    if is_monotone_dnf(cnf) and rank > math.log(n):
       ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8e369c5b.py", line 41, in is_monotone_d
    if any(var.startswith('-') for var in clause):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8e369c5b.py", line 41, in <genexpr>
    if any(var.startswith('-') for var in clause):
           ^^^^^^^^^^^^^^
AttributeError: 'int' object has no attribute 'startswith'
```

Judge reasoning

Test crashed due to invalid type in clause processing, preventing result evaluation | next: Fix data input format to ensure clauses are string variables, then retest

---

Symmetric Power Rank Gap in CNF Incidence Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Boolean Circuit Complexity
- **Recorded**: 2026-05-08 22:53 UTC

- **Entry ID**: `2e234361503d`

Statement

For a random 3-CNF formula Φ on n variables, the rank of the symmetric square of its incidence matrix M satisfies rank(M^⊗2) ≥ Ω(n^{1.5}), whereas for a random 2-CNF formula Ψ on n variables, rank(M^⊗2) = O(n^{1.2}) with probability ≥ 0.95 over 30 seeds.

Rationale

Symmetric powers of incidence matrices encode combinatorial structure that may correlate with circuit complexity. The gap in rank growth between 3-CNF and 2-CNF could reflect inherent differences in their algebraic representations under Schur-Weyl duality.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a13cfc5c.py", line 75, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a13cfc5c.py", line 52, in run_trial
    rank_M3 = rank(M3)
              ^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a13cfc5c.py", line 30, in rank
    A = gaussian_elimination(A)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a13cfc5c.py", line 24, in gaussian_elim
    factor = A[j][i] / A[i][i]
             ~~~~~~~~^~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with division by zero error, preventing data collection. Cannot assess conjecture validity without successful runs. | next: Debug Gaussian elimination implementation to handle singular matrices, then re-run with 30 seeds

---

Real Dimension Gap in Communication Matrices for AC^0 PARITY

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × Communication Complexity
- **Recorded**: 2026-05-08 23:16 UTC
- **Entry ID**: `54a84a6f394b`

Statement

For any AC^0 circuit C computing the parity function on n bits, the real dimension of the solution set to the system of equations defined by the communication matrix of C is Ω(√n).

Rationale

The real dimension of the solution set captures the geometric complexity of the communication matrix, which may be tied to the circuit's size. High real dimension could imply that the function requires larger circuits.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d1a3e697.py", line 107, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d1a3e697.py", line 76, in run_trial
    dim = rank(A)
          ^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d1a3e697.py", line 52, in rank
    rref = gaussian_elimination(A)
           ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d1a3e697.py", line 25, in gaussian_elim
    factor = A[j][i] / A[i][i]
             ~~~~~~~~^~~~~~~~~
ZeroDivisionError: division by zero
```

Judge reasoning

Test crashed with division-by-zero error, preventing dimension calculation. The crash indicates a potential flaw in the matrix construction or Gaussian elimination implementation. | next: Debug the Gaussian elimination code to handle singular matrices, then re-run with stabilized numerical methods

---

Noncommutative Fourier Spectrum Gap in Read-Twice BP for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative harmonic analysis × Read-twice branching programs for IP_2
- **Recorded**: 2026-05-08 23:44 UTC
- **Entry ID**: `44a8ac95f70e`

Statement

For any read-twice BP P with n variables, the spectral norm of its Fourier transform over the symmetric group S_n is O(log n), but for the IP_2 problem, it is Ω(n).

Rationale

Noncommutative harmonic analysis decomposes BP transition matrices into irreducible representations, revealing structural asymmetries. The spectral norm captures the 'spread' of Fourier coefficients, which should be small for structured BPs but large for inherently complex problems like IP_2.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `-9`, elapsed: `216.07s`

```
(no output)
```

Judge reasoning

Test crashed before producing data, preventing evaluation of support fraction or counterexamples. | next: Investigate crash logs for test code or runtime environment issues

---

Hyperbolicity Lower Bound on Tseitin Resolution Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Hyperbolic Geometry × Resolution Length
- **Recorded**: 2026-05-08 23:56 UTC
- **Entry ID**: `844cf870afe5`

Statement

For a Tseitin formula on a graph G, its resolution length is ≥ 2^{Ω(δ(G))}, where δ(G) is Gromov hyperbolicity. For graphs with δ(G) ≤ 1 (tree-like), resolution length is polynomial in n.

Rationale

Hyperbolicity captures geometric constraints on shortest paths; graphs with higher δ(G) force resolution proofs to traverse exponentially more clauses due to non-tree-like detours, amplifying proof complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06a868ed.py", line 133, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06a868ed.py", line 105, in run_trial
    delta_G = gromov_hyperbolicity(graph)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06a868ed.py", line 76, in gromov_hyperb
   delta = (triangle_distance(a, b, c) + triangle_distance(a, c, b) - triangle_distance(b, c, a)) / 2
            ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06a868ed.py", line 70, in triangle_dist
   return min(len(graph[a] & graph[b]), len(graph[b] & graph[c]), len(graph[c] & graph[a]))
                  ~~~~~~~~~^~~~~~~~~
TypeError: unsupported operand type(s) for &: 'list' and 'list'
```

Judge reasoning

Test crashed due to TypeError in hyperbolicity calculation, preventing data collection | next: Fix the list intersection operator error in gromov_hyperbolicity implementation